

FORMATION OF GERM LAYERS
AND EARLY TISSUE AND
ORGAN DIFFERENTIATION:
THIRD WEEK

- **The third week** of embryonic development coincides:
 - with the week following the first missed menstrual period;
 - that is, 5 weeks after the first day of the last normal menstrual period.

Cessation of menstruation is often the first indication that a woman may be pregnant

≈ **3 weeks after conception**, OR

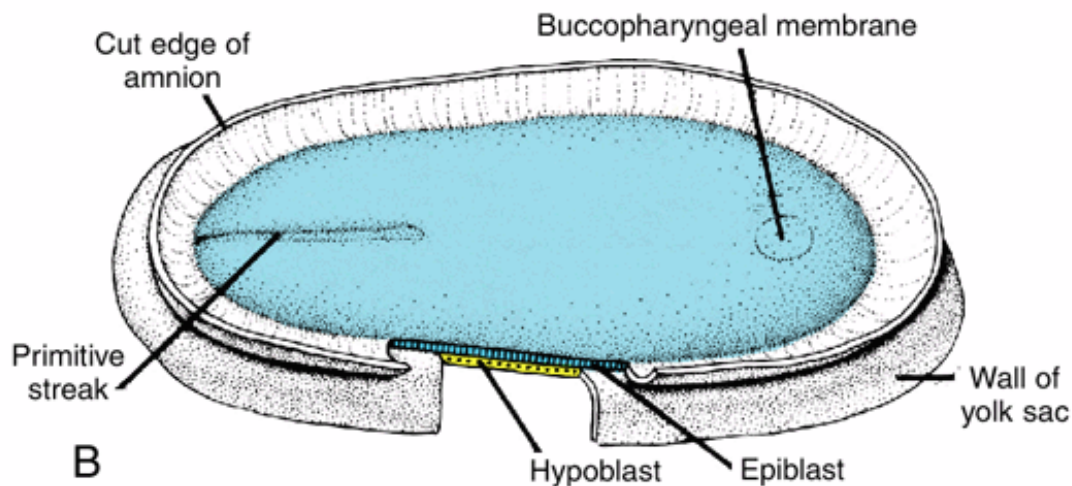
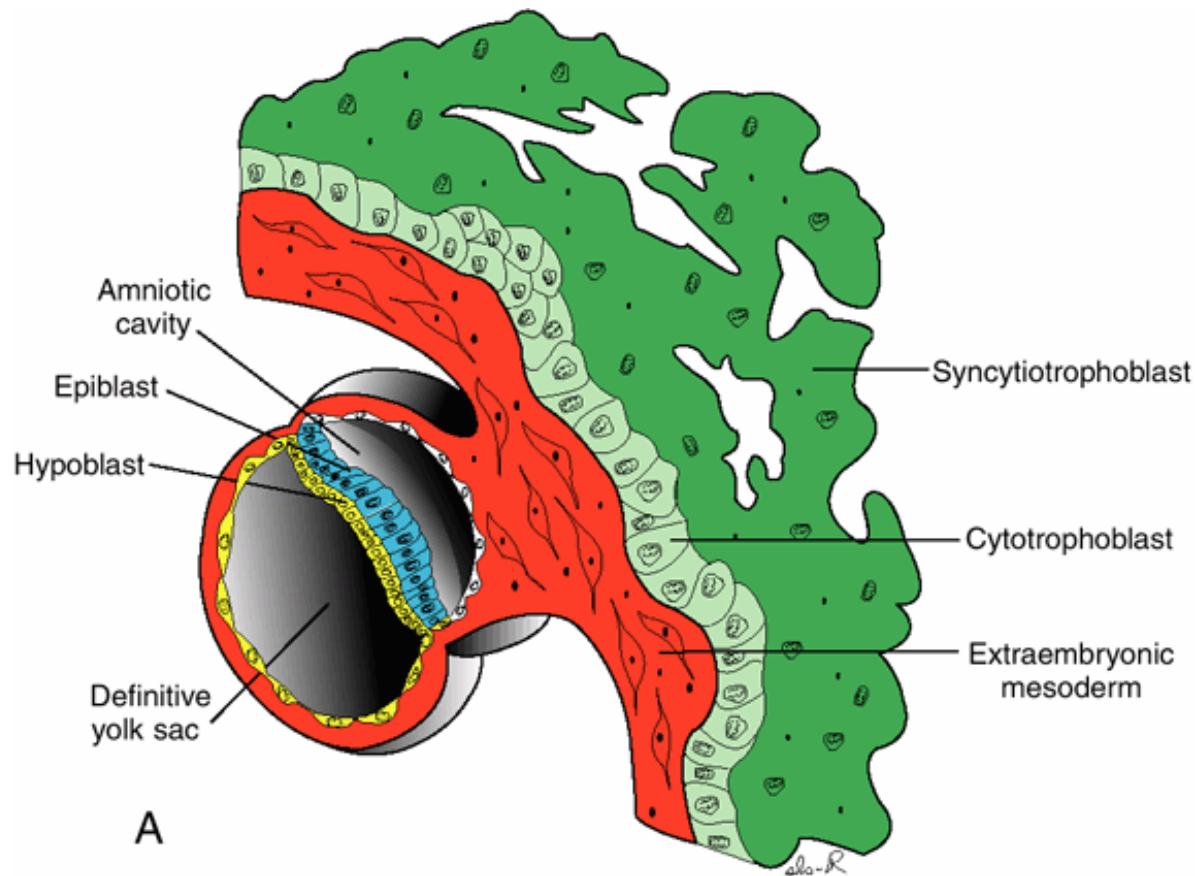
≈ **5 weeks after** the LNMP, a normal pregnancy can be detected with **ultrasonography**

Pregnancy Symptoms

- Frequent symptoms of pregnancy are *nausea and vomiting*, which may occur by the end of the third week; however, the time of onset of these symptoms varies.
- **Vaginal bleeding** at the expected time of menstruation does not rule out pregnancy because there may be a slight loss of blood from the implantation site of the blastocyst.
- **Implantation bleeding** results from leakage of blood into the uterine cavity from disrupted lacunar networks in the implanted blastocyst.
 - ✓ *When this bleeding is interpreted as menstruation, an error occurs in determining the expected date of confinement and delivery date of the baby.*

- Rapid development of the embryo from the embryonic disc during the **third week** is characterized by:

1. Differentiation of three germ layers (gastrulation)
2. Appearance of primitive streak
3. Development of notochord



- A. Implantation site at the end of the second week.
- B. Representative view of the germ disc at the end of the second week of development. The amniotic cavity has been opened to permit a view of the dorsal side of the **epiblast**. The **hypoblast** and epiblast are in contact with each other, and the primitive streak forms a shallow groove in the caudal region of the embryo.

GASTRULATION: FORMATION OF GERM LAYERS

Gastrulation:

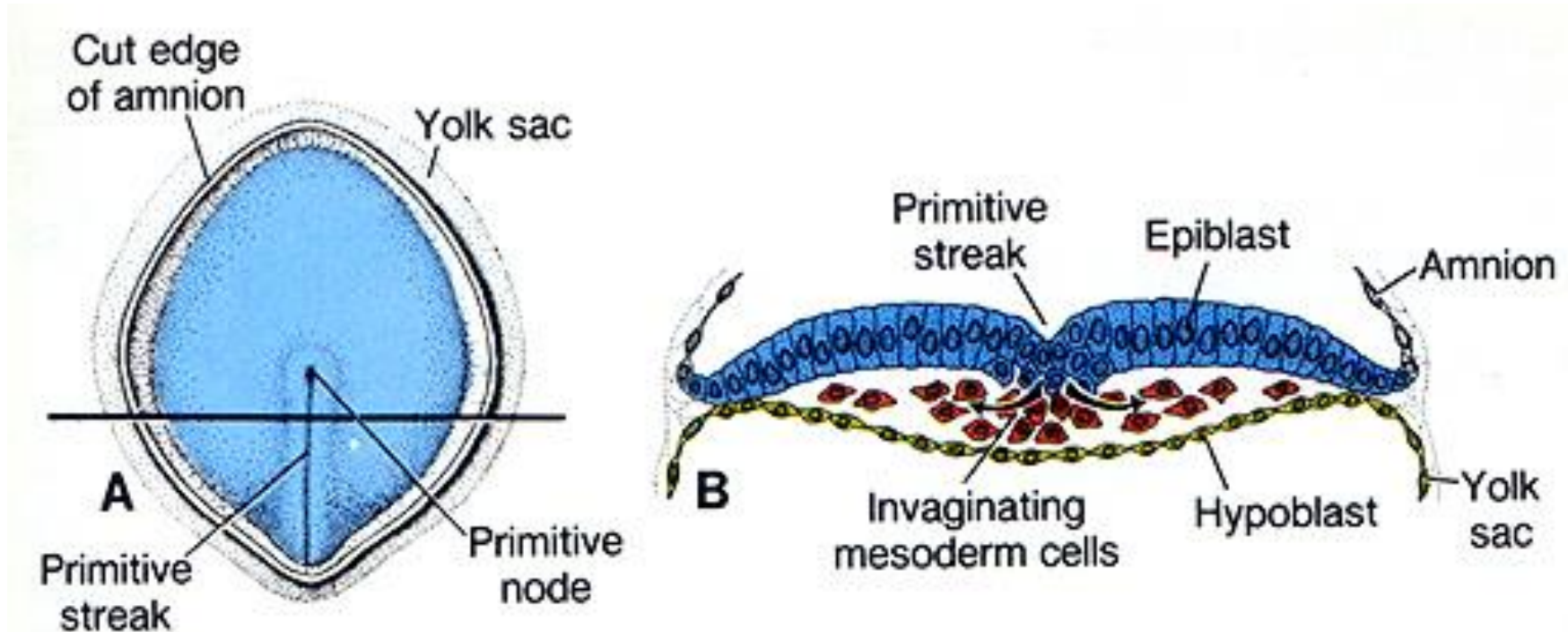
- the formative process by which the three germ layers,
 - which are precursors of all embryonic tissues, and axial orientation are established in embryos
- the process by which the bilaminar disc is converted into **a trilaminar embryonic disc.**
- extensive cell shape changes, rearrangement, movement, and changes in adhesive properties contribute to the **process of gastrulation.**
- the beginning of **morphogenesis** (development of body form)
- **Is the significant event occurring during the 3rd week.**
- Consists of formation of the **primitive streak**, the three germ layers & the **notochord**
- The first morphologic sign of gastrulation begins with formation of the ***primitive streak*** on the surface of the **epiblast** of the embryonic disc
- During this period, the embryo referred to as a **gastrula**

Each of the three germ layers (ectoderm, mesoderm, and endoderm) gives rise to specific tissues and organs:

- **Embryonic ectoderm** gives rise to the epidermis, central and peripheral nervous systems, the eye, and inner ear, and, as neural crest cells, to many connective tissues of the head.
- **Embryonic endoderm** is the source of the epithelial linings of the respiratory and alimentary (digestive) tracts, including the glands opening into the gastrointestinal tract and the glandular cells of associated organs such as the liver and pancreas.
- **Embryonic mesoderm** gives rise to all skeletal muscles, blood cells and the lining of blood vessels, all visceral smooth muscular coats, the serosal linings of all body cavities, the ducts and organs of the reproductive and excretory systems, and most of the cardiovascular system.
 - In the trunk, it is the source of all connective tissues, including cartilage, bones, tendons, ligaments, dermis, and stroma of internal organs.

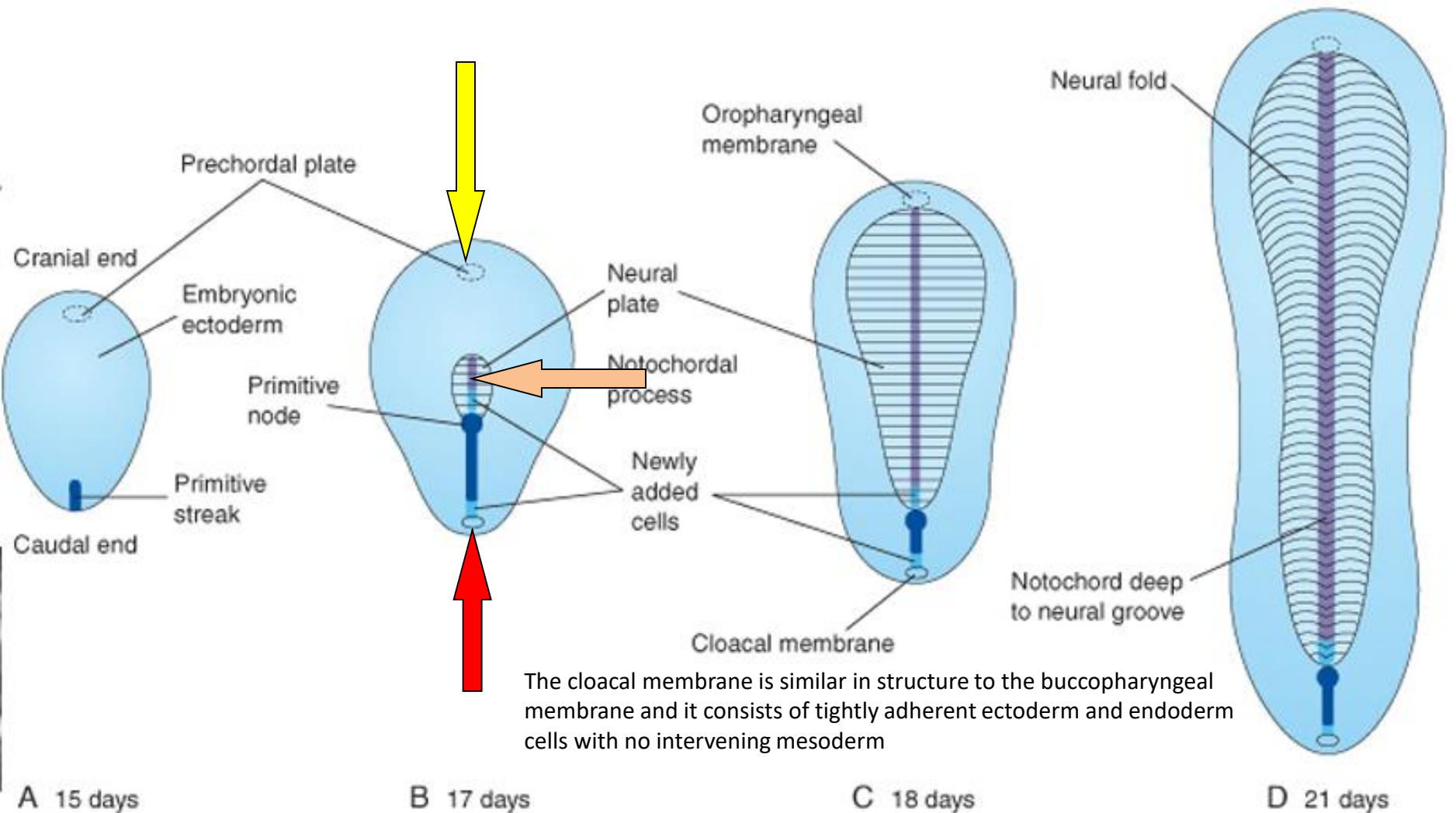
Formation of primitive streak

- epiblastic cells → **primitive streak** (primitive groove)
primitive node (primitive pit)
- { the caudal end
the symmetrical axis
mesoderm



PRIMITIVE STREAK

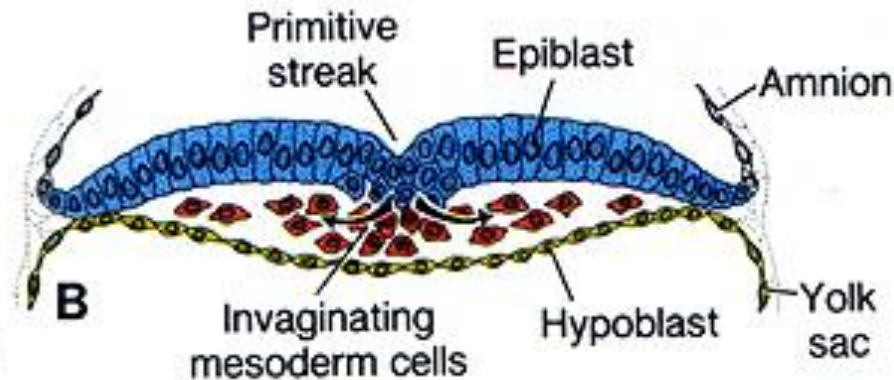
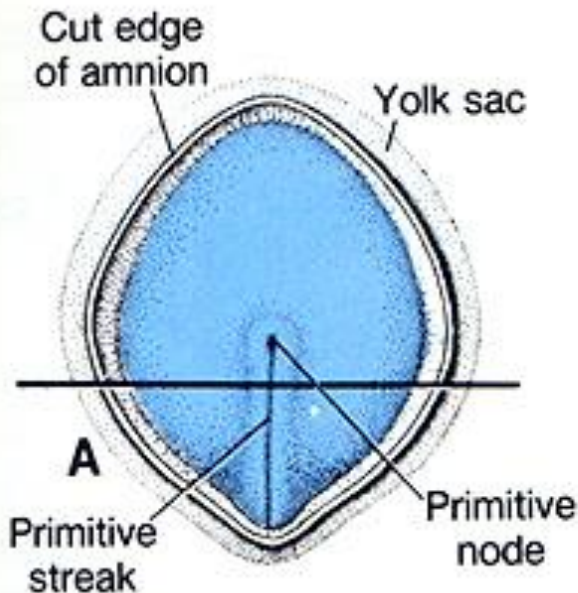
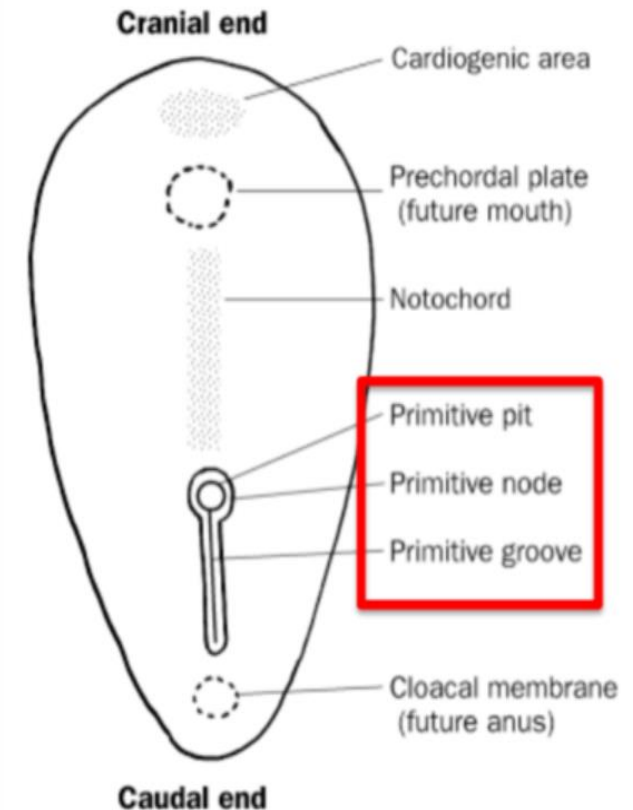
- The first sign of gastrulation
- Forms at the beginning of the **third week**
- formed by **a thickened linear band of epiblast** in the caudal half of the epiblast, in the median plane of the dorsal aspect of the embryonic disc
- and lies along the cranio-caudal axis
- results from the proliferation and movement of cells of the **epiblast** to the median plane of the embryonic disc.
- As the streak elongates by addition of cells to its caudal end, its cranial end proliferates to form a **primitive node**
- a narrow groove-**primitive groove**-develops in the primitive streak that is continuous with a small depression in the primitive node-the **primitive pit**.

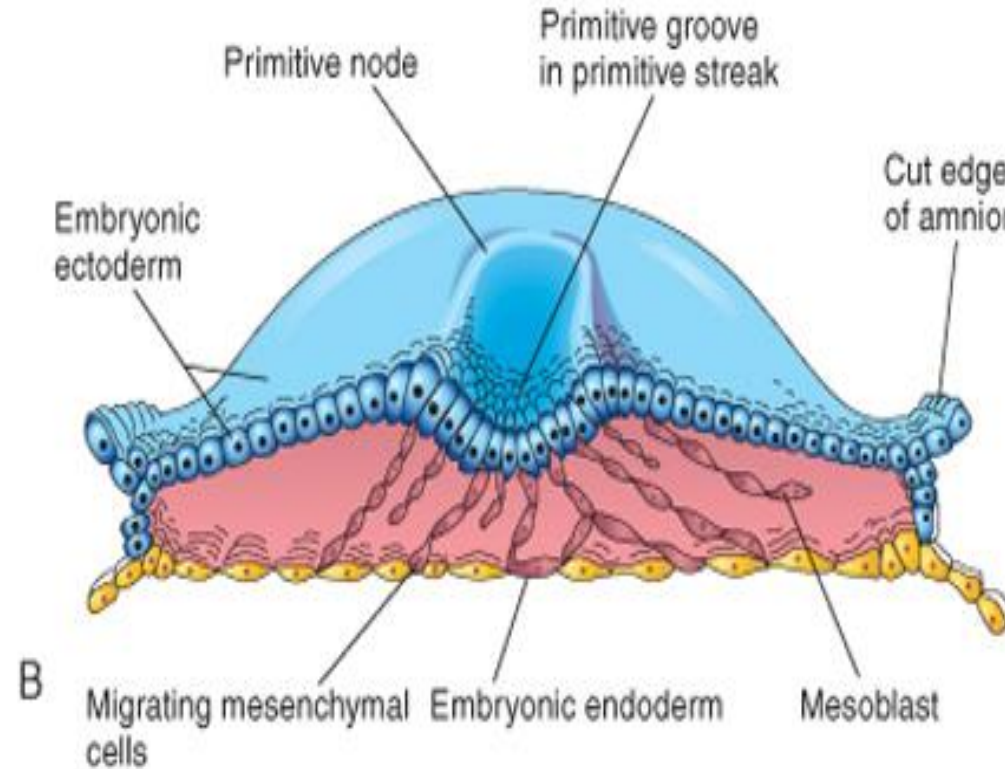
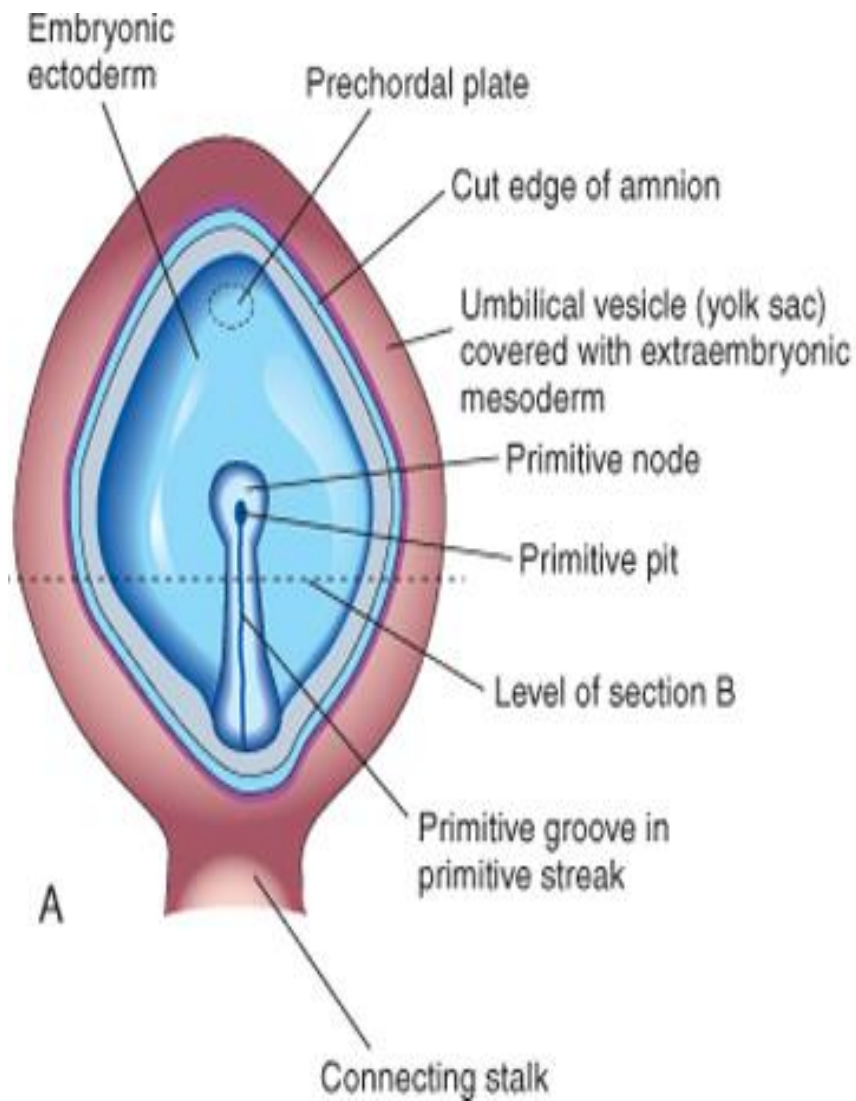


Diagrammatic sketches of dorsal views of the embryonic disc showing how it lengthens and changes shape during the third week. The **primitive streak** lengthens by addition of cells at its caudal end, and the **notochordal process** lengthens by migration of cells from the primitive node. The notochordal process and adjacent mesoderm induce the overlying embryonic ectoderm to form the **neural plate**, the primordium of the central nervous system. Observe that as the notochordal process elongates, the primitive streak shortens. **At the end of the third week, the notochordal process is transformed into the notochord.**

- As the primitive streak appears, it is possible to identify the embryo's:

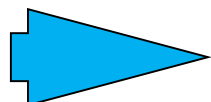
- craniocaudal axis
- its cranial and caudal ends
- its dorsal and ventral surfaces
- and its right and left sides.





The primitive groove and pit result from the invagination (inward movement) of epiblastic cells.

- Soon after the primitive streak appears, cells leave its deep surface and form **mesenchyme**
- **Mesenchymal cells** are ameboid and actively phagocytic.
 - Mesenchyme forms the supporting tissues of the embryo, such as most of the CT of the body and the CT framework of glands.
 - Some mesenchyme forms **mesoblast** (undifferentiated mesoderm), which forms the **intraembryonic, or embryonic, mesoderm**.
- Cells from the epiblast as well as from the primitive node and other parts of the primitive streak displace the hypoblast, forming the **embryonic endoderm** in the roof of the umbilical vesicle.
- The cells remaining in the epiblast form the **embryonic ectoderm**.
- Mesenchymal cells derived from the primitive streak migrate widely.
 - These pluripotential cells have the potential to proliferate and differentiate into diverse types of cells, such as
 - fibroblasts,
 - chondroblasts, and
 - osteoblasts.



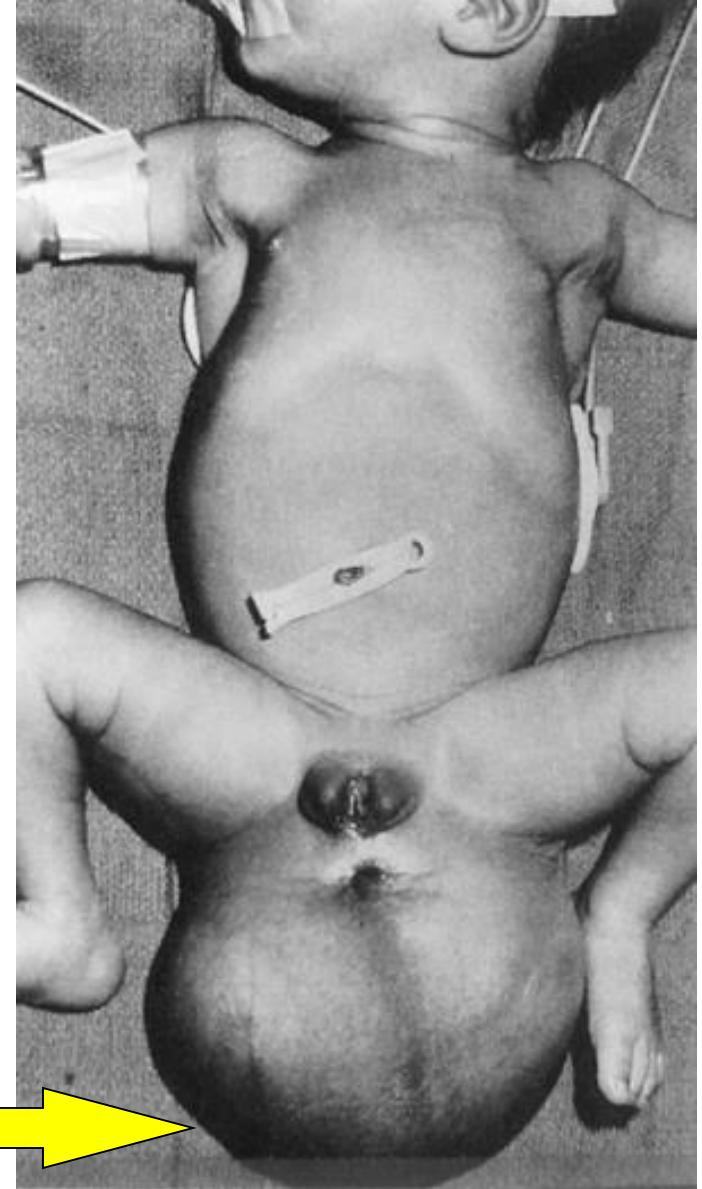
In summary, cells of the epiblast, through the process of gastrulation, give rise to all three germ layers in the embryo, the primordia of all its tissues and organs.

Fate of the Primitive Streak

- The primitive streak actively:
 - forms mesoderm by the ingression of cells until the early part of the **4th week**
 - Diminishes in relative size and becomes an insignificant structure in the sacrococcygeal region of the embryo
 - Normally, the primitive streak undergoes degenerative changes and disappears by **the end of the 4th week**
 - Remnants of the primitive streak may persist and give rise to a **sacrococcygeal teratoma**
 - Because they are derived from **pluripotent** primitive streak cells, these tumors contain tissues derived from all three germ layers in incomplete stages of differentiation



Female infant with a large **sacroccygeal teratoma** that developed from ***remnants of the primitive streak***. The tumor, a neoplasm made up of several different types of tissue, was surgically removed.



Sacroccygeal teratoma resulting from remnants of the primitive streak. These tumors may become malignant and are most common in ***female fetuses***.

NOTOCHORDAL PROCESS AND NOTOCHORD

- Some mesenchymal cells that migrate cranially from the primitive node and pit, forms a median cellular cord, the **notochordal process**
- This process soon acquires a lumen, the **notochordal canal**.
- The notochordal process grows cranially between the ectoderm and endoderm until it reaches the **prechordal plate (the future site of the mouth)**,
 - In this area the ectoderm is attached directly to the endoderm without intervening mesoderm.
 - This area is known as the **oropharyngeal membrane**, and it will break down to become the **mouth**.

■ The prechordal plate:

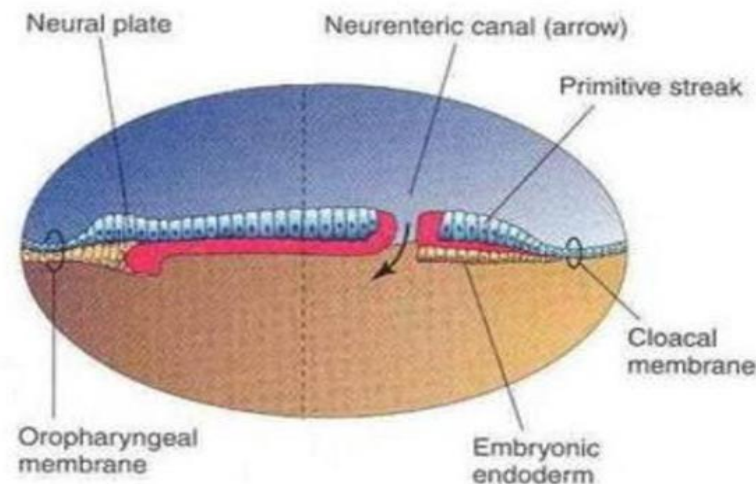
- is the primordium of the **oropharyngeal membrane**, located at the future site of the oral cavity
- Prechordal mesoderm is a mesenchymal population rostral to the notochord and essential in **forebrain and eye induction**.
- At the other end of the primitive streak the ectoderm is also fused directly to the endoderm; this is known as the **cloacal membrane (proctodeum)**, or primordial **anus**.
- The embryonic disc remains bilaminar here and at the **oropharyngeal membrane** because the embryonic ectoderm and endoderm are fused at these sites
- By the **middle of the third week**, intraembryonic mesoderm separates the ectoderm and endoderm everywhere except:
 - ✓ At the **oropharyngeal membrane** cranially
 - ✓ In the median plane cranial to the primitive node, where the **notochordal process** is located
 - ✓ At the **cloacal membrane** caudally

- Some mesenchymal cells from the primitive streak that have mesodermal fates migrate cranially on each side of the notochordal process and around the prechordal plate.
- Here they meet cranially to form cardiogenic mesoderm in the **cardiogenic area** where the **heart primordium** begins to develop at the end of the third week

Notochord

- Mesenchymal cells migrate cranially from the primitive pit towards the prechordal plate, and form a rod like notochordal process
- The notochord will eventually become *the nucleus pulposus of each intervertebral disk*.
- The notochordal process becomes canalized forming a hollow tube, the notochordal canal, communicating with the primitive pit.
- A temporary communication is established between the amniotic cavity and the yolk sac, termed the **neurenteric canal**.

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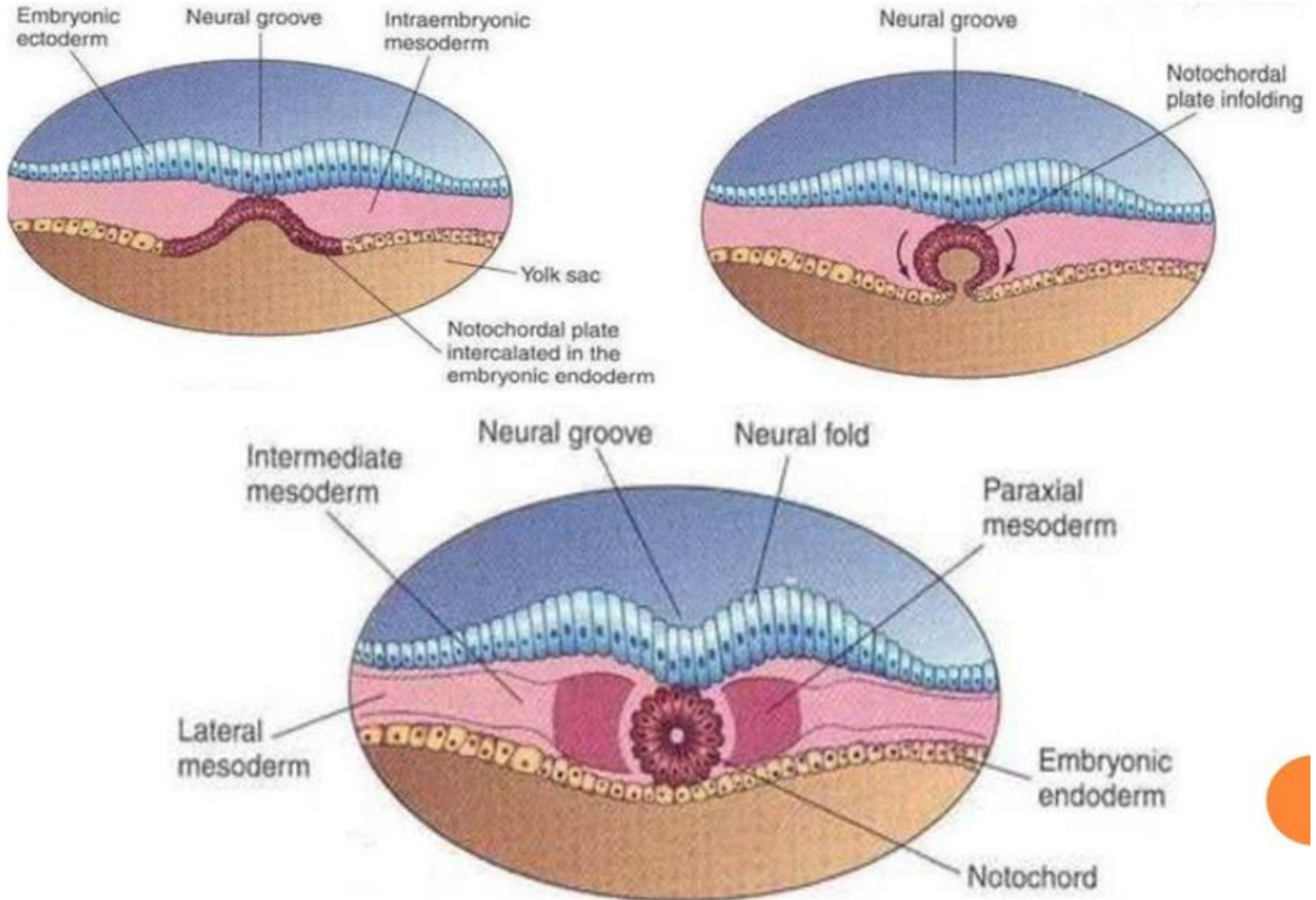


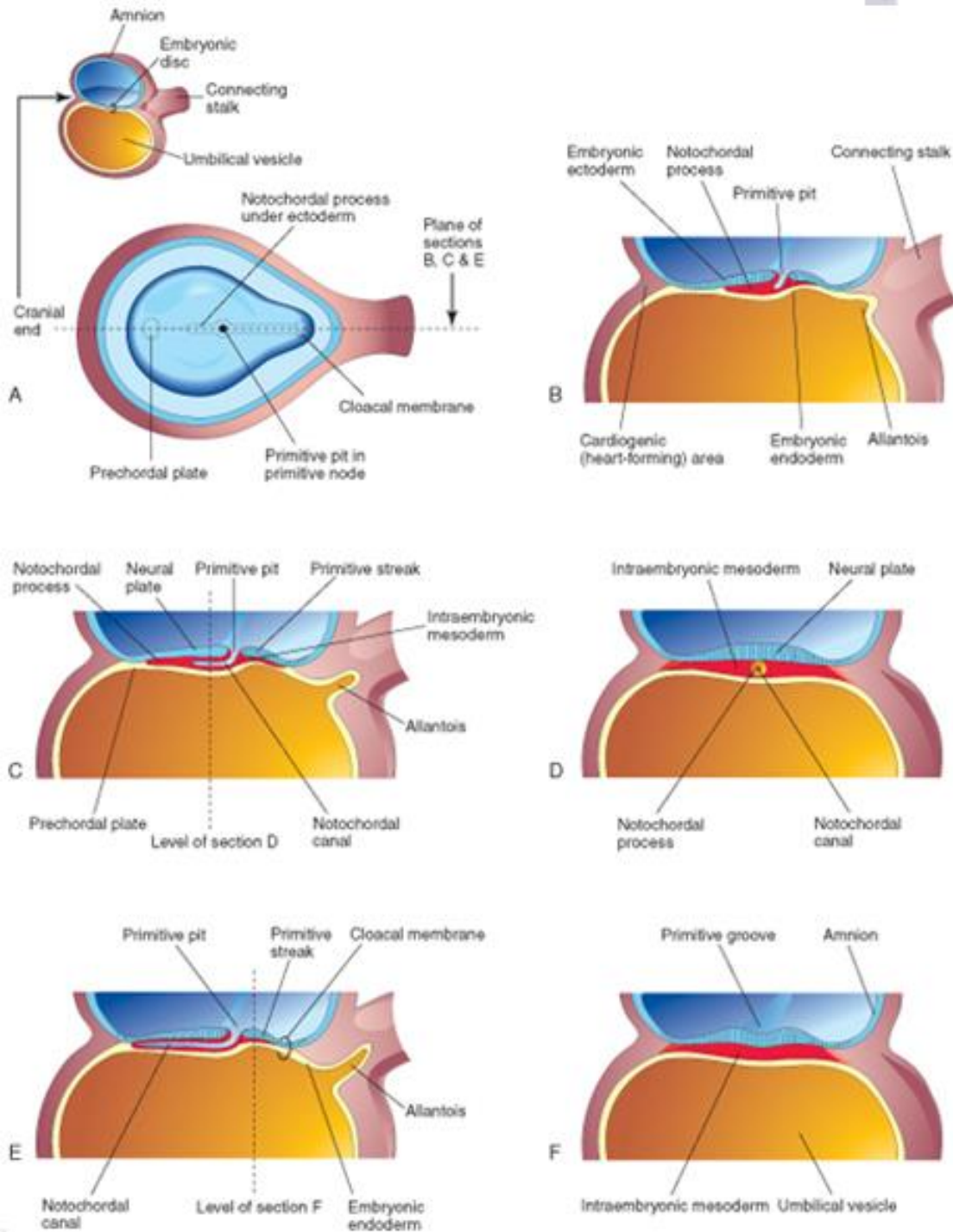
The notochord develops as follows:

- The **notochordal process** elongates by invagination of cells from the primitive pit.
- The primitive pit extends into the notochordal process, forming a **notochordal canal**
- The notochordal process is now a cellular tube that extends cranially from the primitive node to the prechordal plate.
- The floor of the notochordal process fuses with the underlying embryonic endoderm
- The fused layers gradually undergo degeneration, resulting in the formation of openings in the floor of the notochordal process, which brings the notochordal canal into communication with the umbilical vesicle
- The openings rapidly become confluent and the floor of the notochordal canal disappears; the remains of the notochordal process form a flattened, grooved **notochordal plate**

- Beginning at the cranial end of the embryo, the notochordal cells proliferate and **the notochordal plate infolds** to form the **notochord**
- The proximal part of the notochordal canal persists temporarily as the *neurenteric canal*, which forms a transitory communication between the amniotic and umbilical vesicle cavities
- When development of the notochord is complete, the neurenteric canal normally obliterates
- The notochord becomes detached from the endoderm of the umbilical vesicle, which again becomes a continuous layer.

NOTOCHORDAL PLATE FOLDS TO FORM THE NOTOCHORD.





Illustrations of the development of the notochordal process.

The small sketch at the upper left is for orientation.

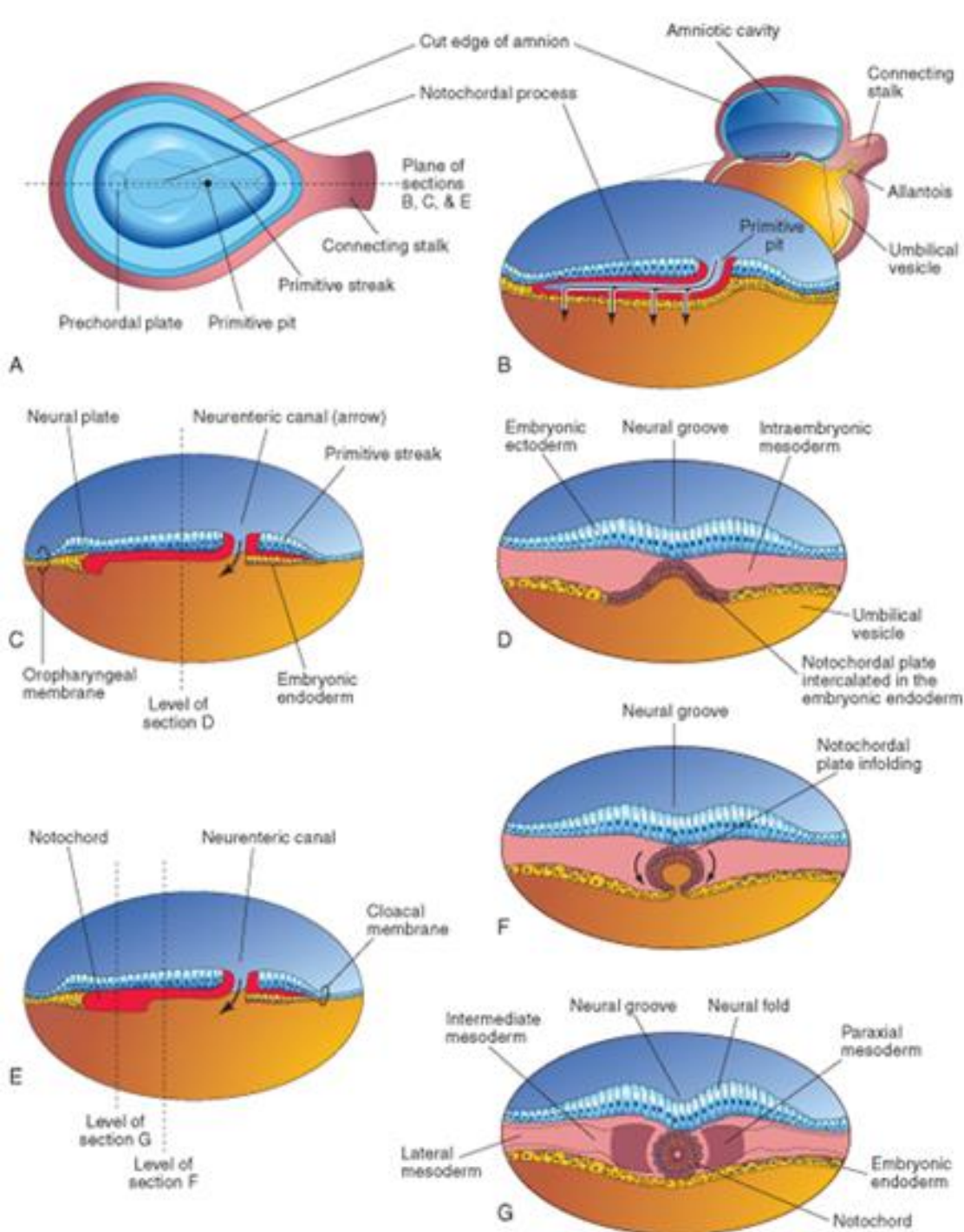
A, Dorsal view of the embryonic disc (approximately 16 days) exposed by removal of the amnion.

The notochordal process is shown as if it were visible through the embryonic ectoderm.

B, C, and E, Median sections at the plane shown in **A**, illustrating successive stages in the development of the notochordal process and canal.

The stages shown in **C** and **E** occur at approximately 18 days.

D and F, Transverse sections through the embryonic disc at the levels shown in **C** and **E**.



gastrulation, or formation of the germ layers, continues in caudal segments while cranial structures are differentiating, causing the embryo to develop cephalocaudally.

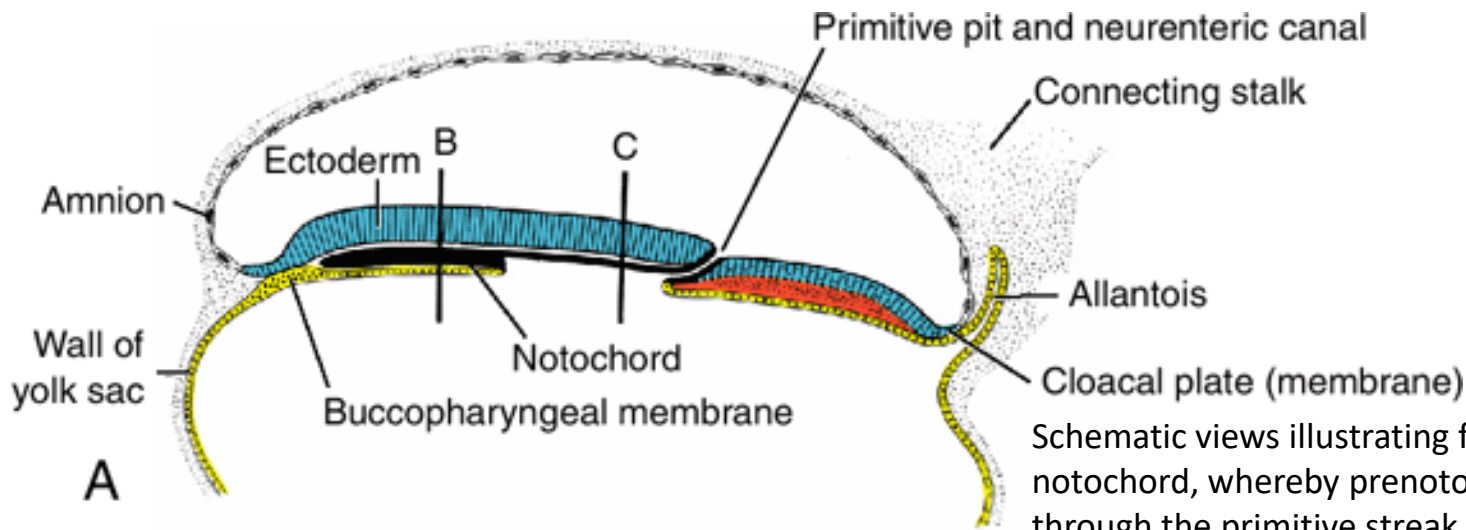
Illustrations of the further development of the notochord by transformation of the notochordal process.

A, Dorsal view of the embryonic disc (approximately 18 days), exposed by removing the amnion.

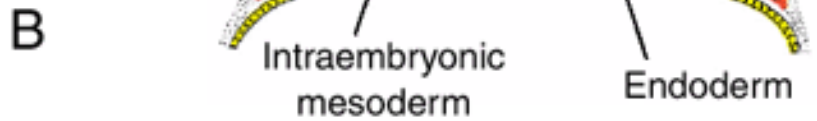
B, Three-dimensional median section of the embryo.

C and **E**, Similar sections of slightly older embryos.

D, **F**, and **G**, Transverse sections of the trilaminar embryonic disc at the levels shown in **C** and **E**.

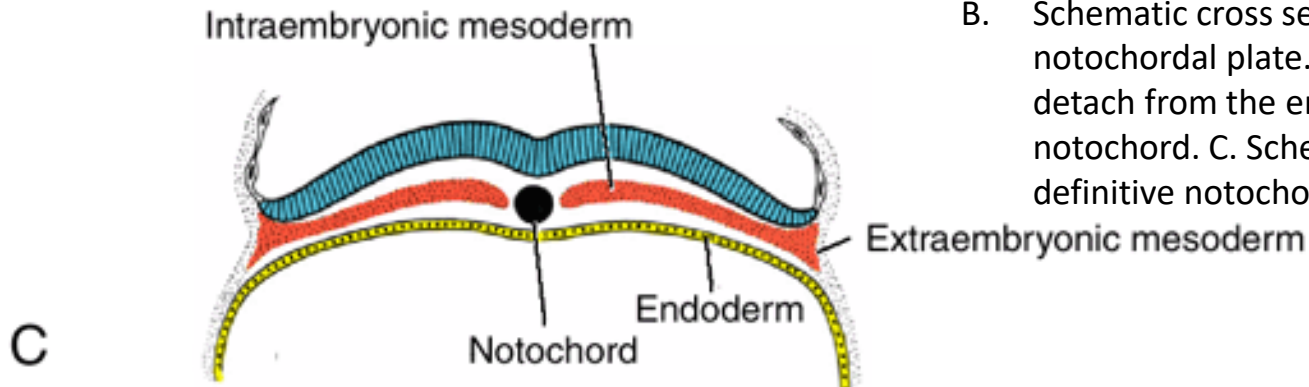


Schematic views illustrating formation of the notochord, whereby prenotochordal cells migrate through the primitive streak, become intercalated in the endoderm to form the notochordal plate, and finally detach from the endoderm to form the definitive notochord. Because these events occur in a cranial-to-caudal sequence, portions of the definitive notochord are established in the head region first.



A. Drawing of a sagittal section through a 17-day embryo. The most cranial portion of the definitive notochord has formed, while prenotochordal cells caudal to this region are intercalated into the endoderm as the notochordal plate.

B. Schematic cross section through the region of the notochordal plate. Soon the notochordal plate will detach from the endoderm to form the definitive notochord. C. Schematic view showing the definitive notochord.

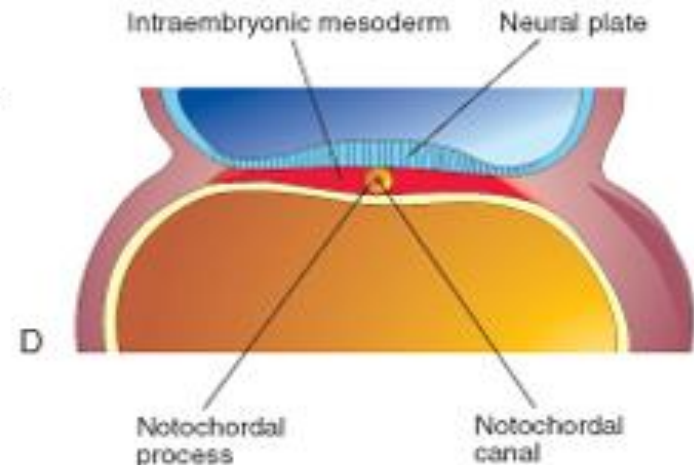
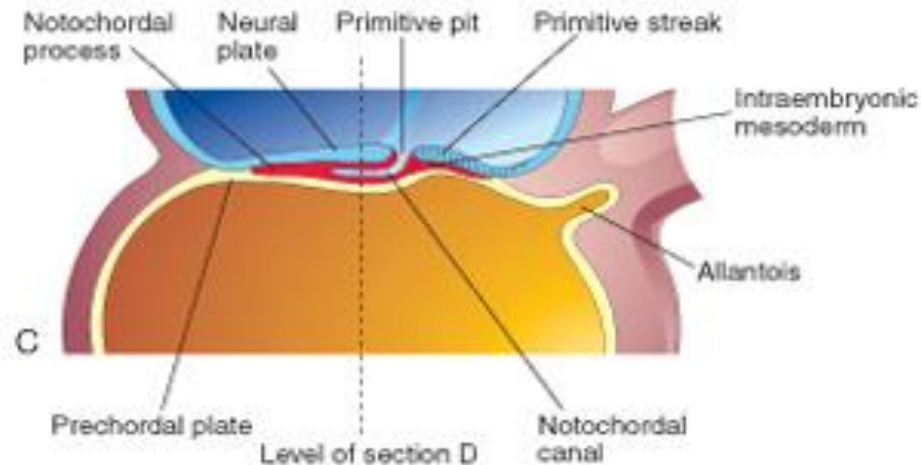
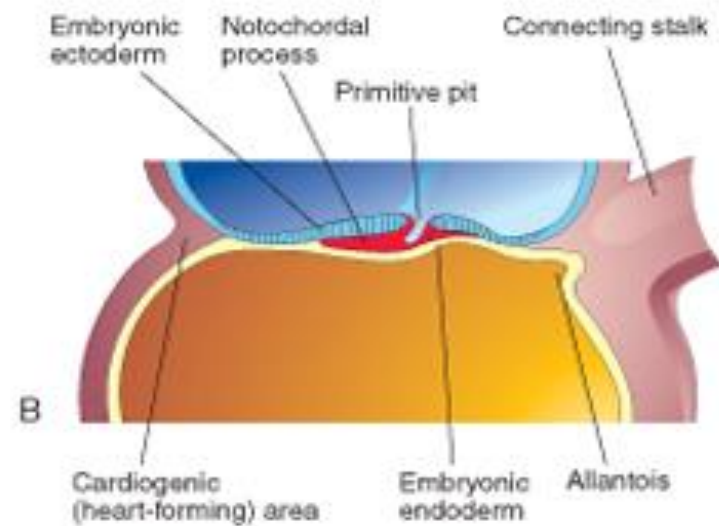
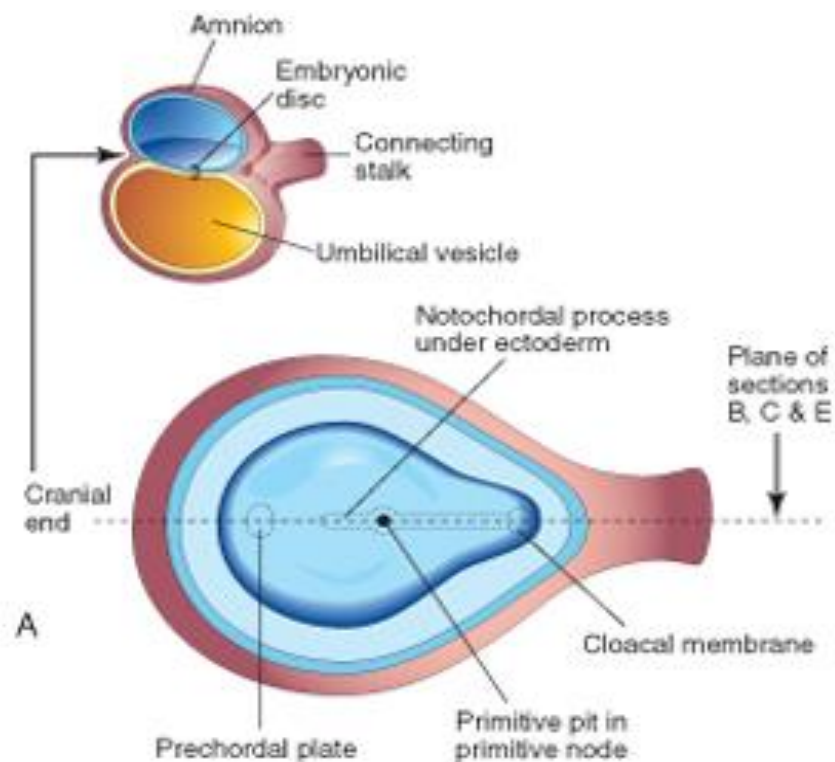


■ **Functions of Notochord**

- Defines primordial axis of the embryo
- Provides rigidity to the embryo
- Serves as a basis for the development of the axial skeleton
- Indicates the future site of the vertebral bodies/column
- Regulates differentiation of surrounding structures including the overlying ectoderm (neural plate) and mesoderm (somites).

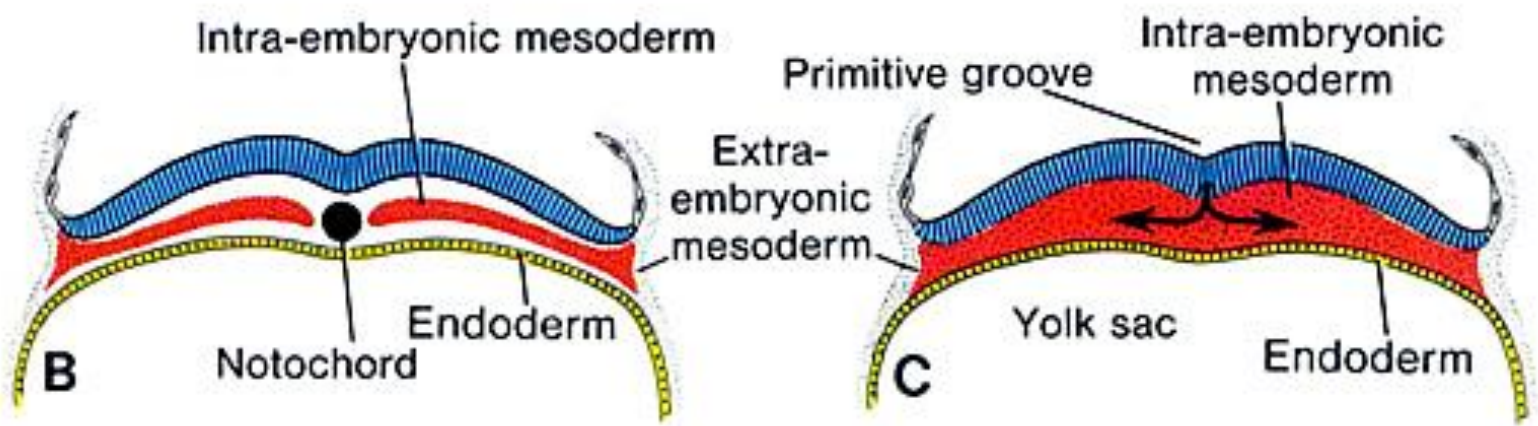
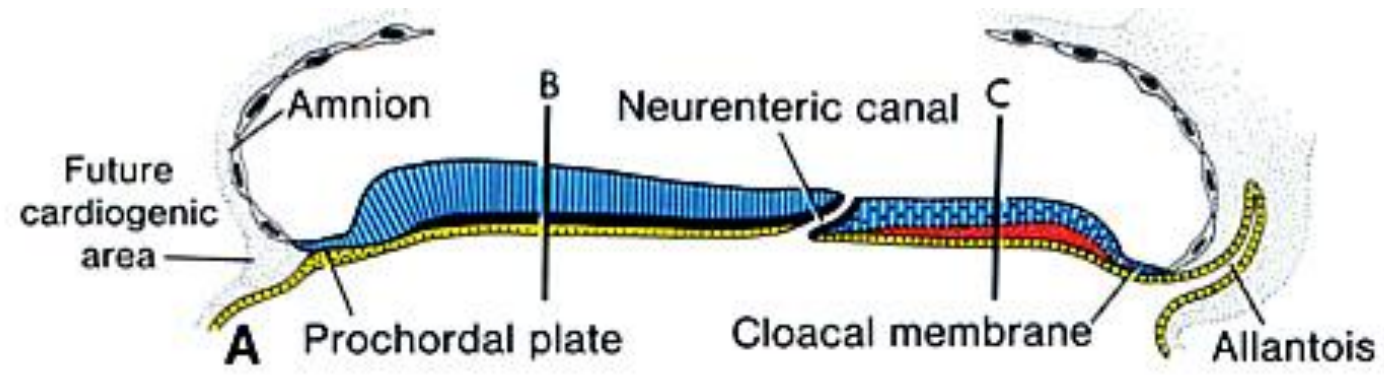
- **Fate of Notochord**

- Degenerates and disappears as the bodies of the vertebrae develop, but it persists as the nucleus pulposus of each [intervertebral disc](#)
- Remnants of notochordal tissue give rise to tumors called **Chordomas**
- Both benign and malignant tumors (**chordomas**) may form from **vestigial remnants of notochordal tissue**
- Approximately **one third of chordomas** occur at the base of the **cranium** and extend to the **nasopharynx**
- Chordomas grow slowly and malignant forms infiltrate bone.



• **Formation of mesoderm**

- 1) Cells of the **epiblast** ↑ primitive streak
hypoblast and spread between epiblast and
intraembryonic mesoderm.



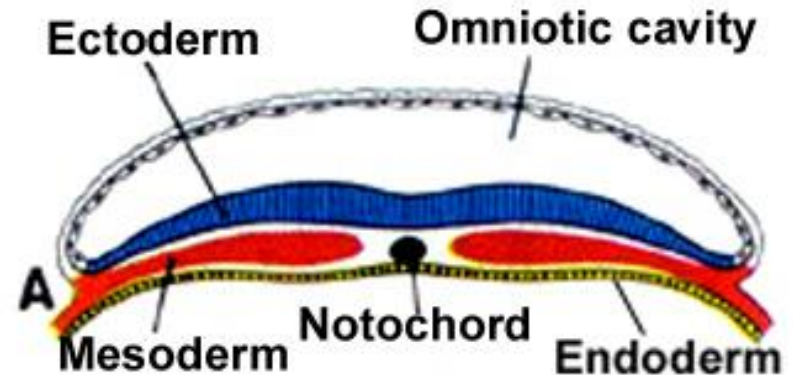
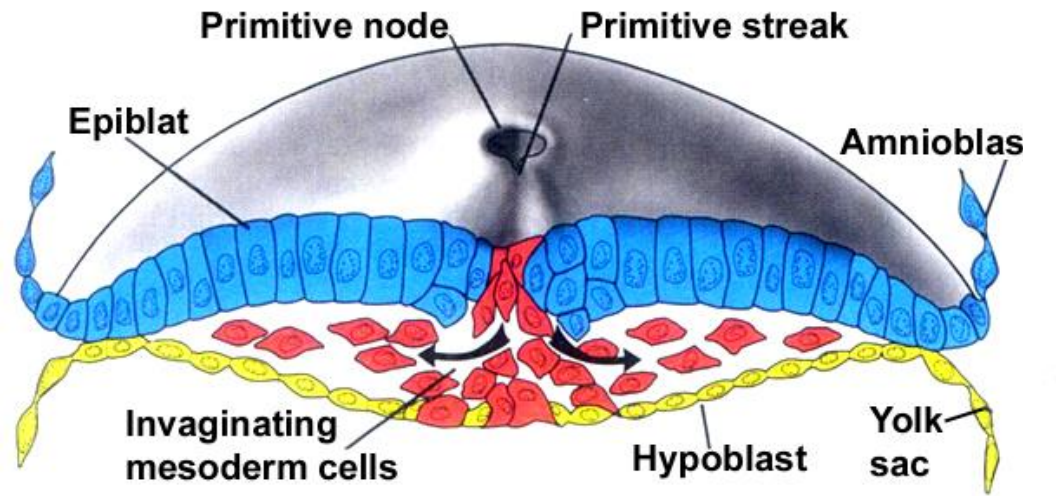
- Formation of trilaminar germ disc
- Epiblast cells →

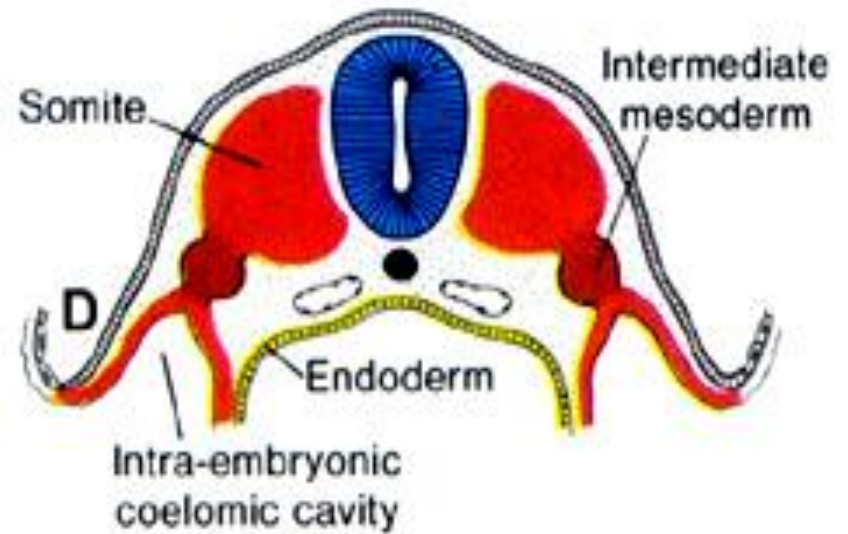
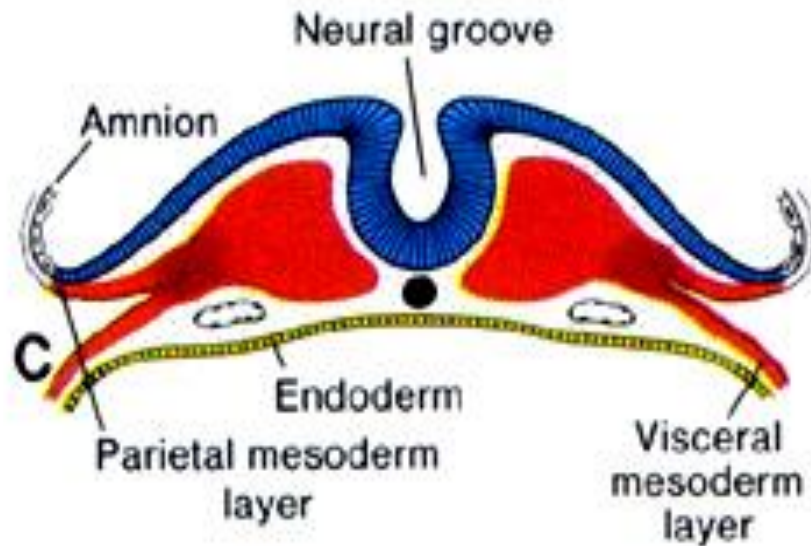
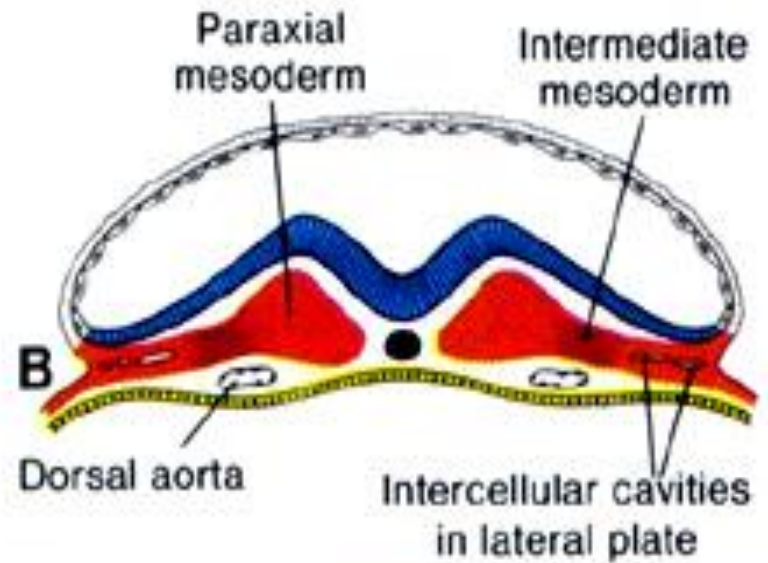
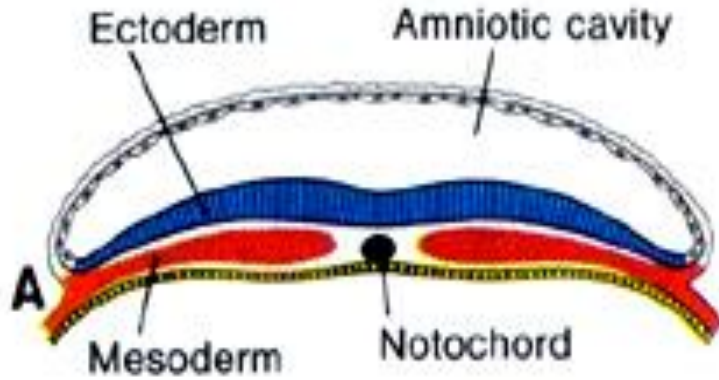
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ectoderm
mesoderm
endoderm

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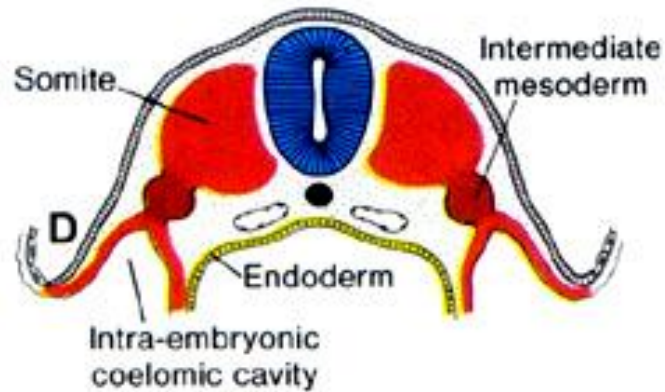
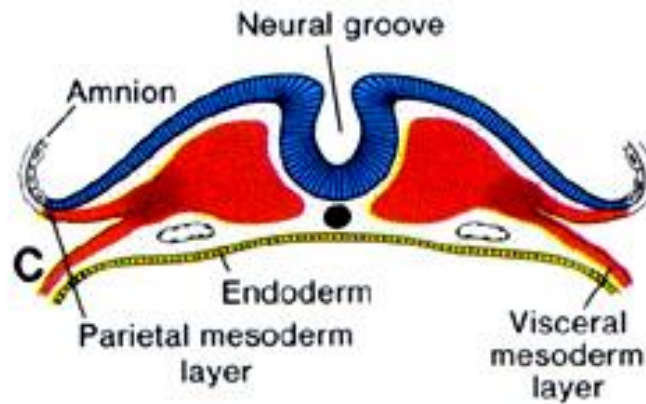
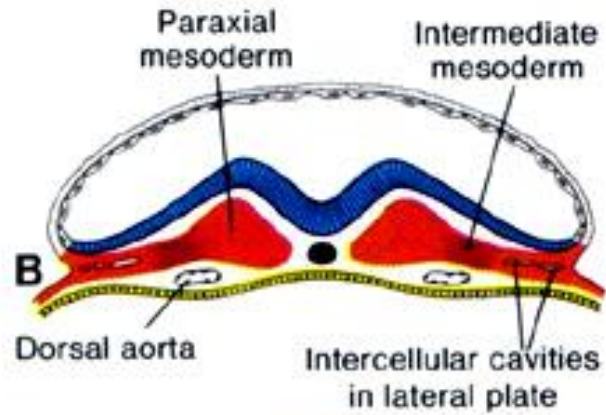
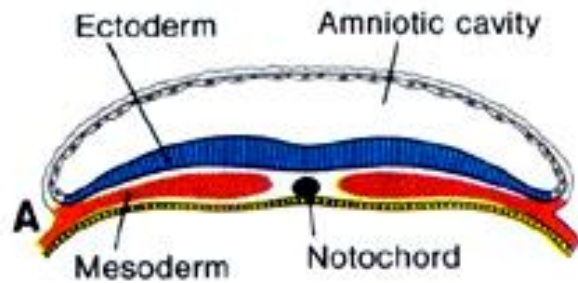
 trilaminar germ disc





DEVELOPMENT OF THE INTRAEMBRYONIC COELOM

- The primordium of the intraembryonic coelom (embryonic body cavity)
 - appears as isolated **coelomic spaces** in the lateral mesoderm and **cardiogenic (heart-forming) mesoderm**.
 - These spaces soon coalesce to form a single horseshoe-shaped cavity, **the intraembryonic coelom**
 - The cavity divides the lateral mesoderm into two layers:
 - A **somatic or parietal layer** of lateral mesoderm located beneath the ectodermal epithelium and continuous with the extraembryonic mesoderm covering the amnion
 - A **splanchnic or visceral layer** of lateral mesoderm located adjacent to the endoderm and continuous with the extraembryonic mesoderm covering the umbilical vesicle



- The **somatic mesoderm** and **overlying ectoderm** form the embryonic body wall or **somatopleure**
- the **splanchnic mesoderm** and **underlying embryonic endoderm** form the embryonic gut or **splanchnopleure**.
- During the **2nd month**, the intraembryonic coelom is divided into three body cavities:
 - **Pericardial cavity**
 - **Pleural cavities**
 - **Peritoneal cavity**

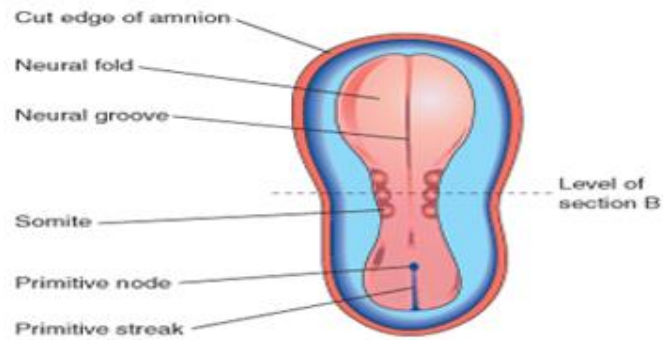
NEURULATION: Formation of the neural tube

- The processes of the formation of the neural plate and neural folds and closure of the folds to form the **neural tube**

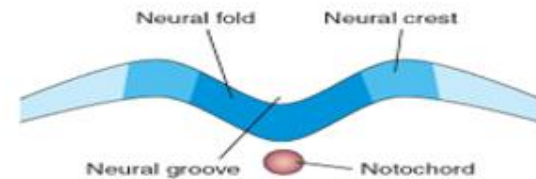
Neural plate and Neural tube:

- the overlying embryonic ectoderm located at or adjacent to the midline to thicken
- and form an elongated plate of thickened epithelial cells, the **neural plate**, because of inductive effect of the notochord
- The ectoderm of the neural plate (**neuroectoderm**) gives rise to the CNS-the brain and spinal cord.
- **Neuroectoderm** also gives rise to various other structures, for example, the retina.
- It appears rostral to the primitive node and dorsal to the notochord and the mesoderm adjacent to it.
- As the notochord elongates, the neural plate broadens and eventually extends cranially as far as the oropharyngeal membrane
- Eventually the neural plate extends beyond the notochord.

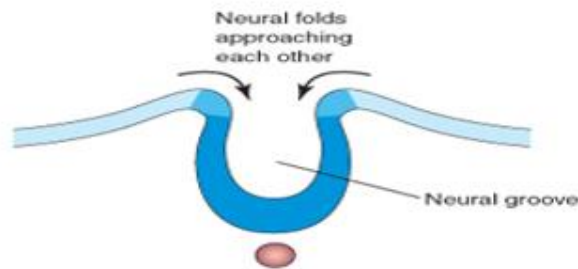
- On $\approx$ the **18th day**, the neural plate invaginates along its central axis to form a longitudinal median **neural groove**
- Neural groove has neural folds on each side.
- **The neural folds:**
 - become particularly prominent at the cranial end of the embryo
 - and are the first signs of brain development.
 - By the end of the **3rd week**, the neural folds have begun to move together and fuse, converting the neural plate into a **neural tube**
 - **The neural tube** soon separates from the surface ectoderm as the neural folds meet.
 - **Neural crest cells** undergo an epithelial to mesenchymal transition and migrate away as the neural folds meet and the free edges of the surface ectoderm (non-neural ectoderm)
 - Neurulation is completed during the **4th week** when closure of the **caudal neuropore occurs**.
 - During neurulation, the embryo may be referred to as a **neurula**.



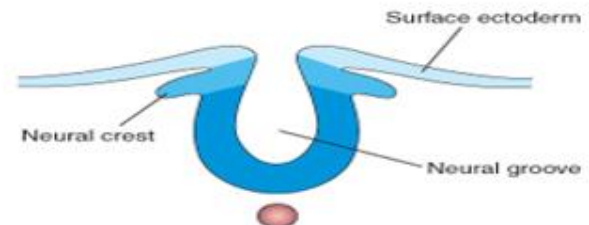
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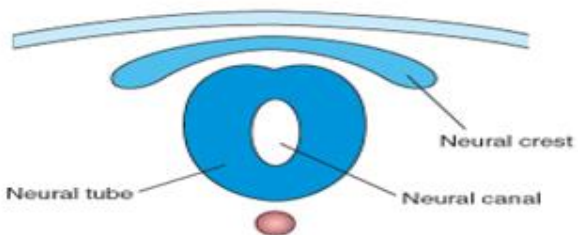
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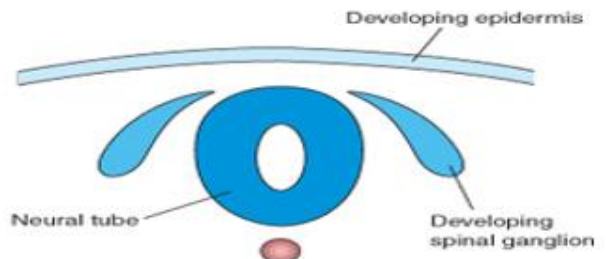
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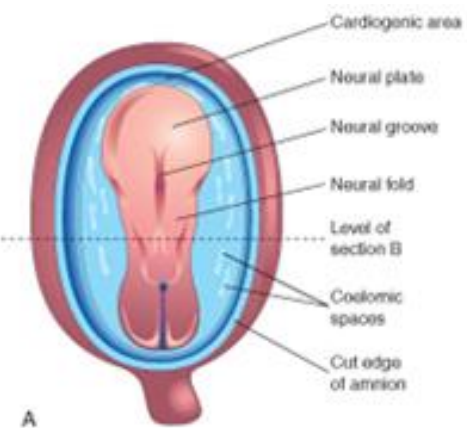
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Neural tube consists of a cranial open is the anterior (rostral) neuropore, and the caudal open is the posterior (caudal) neuropore.

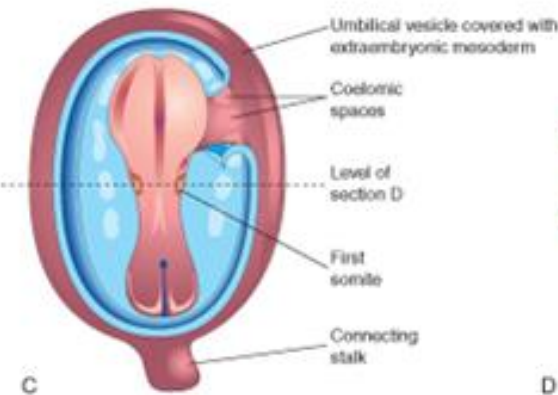
The anterior neuropore closes on or before day 26 and the caudal neuropore closes before the end of the fourth week.

Neural Crest Formation

- As the neural folds fuse to form the neural tube, some neuroectodermal cells lying along the inner margin of each neural fold lose their epithelial affinities and attachments to neighboring cells
- As the neural tube separates from the surface ectoderm, **neural crest cells** form a flattened irregular mass, **the neural crest**, between the neural tube and the overlying surface
- The neural crest soon separates into right and left parts that shift to the dorsolateral aspects of the neural tube



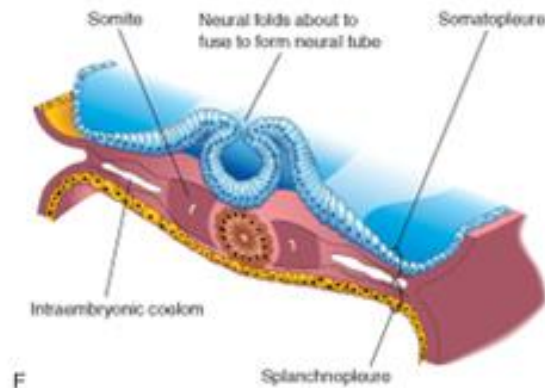
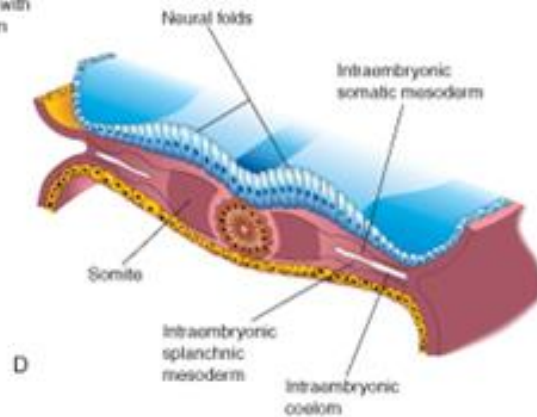
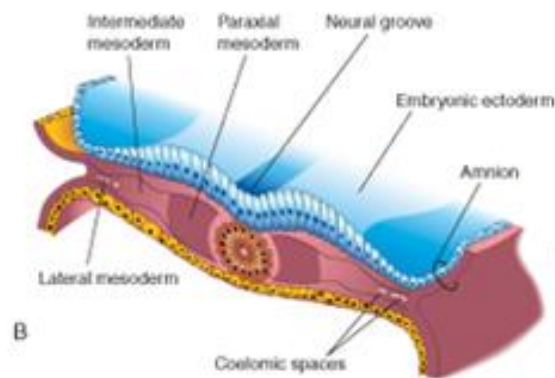
18 days



20 days



21 days



Drawings of embryos of 19 to 21 days illustrating **development of the somites and intraembryonic coelom**.

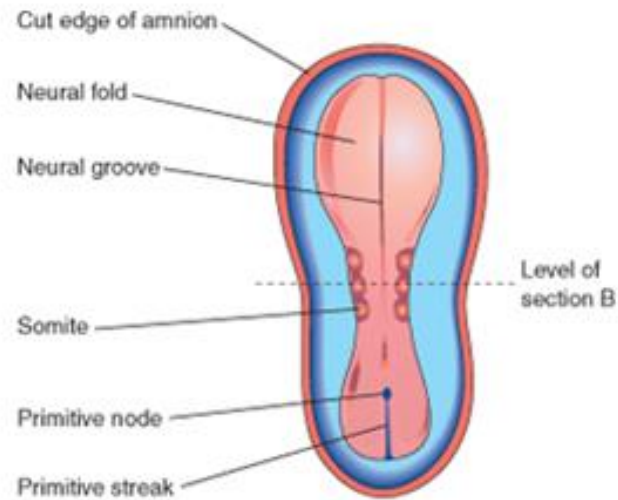
A, C, and E, Dorsal views of the embryo, exposed by removal of the amnion.

B, D, and F, Transverse sections through the embryonic disc at the levels shown.

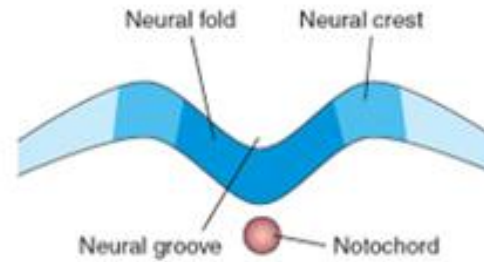
A, Presomite embryo of approximately 18 days.

C, An embryo of approximately 20 days showing the first pair of somites. Part of the somatopleure on the right has been removed to show the coelomic spaces in the lateral mesoderm.

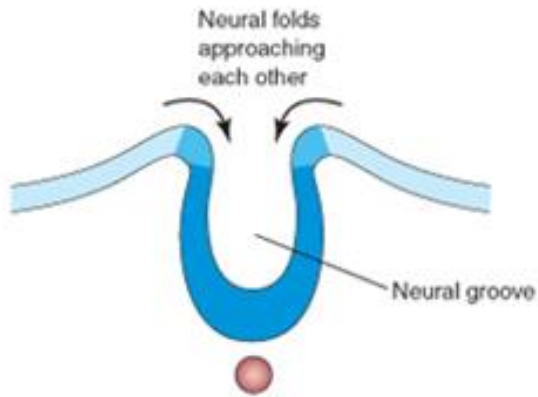
E, A three-somite embryo (approximately 21 days) showing the horseshoe-shaped intraembryonic coelom, exposed on the right by removal of part of the somatopleure.



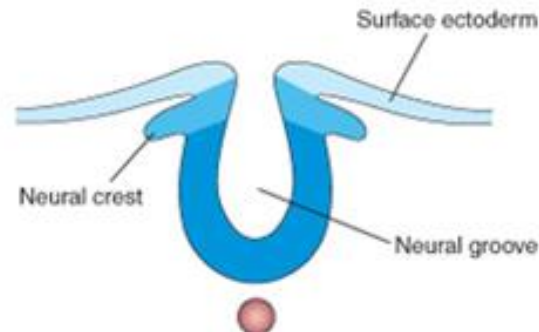
A



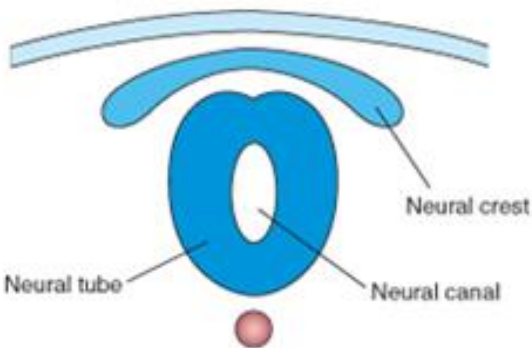
B



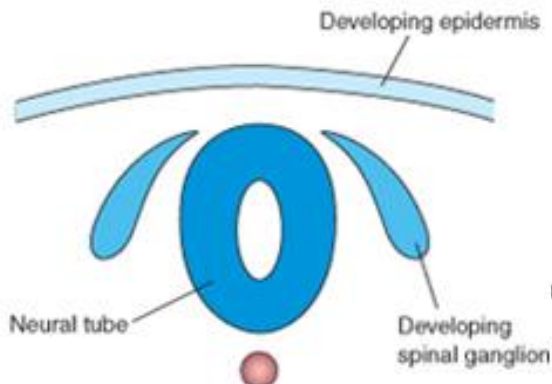
C



D



E



F

Diagrammatic transverse sections through progressively older embryos illustrating formation of the

neural groove,
neural folds,
neural tube,
and
neural crest.

anterior neuropore

fail to close



anencephaly

posterior neuropore



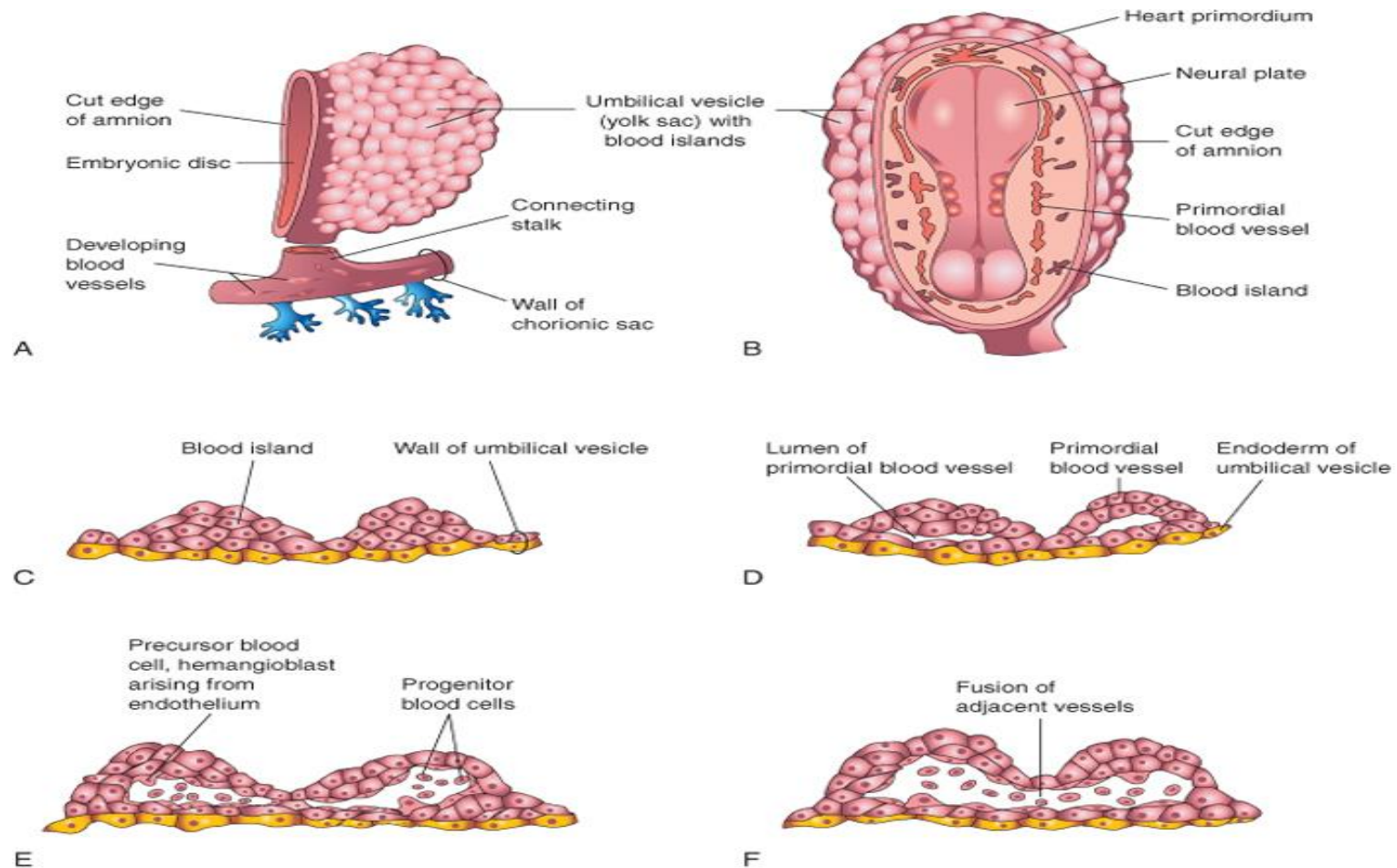
myeloschisis

(spina bifida)



EARLY DEVELOPMENT OF THE CARDIOVASCULAR SYSTEM

- At the beginning of the 3rd week, *vasculogenesis* and *angiogenesis* (blood vessel formation) begins in the extraembryonic mesoderm of the umbilical vesicle, connecting stalk, and chorion
- The early formation of the cardiovascular system is correlated with the urgent need for blood vessels to bring oxygen and nourishment to the embryo from the maternal circulation through the placenta.
- During the 3rd week, a primordial uteroplacental circulation develops



- **Angiogenesis** is the process in which new blood vessels develop from pre-existing blood vessels.
- **Vasculogenesis** is the formation of new blood vessels from angioblasts and endothelial cells.

Vasculogenesis and Angiogenesis

- The formation of the embryonic vascular system involves two processes:
 - 1. Vasculogenesis:** is the formation of new vascular channels by assembly of individual cell precursors called **angioblasts** and **endothelial cells**.
 - 2. Angiogenesis:** is the formation of new vessels by budding and branching from preexisting vessels.
- During the blood vessel formation (*vasculogenesis*):
 - Mesenchymal cells (mesoderm derived) differentiate into endothelial cell precursors-**angioblasts** (vessel-forming cells)
 - **Angioblasts** aggregate to form isolated **angiogenic cell** clusters called **blood islands**- associated with the umbilical vesicle or endothelial cords within the embryo.

- Small cavities appear within the blood islands and endothelial cords by confluence of intercellular clefts.
- Angioblasts flatten to form endothelial cells that arrange themselves around the cavities in the blood island to form the endothelium.
- These endothelium-lined cavities soon fuse to form networks of endothelial channels (vasculogenesis).
- Vessels sprout into adjacent areas by endothelial budding and fuse with other vessels.

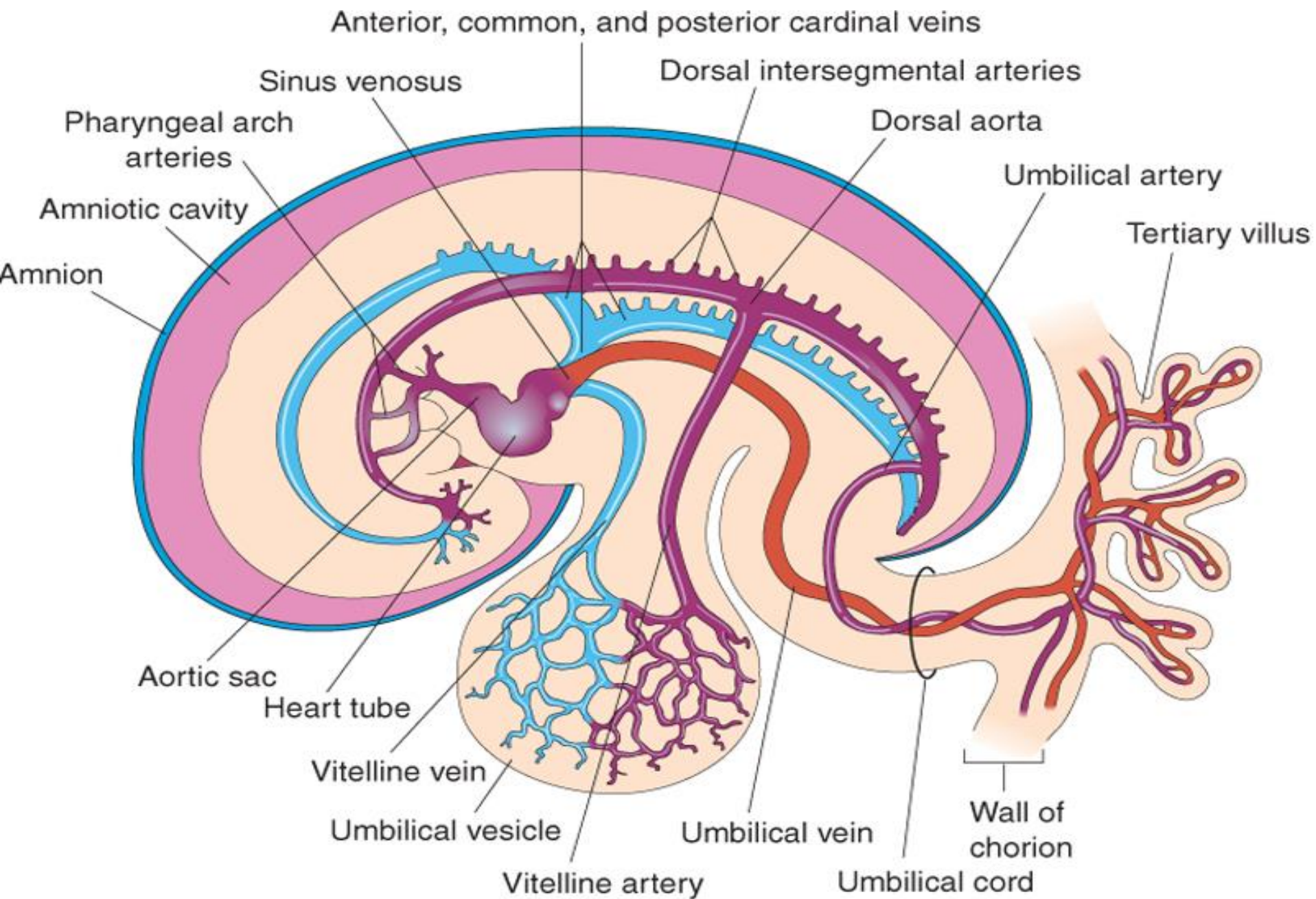
Questions: What is the difference between Vasculogenesis and angiogenesis?

Ans. Vasculogenesis is defined as the differentiation of precursor cells (angioblasts) into endothelial cells and the **de novo** formation of a primitive vascular network whereas **angiogenesis** is defined as the growth of new capillaries from pre-existing blood vessels (Risau, 1997).

- **Blood cells** develop from the endothelial cells of vessels as they develop on the umbilical vesicle and allantois at the end of the third week and later in specialized sites along the dorsal aorta.
- Blood formation (**hematogenesis**) does not begin in the embryo **until the fifth week**.
- It occurs first along the aorta and then in various parts of the embryonic mesenchyme, mainly, **the liver**, and **later in the spleen, bone marrow**, and **lymph nodes**.
- Fetal and adult erythrocytes are derived from different hematopoietic progenitor cells (*hemangioblasts*).
- The mesenchymal cells surrounding the primordial endothelial blood vessels differentiate into the muscular and connective tissue elements of the vessels.

The Primordial Cardiovascular System

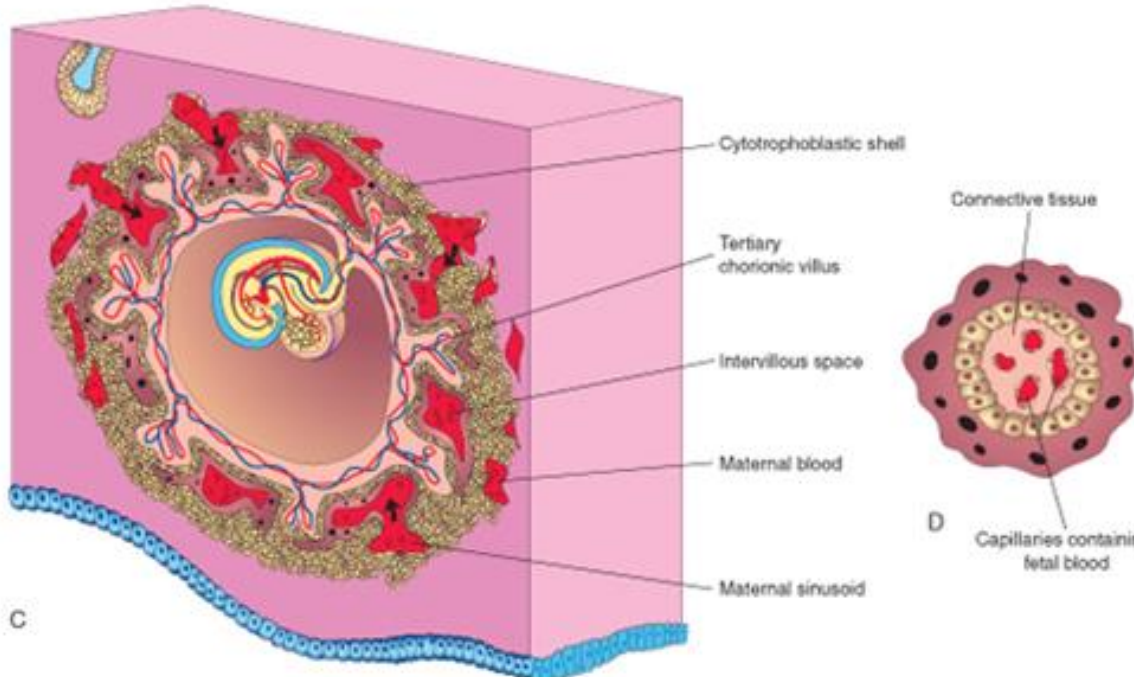
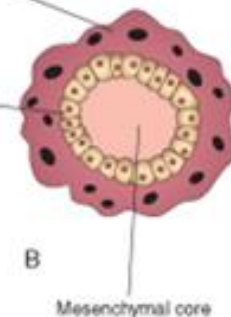
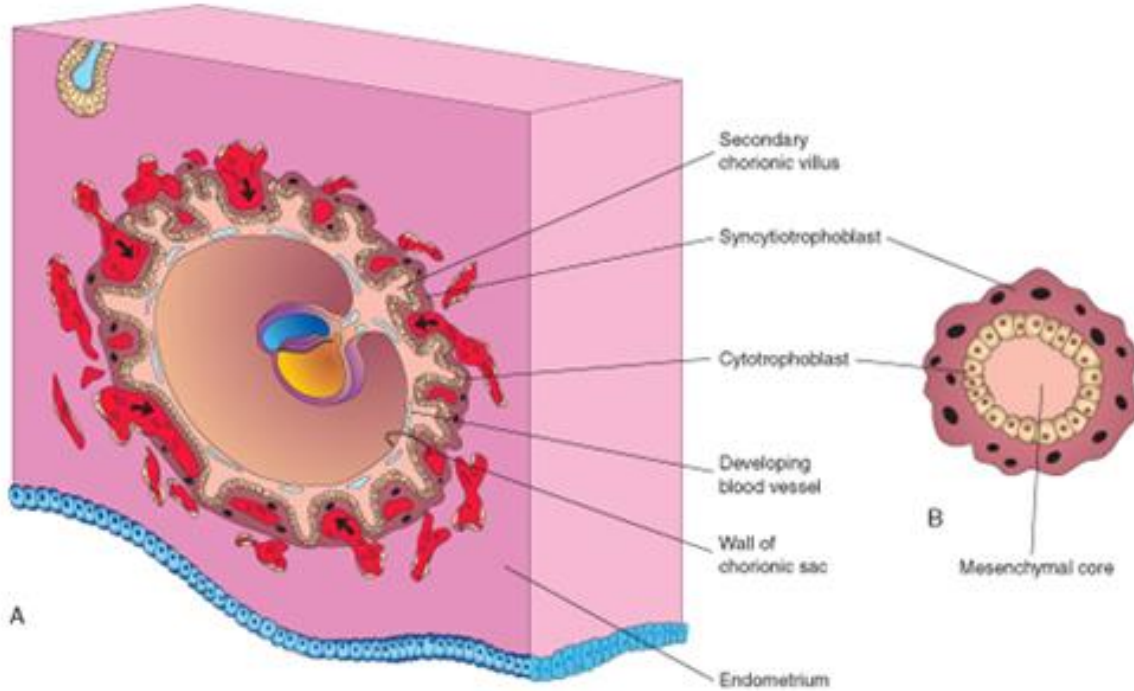
- The heart and great vessels form from mesenchymal cells in the cardiogenic area
- Paired, longitudinal endothelial-lined channels-the endocardial heart tubes-develop during the third week and fuse to form a primordial heart tube.
- The tubular heart joins with blood vessels in the embryo, connecting stalk, chorion, and umbilical vesicle to form a primordial cardiovascular system
- By the end of the third week, the blood is circulating and the heart begins to beat on the 21st or 22nd day.
- *The cardiovascular system is the first organ system to reach a functional state*



DEVELOPMENT OF CHORIONIC VILLI

- Shortly after **primary chorionic villi** appear at the end of the second week, they begin to branch.
- Early in the third week, mesenchyme grows into these primary villi, forming **a core of mesenchymal tissue**.
- The villi at this stage-**secondary chorionic villi**-cover the entire surface of the chorionic sac.
- Some mesenchymal cells in the villi soon differentiate into capillaries and blood cells.
- They are **called tertiary chorionic villi** when blood vessels are visible in them.
- The capillaries in the chorionic villi fuse to form **arteriocapillary networks**, which soon become connected with the embryonic heart through vessels that differentiate in the mesenchyme of the chorion and connecting stalk.
- By the end of the third week, embryonic blood begins to flow slowly through the capillaries in the chorionic villi.
- Oxygen and nutrients in the maternal blood in the intervillous space diffuse through the walls of the villi and enter the embryo's blood.
- Carbon dioxide and waste products diffuse from blood in the fetal capillaries through the wall of the chorionic villi into the maternal blood.

- Concurrently, cytotrophoblastic cells of the chorionic villi proliferate and extend through the syncytiotrophoblast to form a **cytotrophoblastic shell**, which gradually surrounds the chorionic sac and attaches it to the endometrium.
- Villi that attach to the maternal tissues through the cytotrophoblastic shell are **stem chorionic villi** (anchoring villi).
- The villi that grow from the sides of the stem villi are **branch chorionic villi** (terminal villi).
- It is through the walls of the branch villi that the main exchange of material between the blood of the mother and the embryo takes place.
- The branch villi are bathed in continually changing maternal blood in the intervillous space.



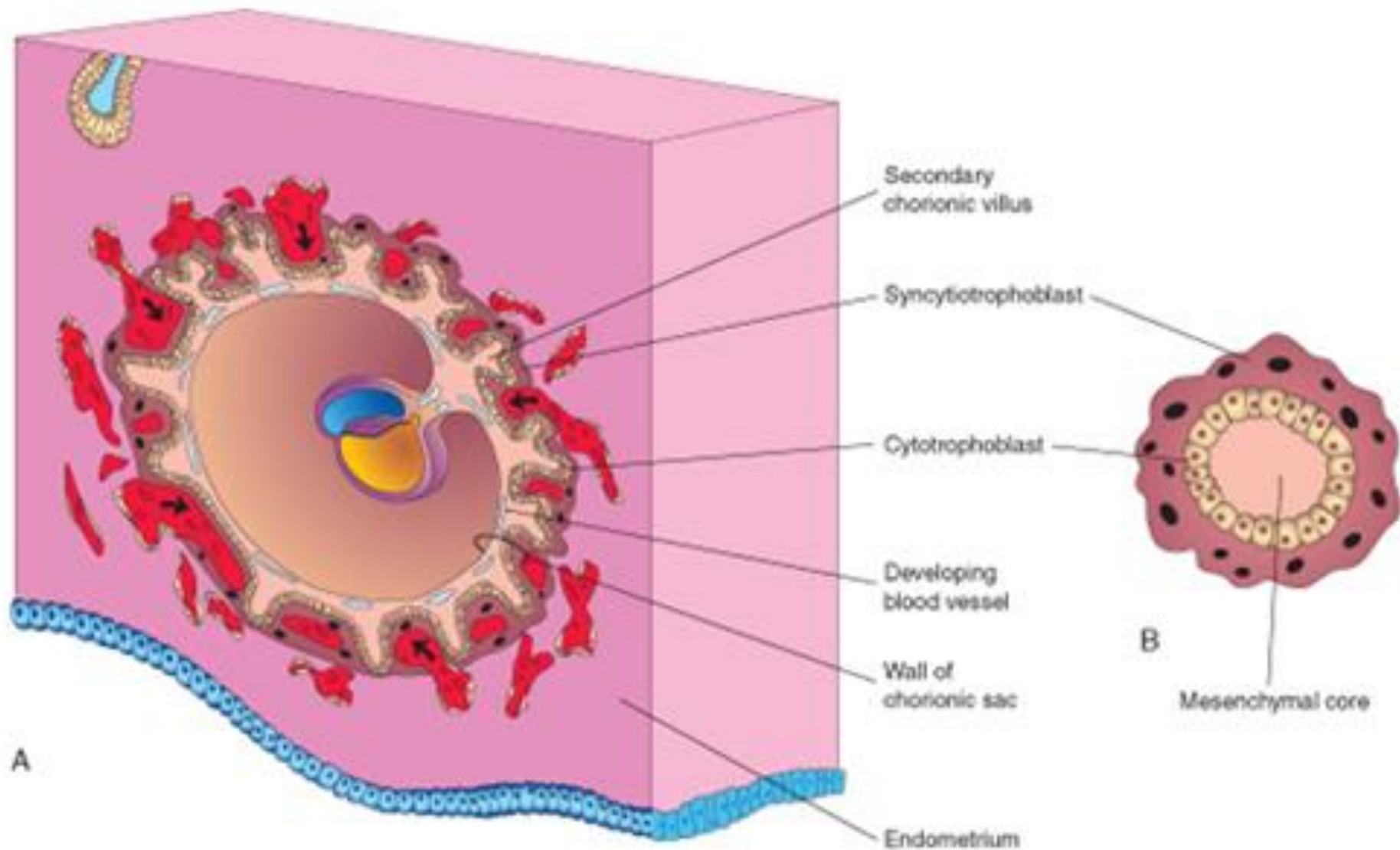
Diagrams illustrating development of secondary chorionic villi into tertiary chorionic villi. Early formation of the placenta is also shown.

A, Sagittal section of an embryo (approximately 16 days).

B, Section of a secondary chorionic villus.

C, Section of an implanted embryo (approximately 21 days).

D, Section of a tertiary chorionic villus. The fetal blood in the capillaries is separated from the maternal blood surrounding the villus by the endothelium of the capillary, embryonic connective tissue, cytotrophoblast, and syncytiotrophoblast.

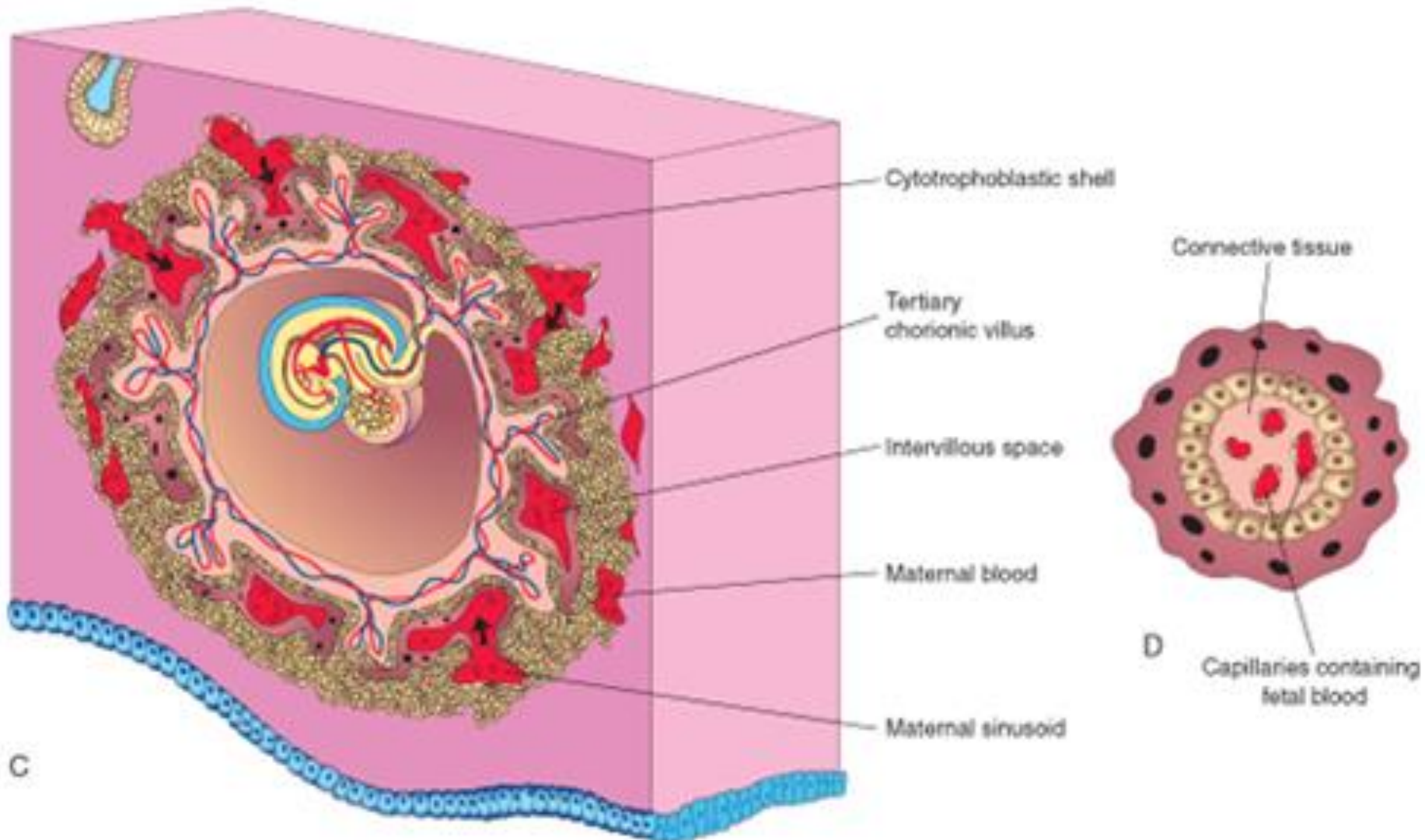


Diagrams illustrating development of secondary chorionic villi into tertiary chorionic villi.

Early formation of the placenta is also shown.

A, Sagittal section of an embryo (approximately 16 days).

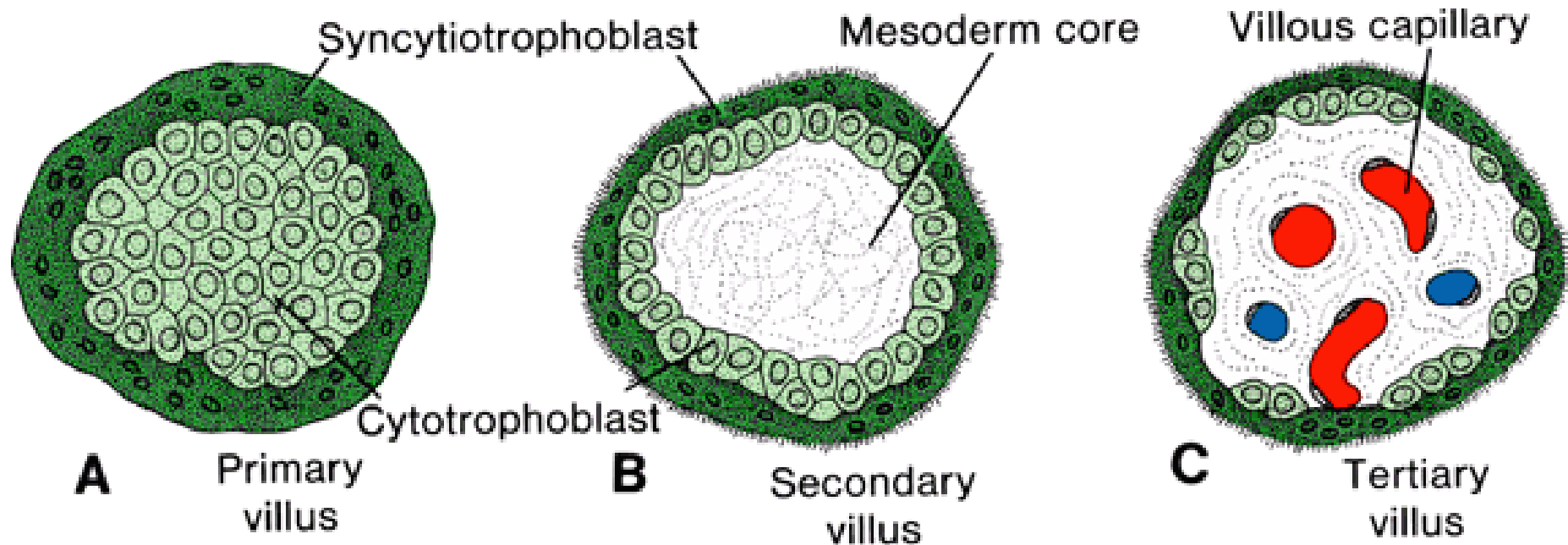
B, Section of a secondary chorionic villus.



Diagrams illustrating development of secondary chorionic villi into tertiary chorionic villi. Early formation of the placenta is also shown.

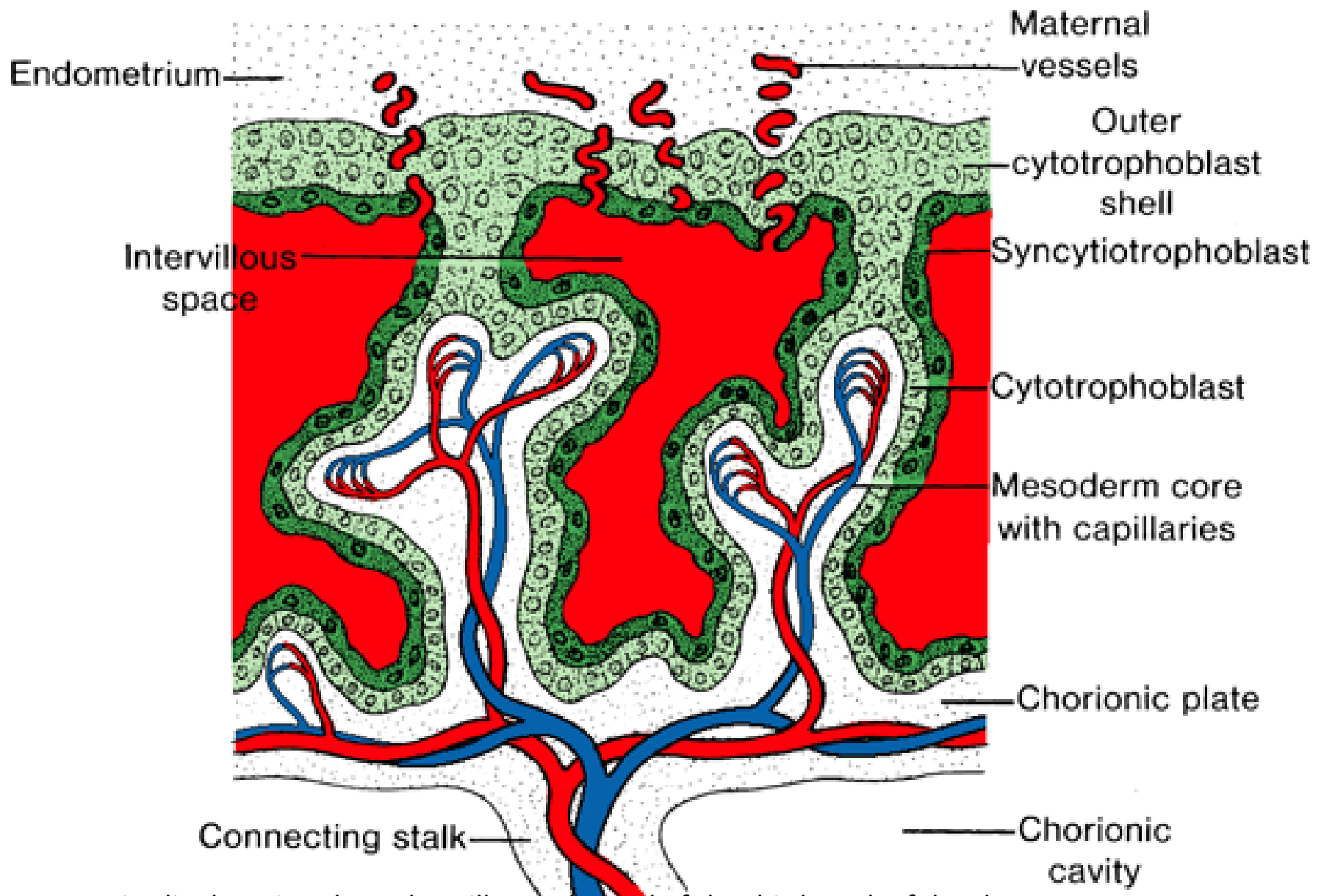
C, Section of an implanted embryo (approximately 21 days).

D, Section of a tertiary chorionic villus. The fetal blood in the capillaries is separated from the maternal blood surrounding the villus by the endothelium of the capillary, embryonic connective tissue, cytotrophoblast, and syncytiotrophoblast.

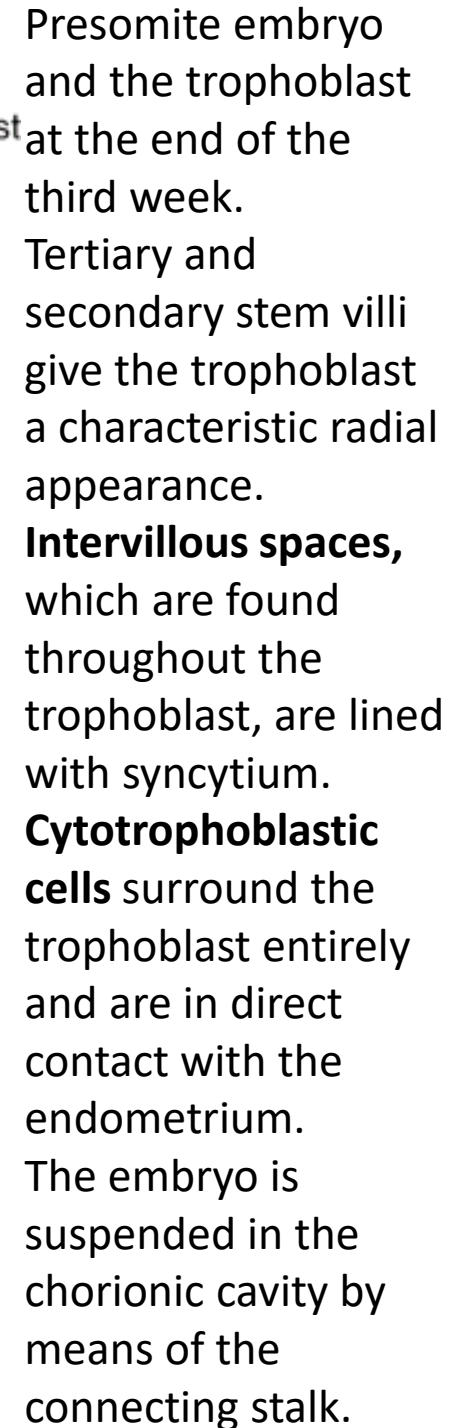


Development of a villus.

- Transverse section of a primary villus showing a core of cytotrophoblastic cells covered by a layer of syncytium.
- Transverse section of a secondary villus with a core of mesoderm covered by a single layer of cytotrophoblastic cells, which in turn is covered by syncytium.
- Mesoderm of the villus showing a number of capillaries and venules.



Longitudinal section through a villus at the end of the third week of development. Maternal vessels penetrate the cytotrophoblastic shell to enter intervillous spaces, which surround the villi. Capillaries in the villi are in contact with vessels in the chorionic plate and in the connecting stalk, which in turn are connected to intraembryonic vessels.



Tertiary and secondary stem villi give the trophoblast a characteristic radial appearance.

Intervillous spaces, which are found throughout the trophoblast, are lined with syncytium.

Cytotrophoblastic cells surround the trophoblast entirely and are in direct contact with the endometrium. The embryo is suspended in the chorionic cavity by means of the connecting stalk.

Organogenetic Period: 4th to 8th Weeks



→ Development of Tissues, Organs and Body Form

→ **ORGANOGENESIS**

→ CRITICAL PERIOD ► Exposure to teratogens during this period major congenital defects

Embryonic Period

Major embryological events:
Organs form
from three primary
germ tissues.
Emergence of the
basic body plan.



3.5 mm

25 days



10 mm

36 days



30 mm

56 days

During embryonic period:

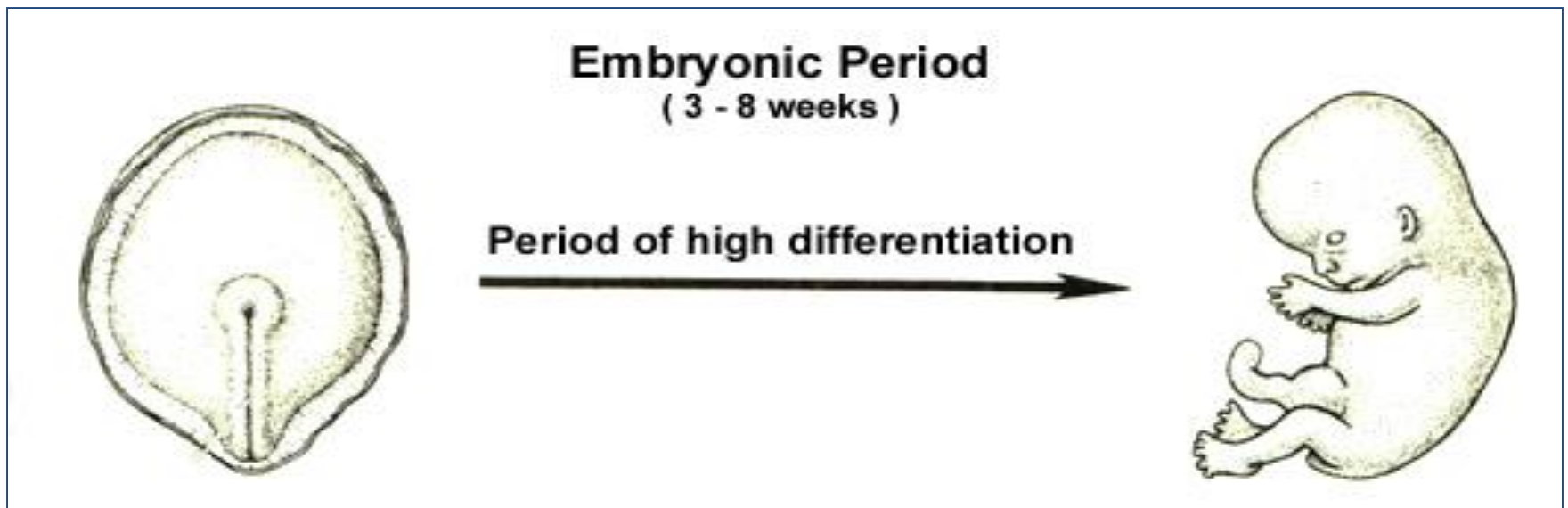
- Primordia of all major organs develop from the 3 germ layers
- the main organ systems have begun to develop;
 - the function of most of them is minimal except for the **cardiovascular system**
- as the tissues and organs form, the shape of the **embryo changes**
- by **the 8th week**, it has a distinctly human appearance
- Because the tissues and organs are differentiating rapidly during the 4th to 8th weeks,
 - exposure of embryos to **teratogens during** this period may cause **major congenital anomalies**
 - **Teratogens** are agents such as drugs and viruses that produce or increase the incidence of congenital anomalies

1. PHASES OF EMBRYONIC DEVELOPMENT:

- Human development may be divided into three phases
 - **The first phase**
 - is **growth phase**, involves cell division and the elaboration of cell products.
 - **The second phase**
 - is **morphogenesis** phase(development of shape, size, or other features of a particular organ or part or the whole of the body).
 - Morphogenesis is an elaborate process during which many complex interactions occur in an orderly sequence.
 - The movement of cells allows them to interact with each other during the formation of tissues and organs.
 - **The third phase**
 - is **differentiation** (maturation of physiologic processes).
 - Completion of differentiation results in the formation of tissues and organs that are capable of performing specialized functions.

FOLDING OF THE EMBRYO

- Is significant event during embryonic period
- Transforms the **flat trilaminar** embryonic disc into a somewhat cylindrical embryo
- Folding occurs in both **the median and horizontal planes**
- and results from rapid growth of the embryo.
- Folding at the cranial and caudal ends and sides of the embryo occurs simultaneously.
- Concurrently, there is relative constriction at the junction of the embryo and umbilical vesicle (yolk sac)



1. Folding of the Embryo in the Median Plane

- Produces **head and tail folds** that result in the cranial and caudal regions moving ventrally as the embryo elongates cranially and caudally

a. Head Fold:

- Occurs by the beginning of the **4th week**
- the neural folds in the cranial region have thickened to form the primordium of the brain.
- the developing brain projects dorsally into the amniotic cavity.
- the developing forebrain grows cranially beyond the **oropharyngeal membrane** and **overhangs the developing heart**.
- As result of head fold:
 - the **septum transversum** (transverse septum), primordial heart, pericardial coelom, and oropharyngeal membrane move onto **the ventral surface of the embryo**

Events in the process of head fold

➤ During folding:

- part of the **endoderm** of the umbilical vesicle is incorporated into the embryo as **the foregut** (primordium of pharynx, esophagus, etc).
- the **oropharyngeal membrane** separates the foregut from the **stomodeum**

➤ After folding:

- the septum transversum lies caudal to the heart to develop into the **central tendon of the diaphragm**
- can affect the arrangement of the embryonic coelom (primordium of body cavities).

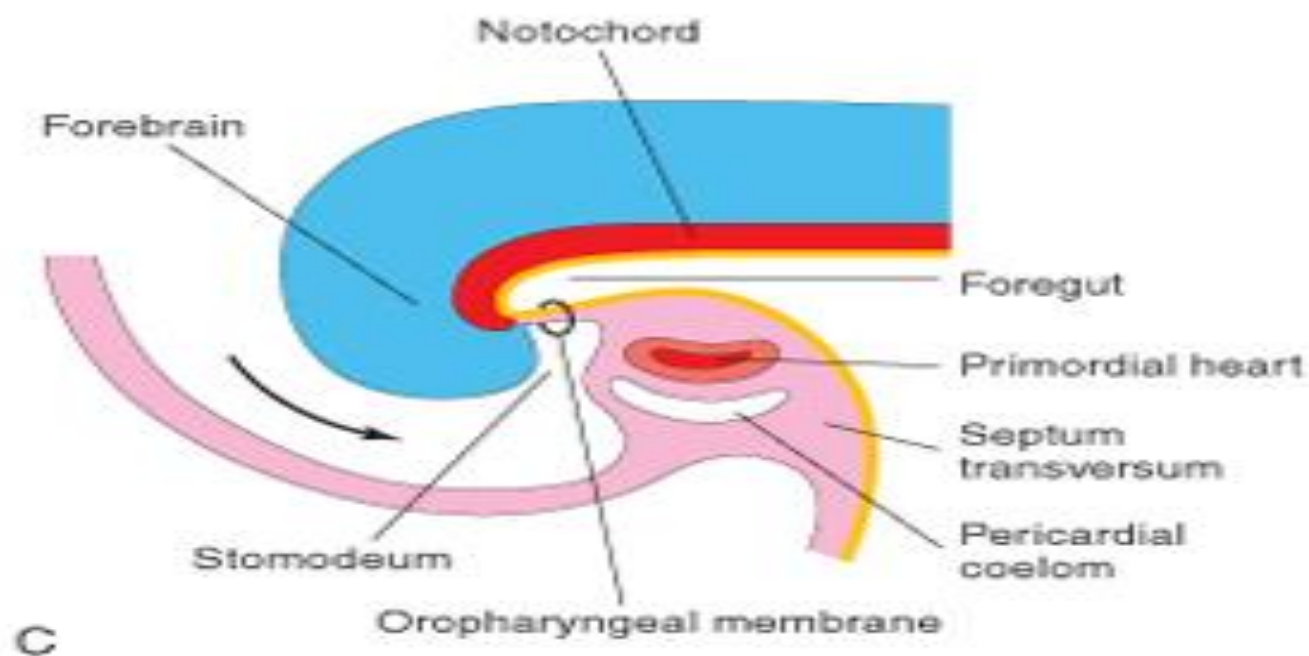
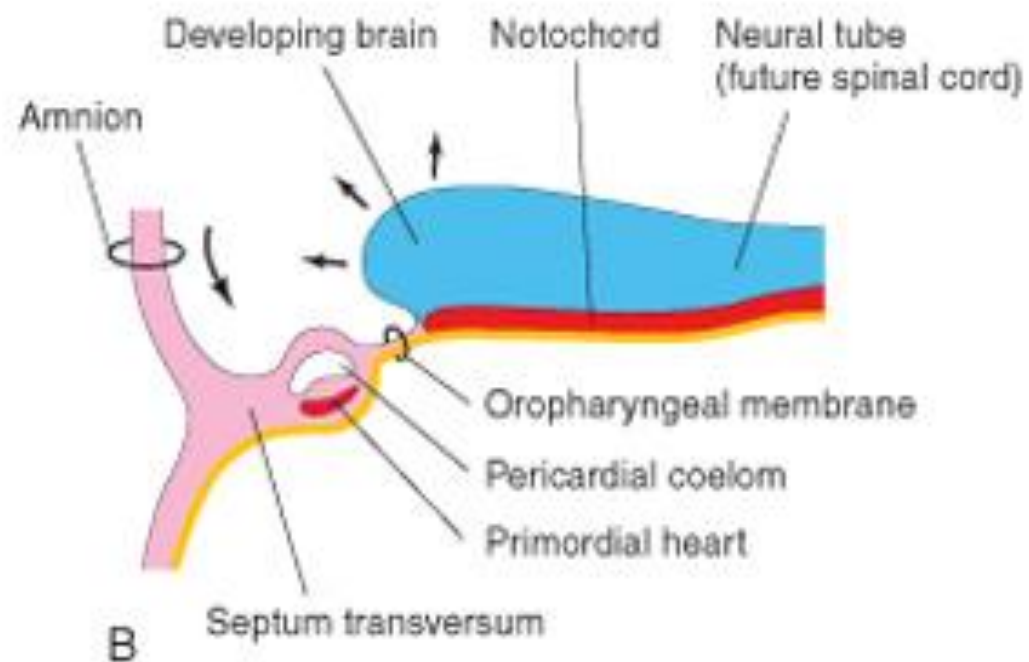
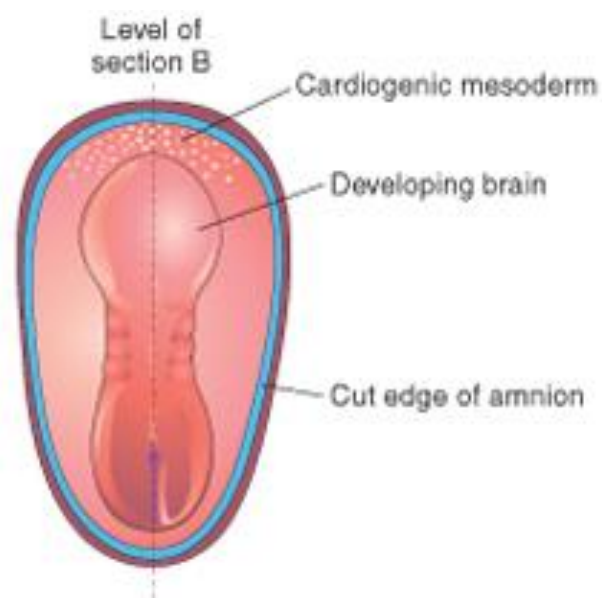
➤ Before folding:

- the coelom consists of a flattened, horseshoe-shaped cavity

➤ After folding:

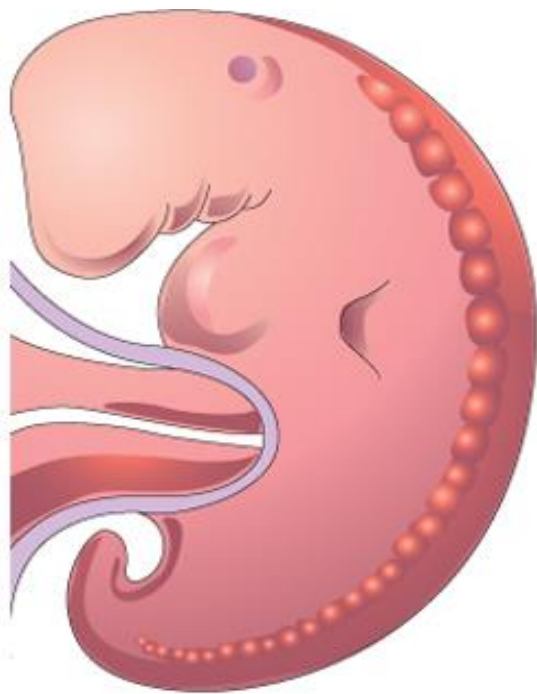
- the **pericardial coelom** lies ventral to the heart and cranial to the **septum transversum**

➤ **At this stage**, the intraembryonic coelom communicates widely on each side with the extraembryonic coelom.

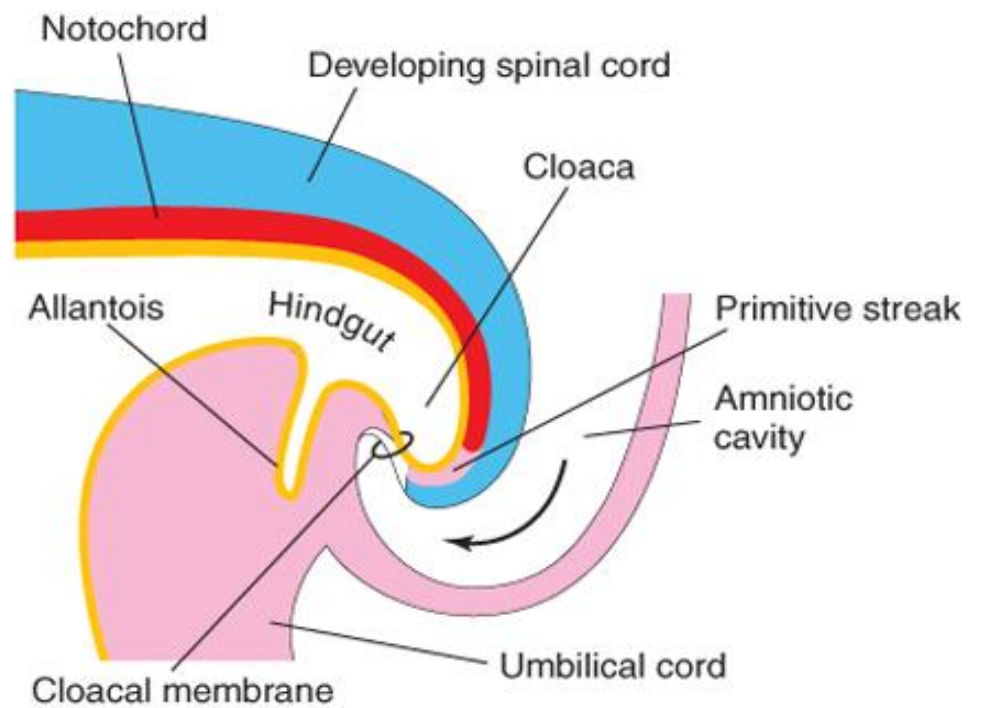


b. Tail Fold:

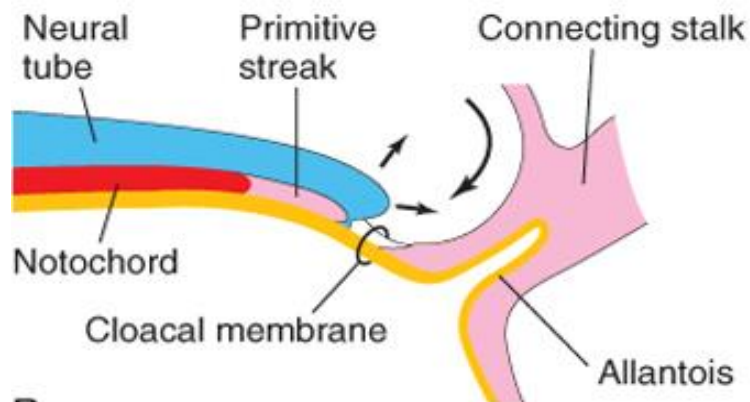
- results primarily from growth of the distal part of the neural tube- the primordium of the spinal cord
- As the embryo grows, the **caudal eminence** (tail region) projects over the **cloacal membrane** (future site of anus).
- **During folding**
 - part of the endodermal germ layer is incorporated into the embryo as the **hindgut** (primordium of descending colon).
 - The terminal part of the hindgut soon dilates slightly to form the **cloaca** (primordium of urinary bladder and rectum)
- **Before folding**
 - the **primitive streak** lies cranial to the cloacal membrane
 - **after folding, it lies caudal to it**
- **The connecting stalk** (primordium of umbilical cord)
 - is now attached to the ventral surface of the embryo,
 - the **allantois** - a diverticulum of the umbilical vesicle- is partially incorporated into the embryo.



A



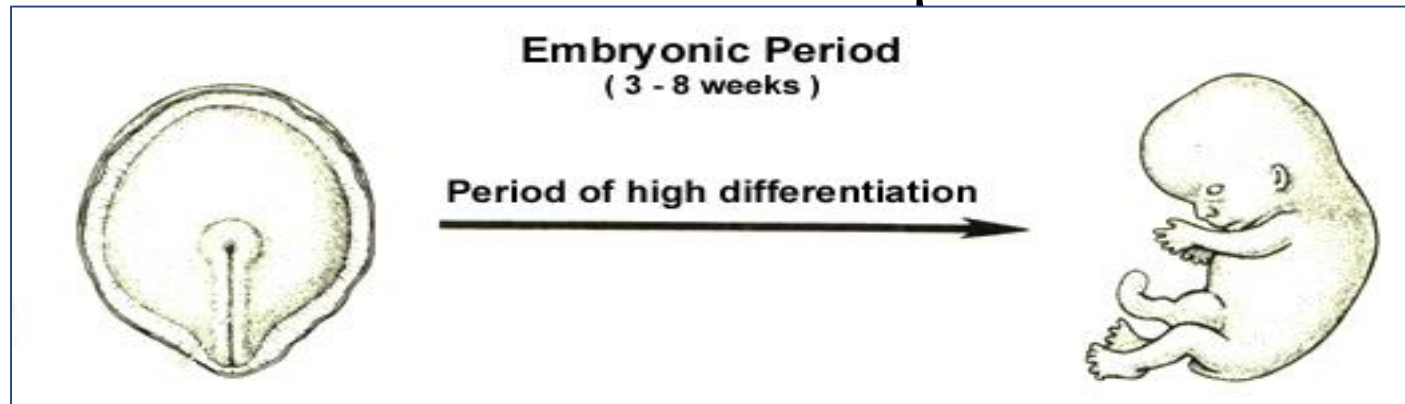
C



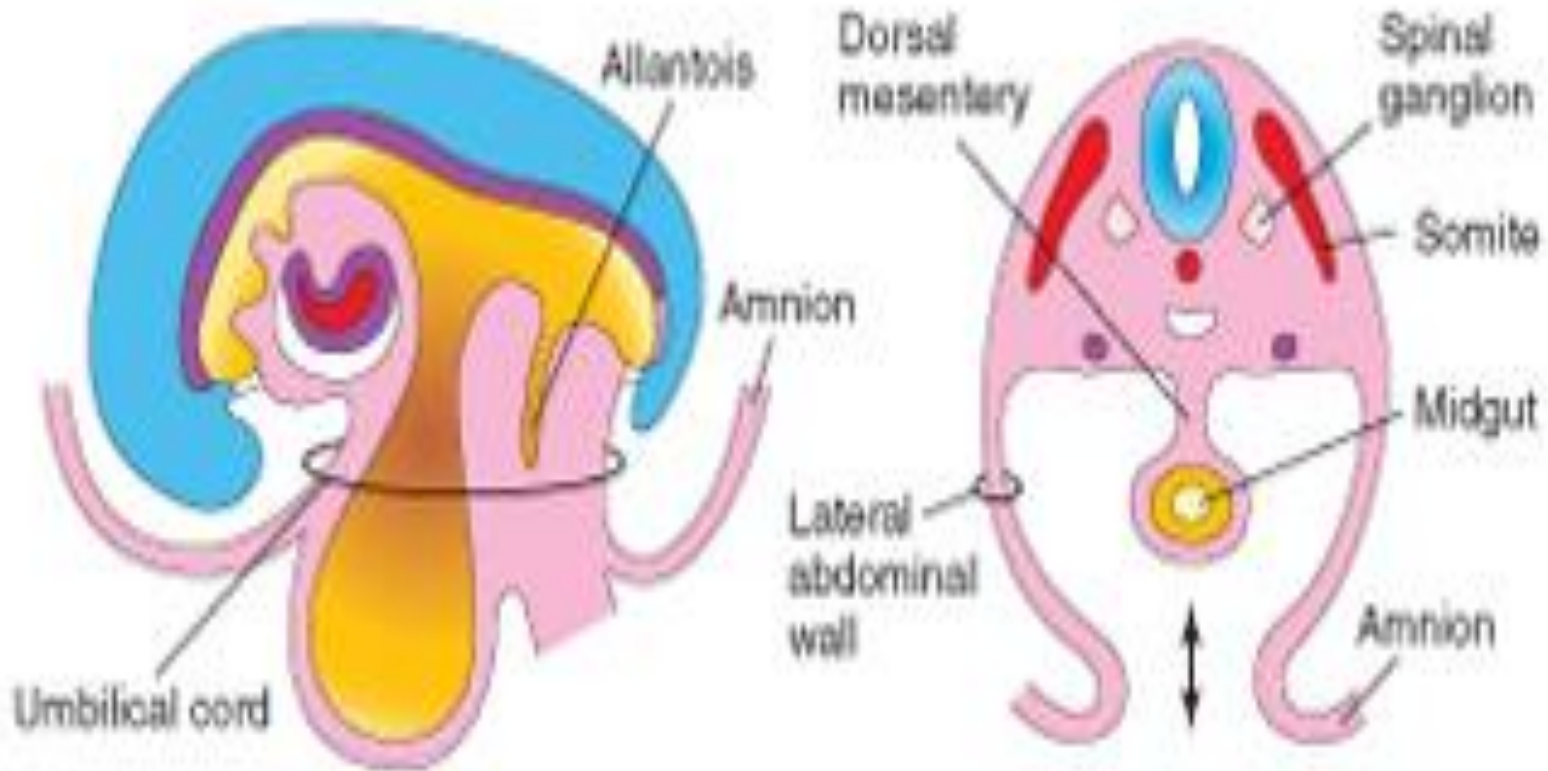
B

Folding of the Embryo in the Horizontal Plane

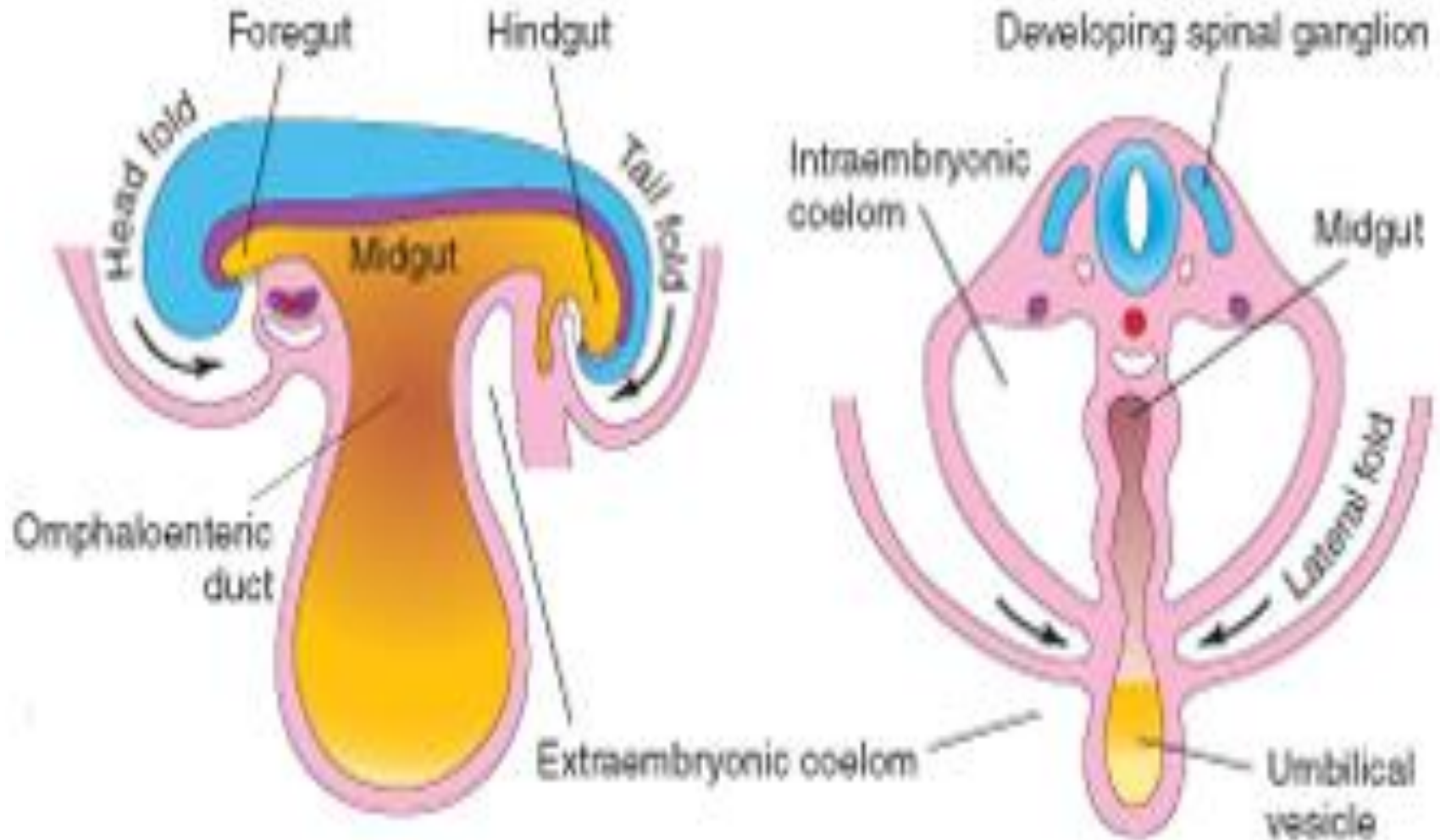
- produces right and left **lateral folds**
- Lateral folding is produced by the **rapidly growing spinal cord and somites.**
- The primordia of the **ventrolateral wall** fold toward the median plane, rolling the edges of the embryonic disc ventrally and forming a **roughly cylindrical embryo**
- Converts the flat embryonic disc into cylindrical human kind
- As the abdominal walls form, part of the endoderm germ layer is incorporated into the embryo as the **midgut**
- Initially, there is a wide connection between the midgut and umbilical vesicle is reduced to an **omphaloenteric duct**



- The region of attachment of the amnion to the ventral surface of the embryo is also reduced to a relatively narrow umbilical region

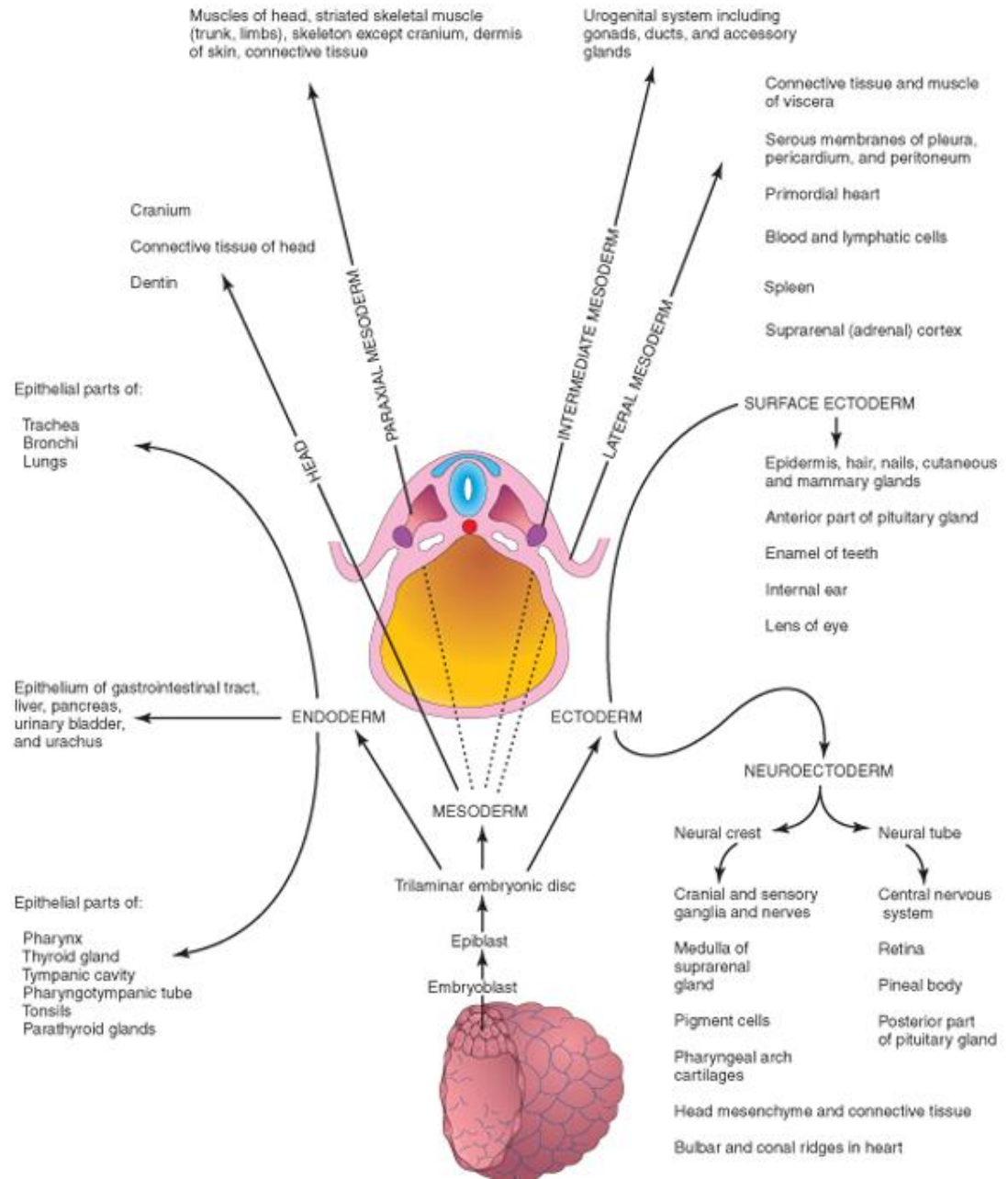


- **As the umbilical cord forms from the connecting stalk,** ventral fusion of the lateral folds reduces the region of communication between the intraembryonic and extraembryonic coelomic cavities to a narrow communication



GERM LAYER DERIVATIVES

1. Ectoderm
2. Mesoderm
3. Endoderm



Differentiation of ectoderm

Ectoderm gives rise to:

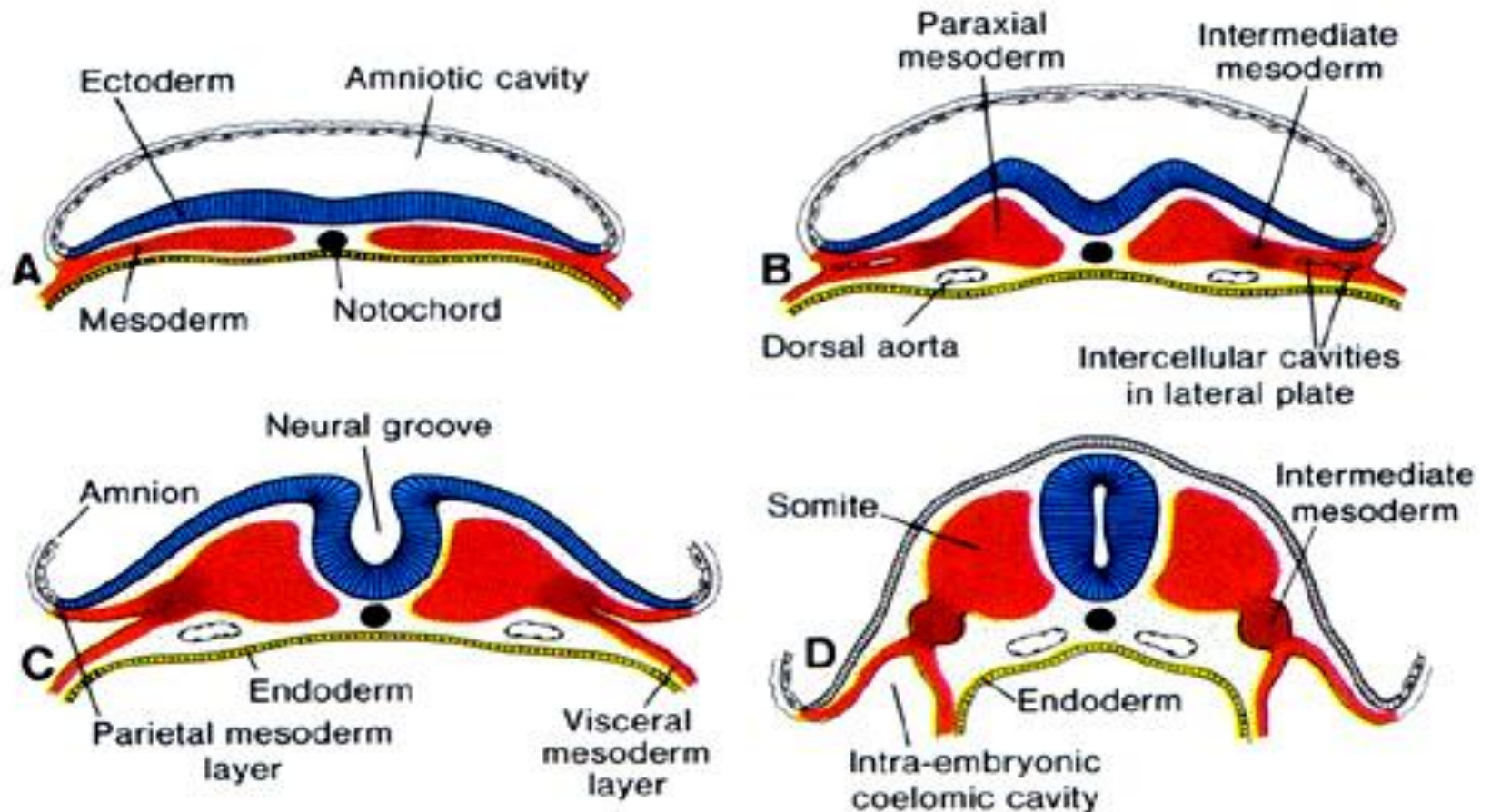
- the central nervous system
- peripheral nervous system
- sensory epithelia of the eye, ear, and nose
- epidermis and its appendages (hair and nails)
- mammary glands
- pituitary gland
- subcutaneous glands and enamel of teeth

Neural crest cells, derived from neuroectoderm, give rise to:

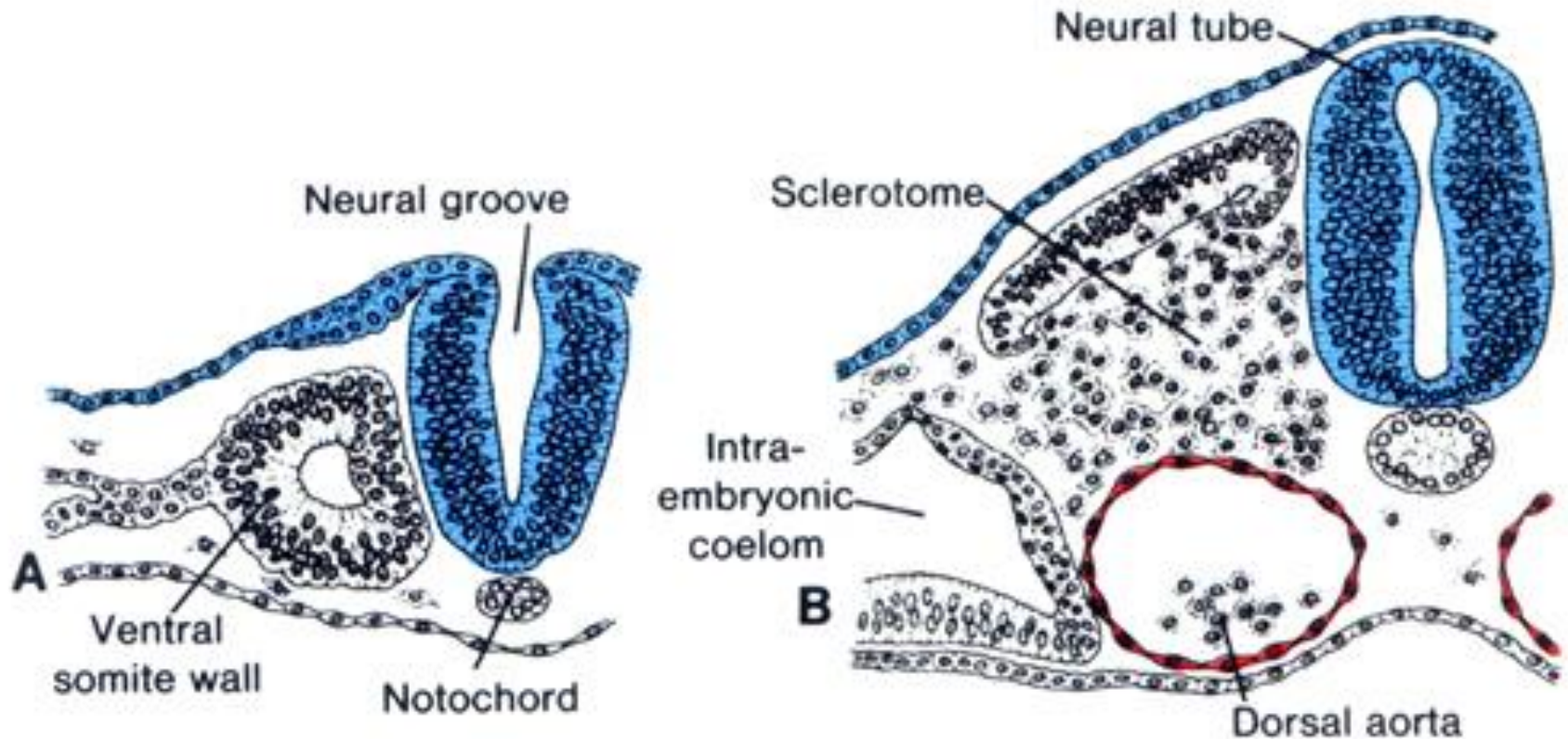
- the cells of the spinal, cranial (cranial nerves V, VII, IX, and X)
- autonomic ganglia
- ensheathing cells of the peripheral nervous system
- pigment cells of the dermis
- muscle, connective tissues, and bone of pharyngeal arch origin
- suprarenal medulla
- meninges (coverings) of the brain and spinal cord

Differentiation of mesoderm

- mesoderm → Paraxial mesoderm → Somite
→ Intermediate mesoderm
→ Lateral mesoderm



- **somites** (3 pairs/day, 42-44 pairs in total)
 - sclerotome → **paraxial bones**
 - dermatome → **dermis**
 - myotome → **muscles**



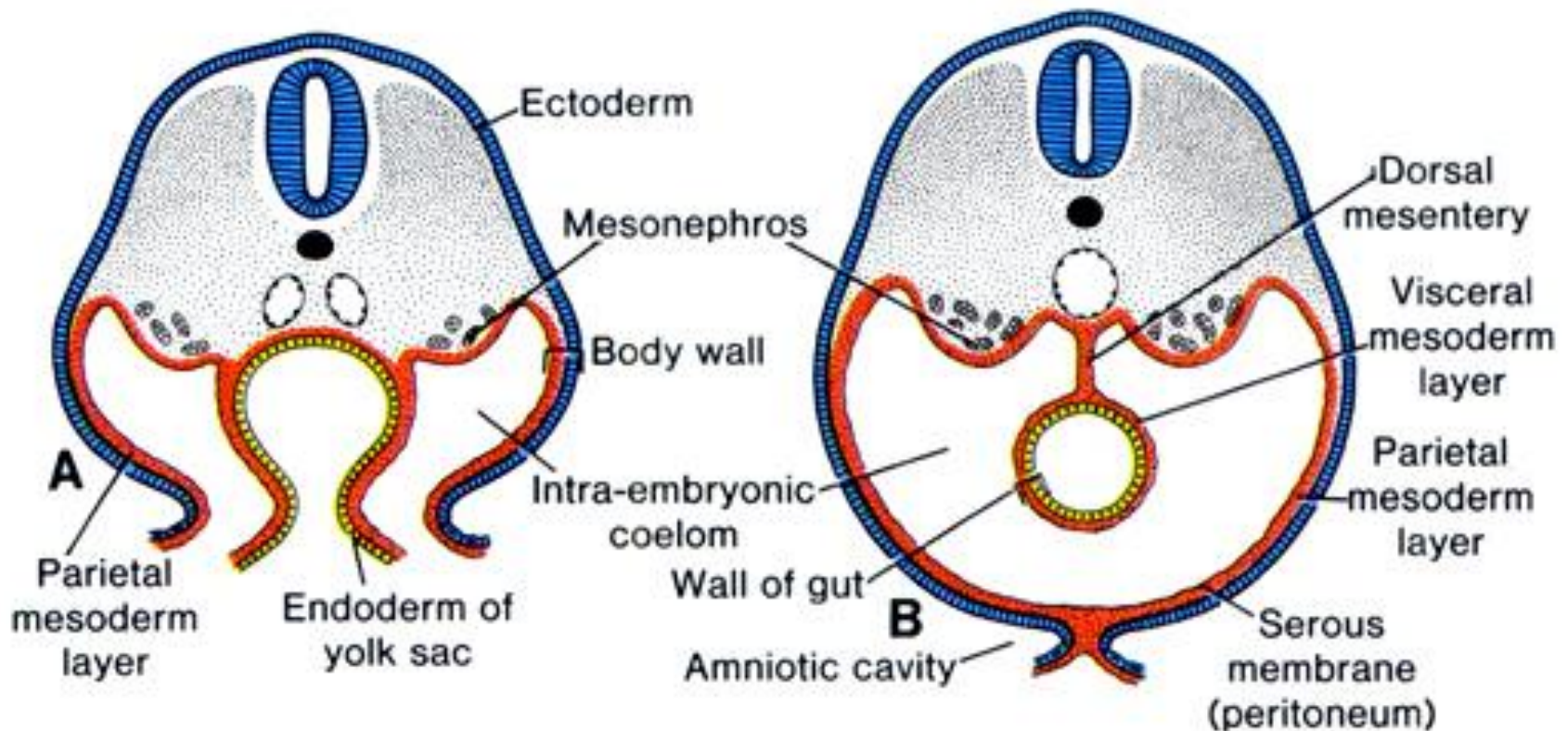
- **Intermediate mesoderm** → **primordium** of the **urogenital system** forming the kidneys and gonads, etc.
- **Lateral mesoderm**
 - 1) Spaces appear and coalesce → **intraembryonic coelom**
→ **body cavities**

- **In summary:**

Mesoderm gives rise:

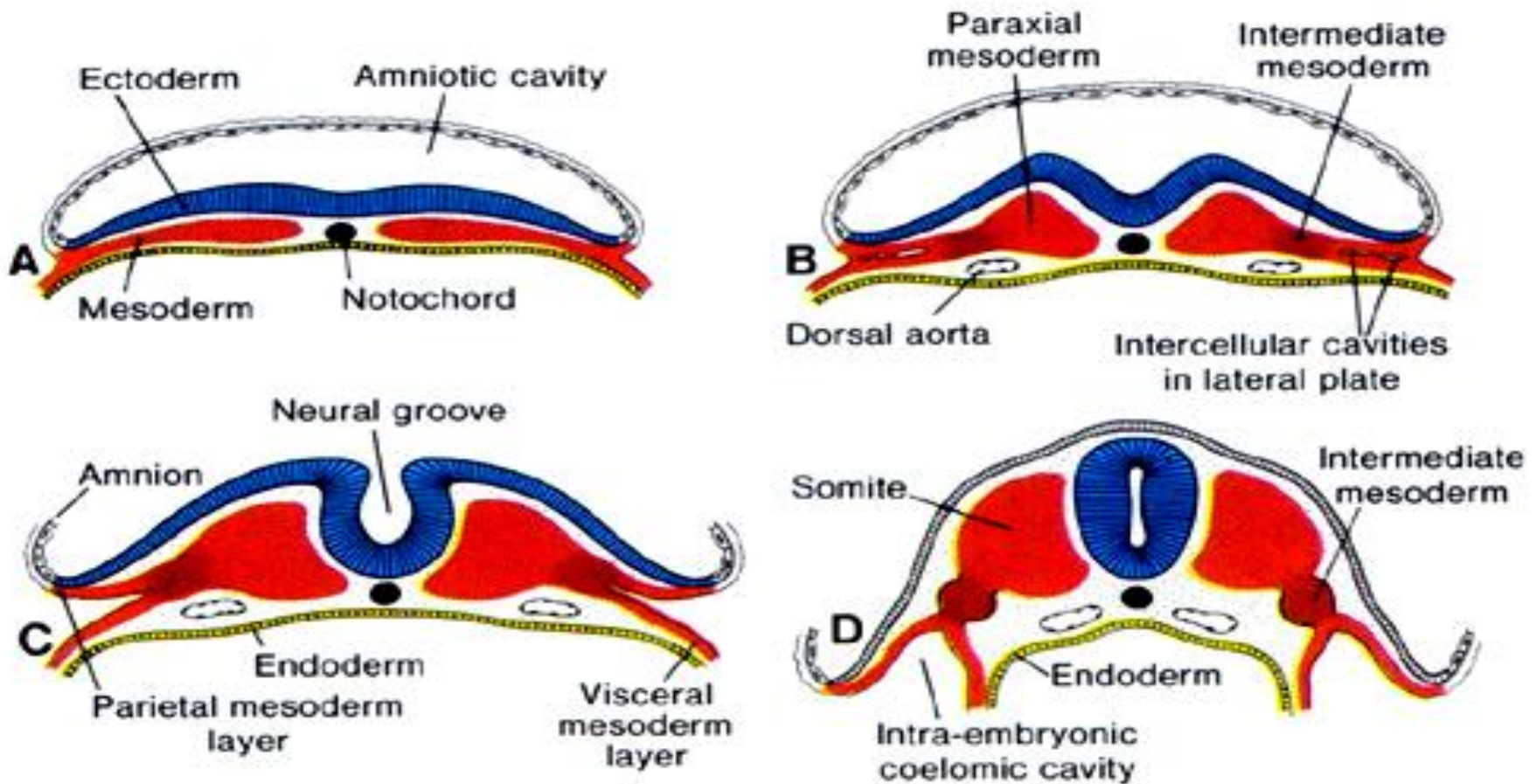
- to connective tissue
- Cartilage, bone, striated and smooth muscles
- heart, blood, and lymphatic vessels
- kidneys; ovaries; testes; genital ducts; serous membranes lining the body cavities (pericardial, pleural, and peritoneal); spleen; and cortex of suprarenal glands

- **Somatic mesoderm** → **bones, muscles and connective tissue of body wall**, and mesothelium lining inner surface of body wall
- **Splanchnic mesoderm** → **muscles and connective tissue of viscera**, and mesothelium covering them.



1. PARAXIAL MESODERM

- By the beginning of the third week, paraxial mesoderm is organized into segments.
- These segments, known as **somitomes**, first appear in the **cephalic region** of the embryo, and their formation proceeds



DEVELOPMENT OF SOMITES

- In addition to the notochord, cells derived from the primitive node **form paraxial mesoderm**. Close to the node, this population appears as a thick, longitudinal column of cells.
- Each column is continuous laterally with the **intermediate mesoderm**, which gradually thins into a layer of lateral mesoderm.
- The **lateral mesoderm** is continuous with the extraembryonic mesoderm covering the umbilical vesicle and amnion.
- Toward the end of the third week, the **paraxial mesoderm** differentiates, condenses, and begins to divide into paired **cuboidal bodies**, the **somites**, which form in a craniocaudal sequence.

- These blocks of mesoderm are located on each side of the developing neural tube.
- About **38 pairs of somites** form during the *somite period of human development* (**days 20 to 30**).
- By the **end of the 5th week**, **42 to 44** pairs of somites are present.
- The somites form distinct surface elevations on the embryo and are somewhat triangular in transverse section
- Because the somites are so prominent during the **4th and 5th weeks**, they are used as one of several criteria for **determining an embryo's age**

Number of Somites Correlated to Approximate Age in Days

Approximate Age (days)	Number of Somites
-----------------------------------	------------------------------

20

1–4

21

4–7

22

7–10

23

10–13

24

13–17

25

17–20

26

20–23

27

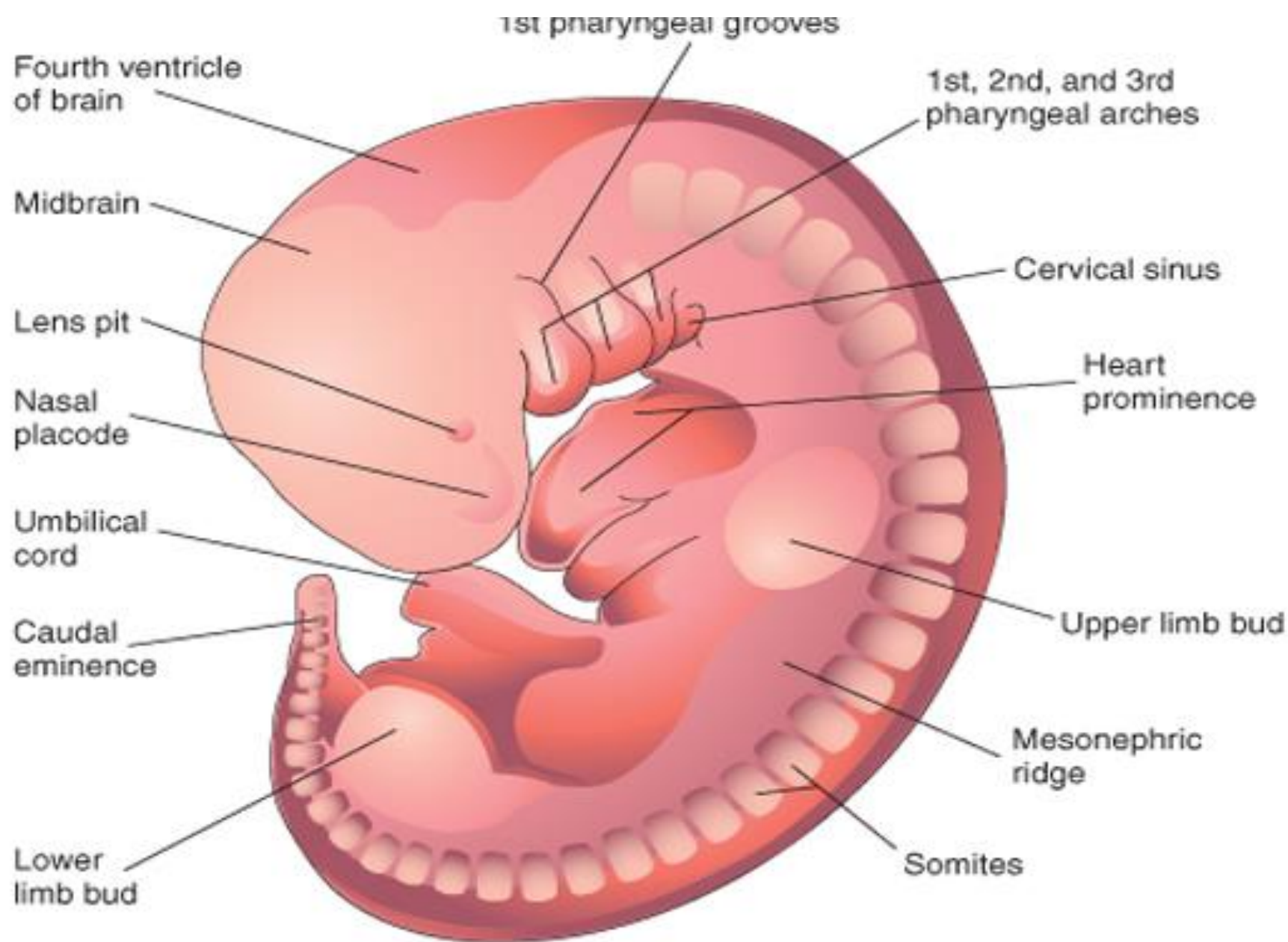
23–26

28

26–29

30

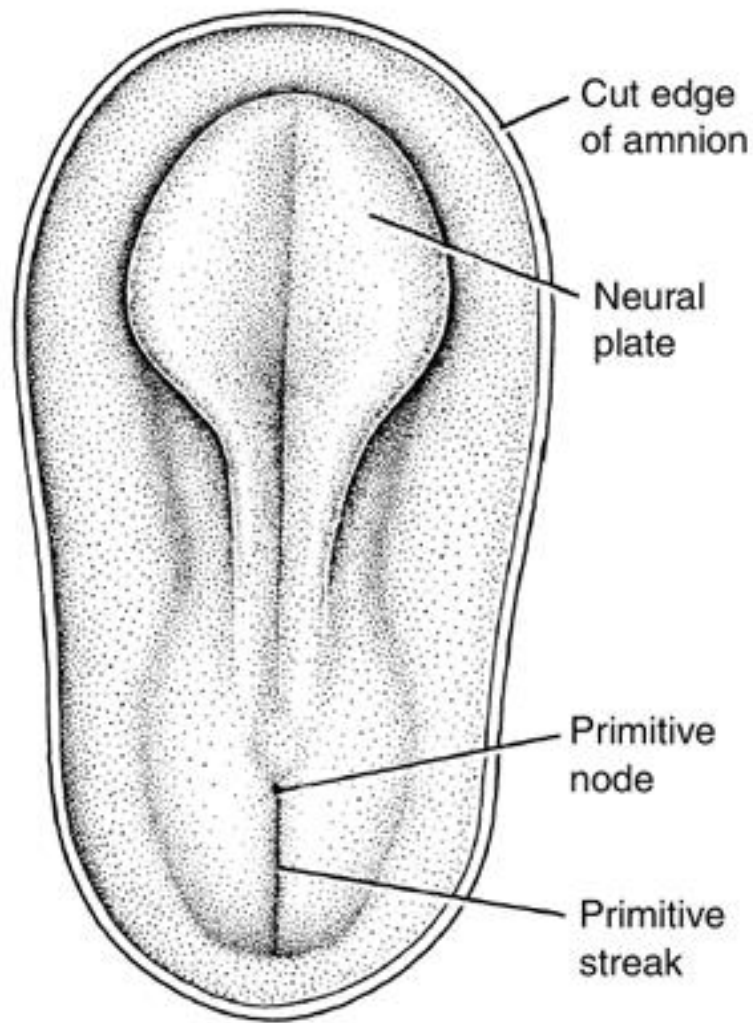
34–35



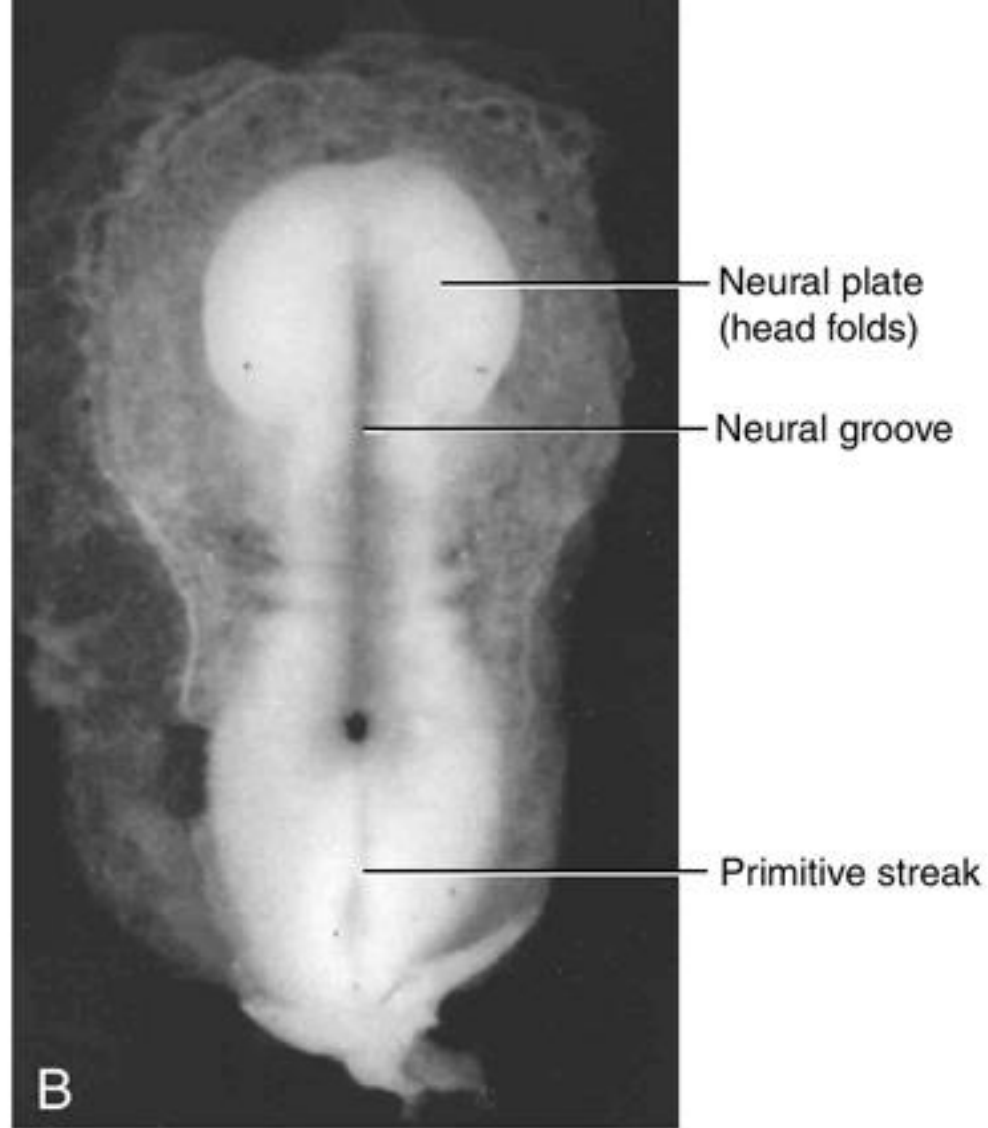
B



Actual size 4.0 mm

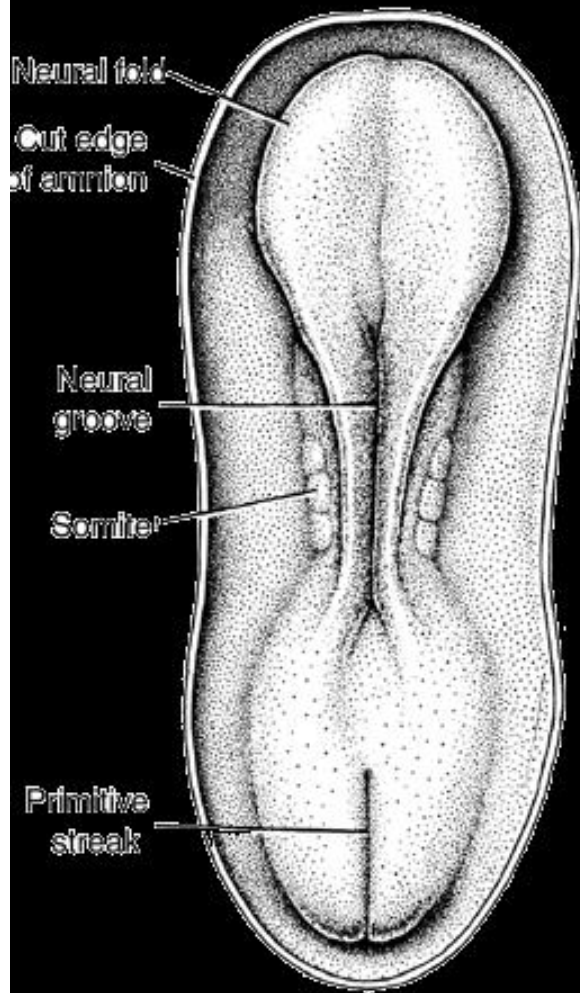


19 days

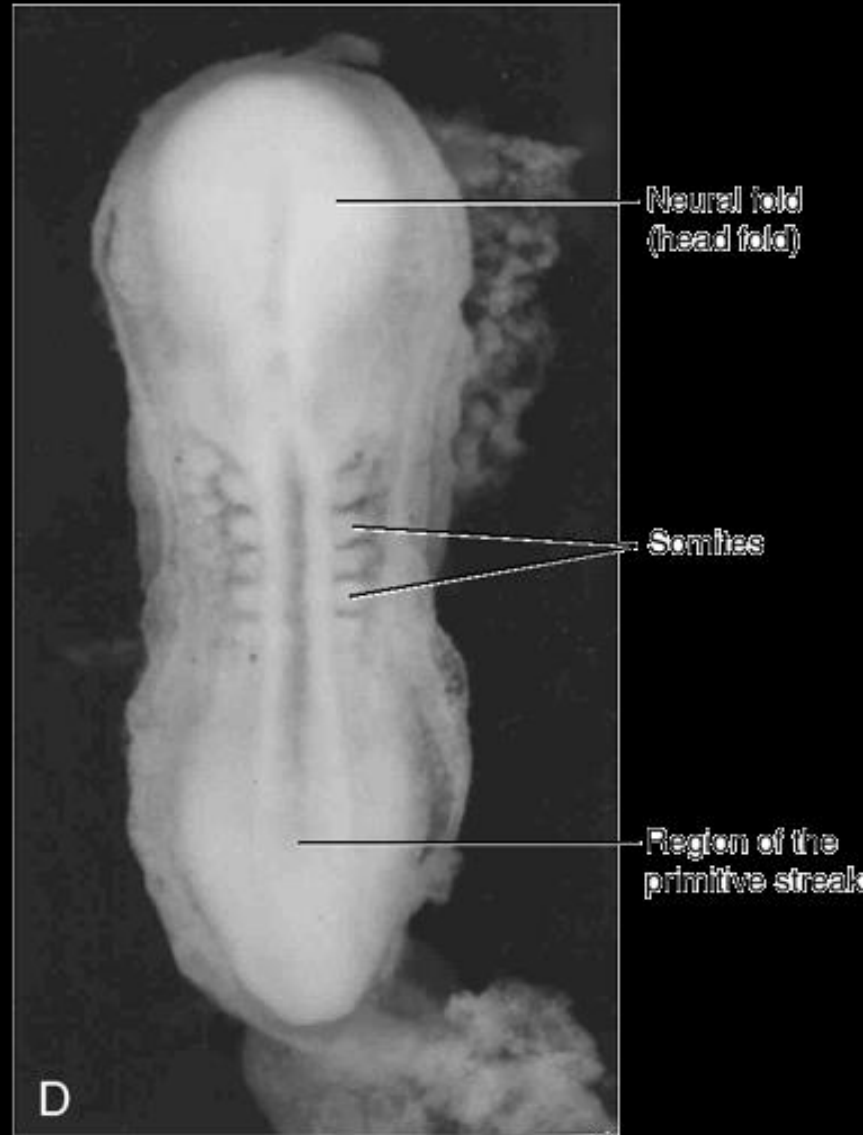


19 days

- A. Dorsal view of a late presomite embryo (approximately **19 days**). The amnion has been removed, and the neural plate is clearly visible.
- B. Dorsal view of a human embryo at **19 days**.



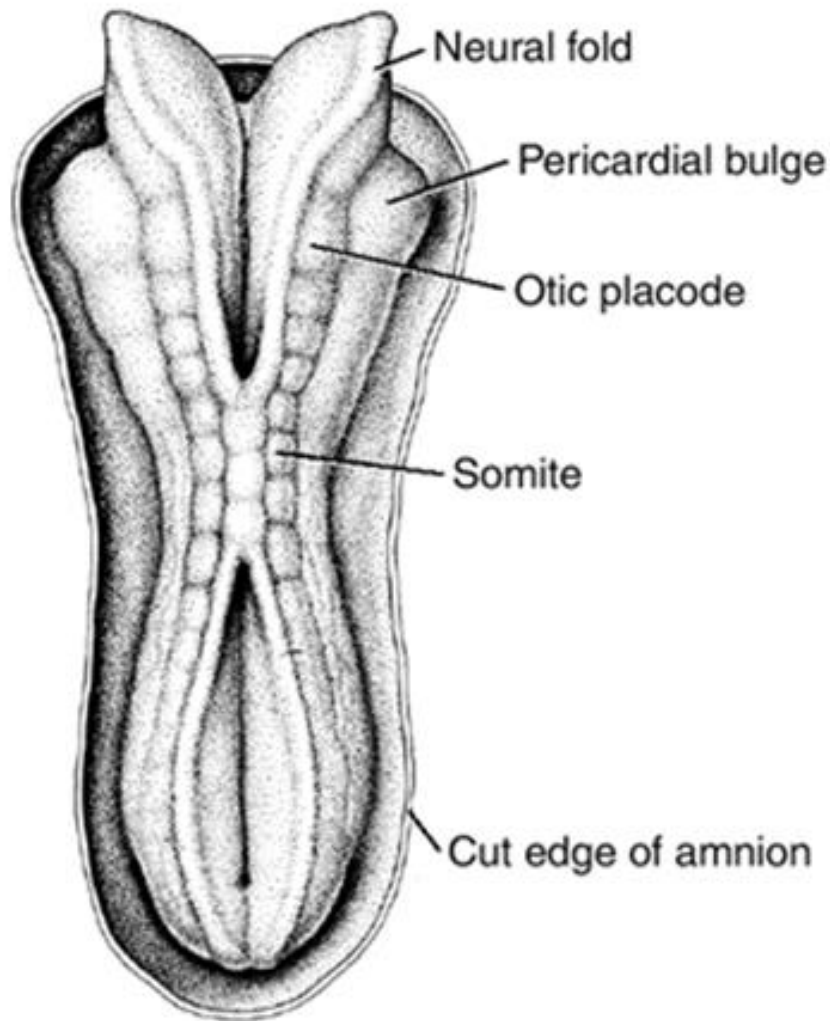
C



D

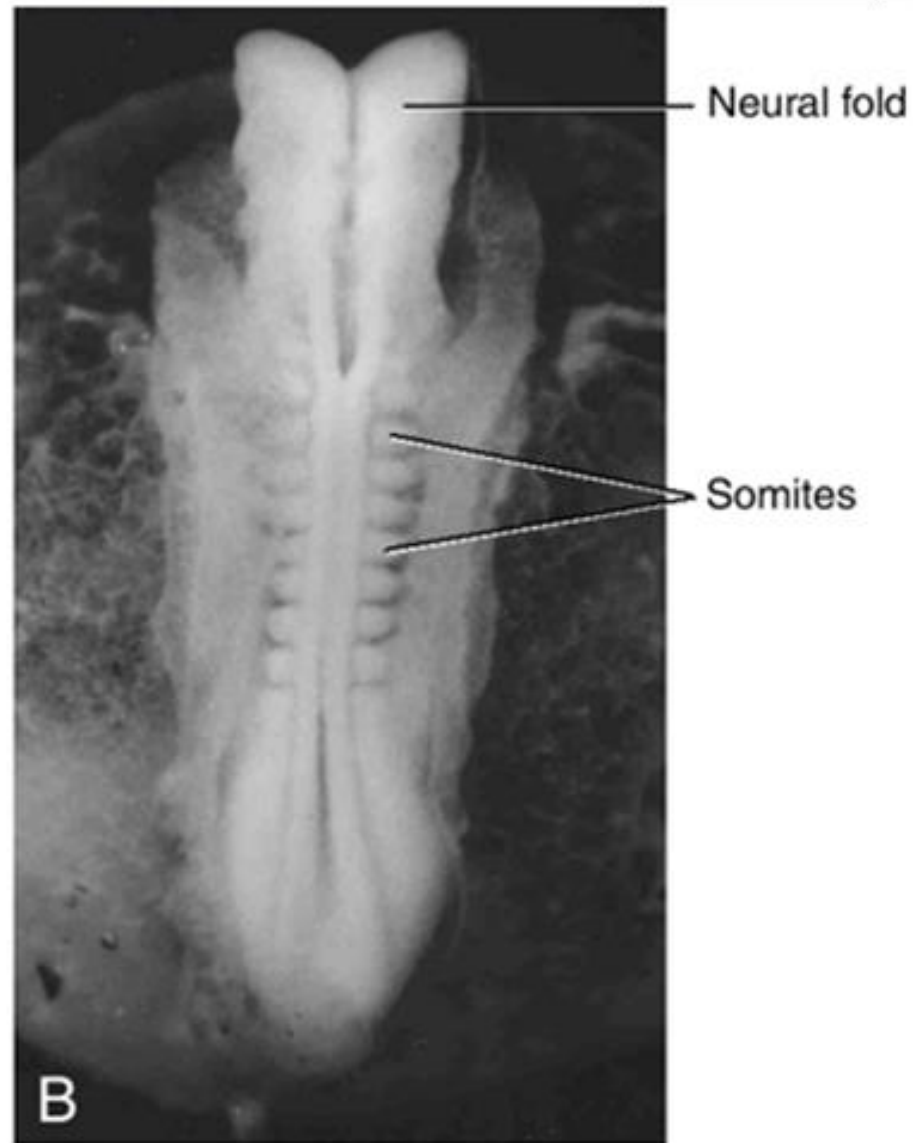
C. Dorsal view of an embryo at approximately **20 days** showing somites and formation of the neural groove and neural folds.

D. Dorsal view of a human embryo at **20 days**.



A

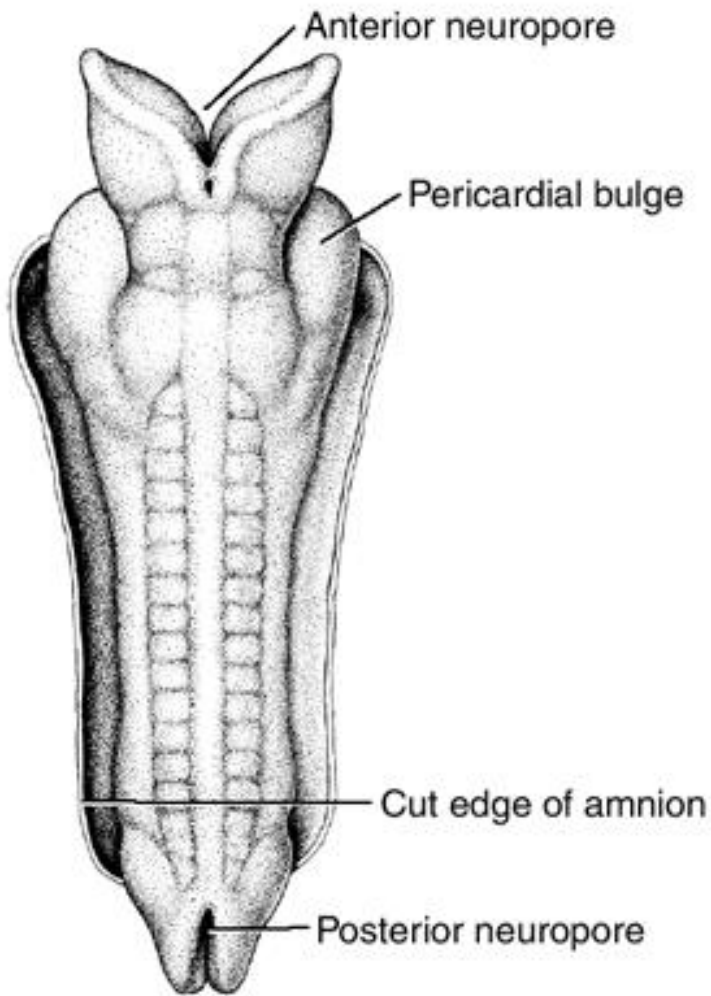
22 days



B

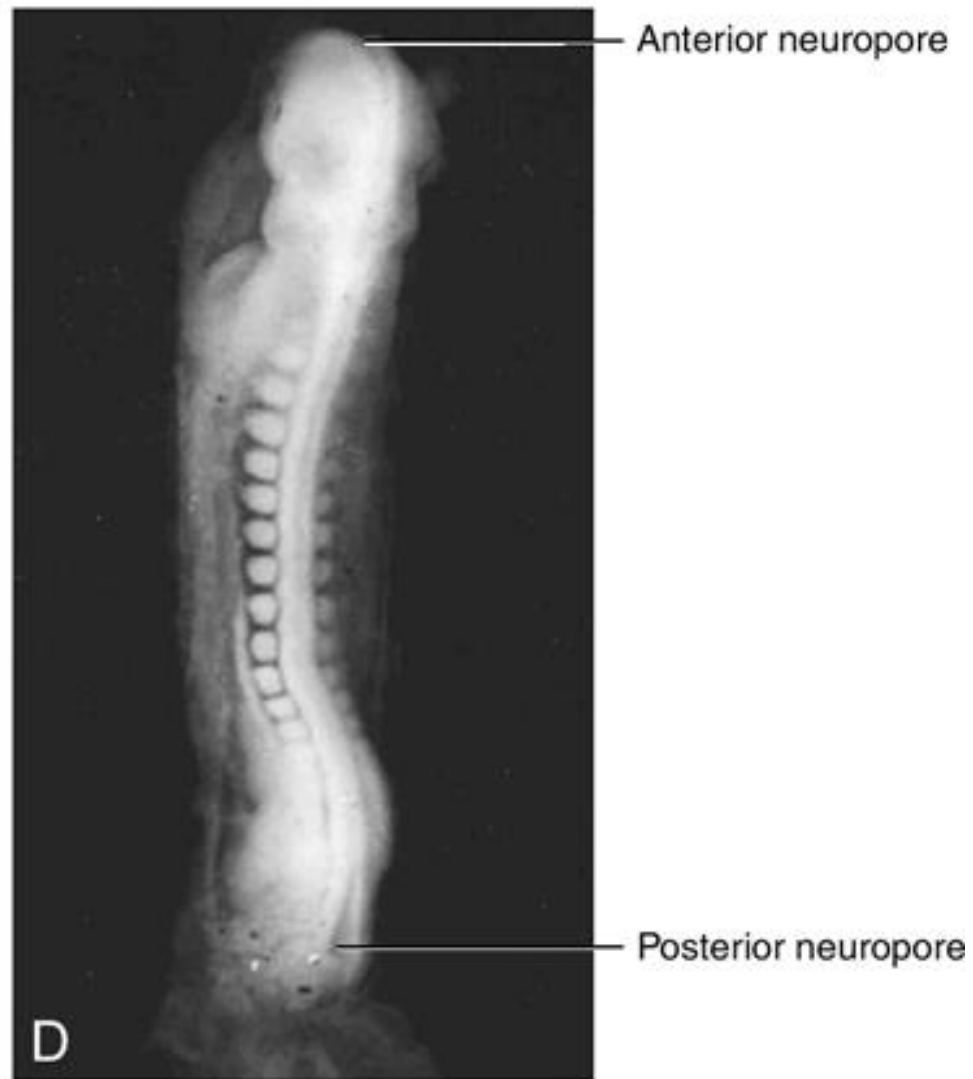
22 days

- A. Dorsal view of an embryo at approximately **day 22**.
 Seven distinct somites are visible on each side of the neural tube.
- B. Dorsal view of a human embryo at **22 days**.



23 days

C

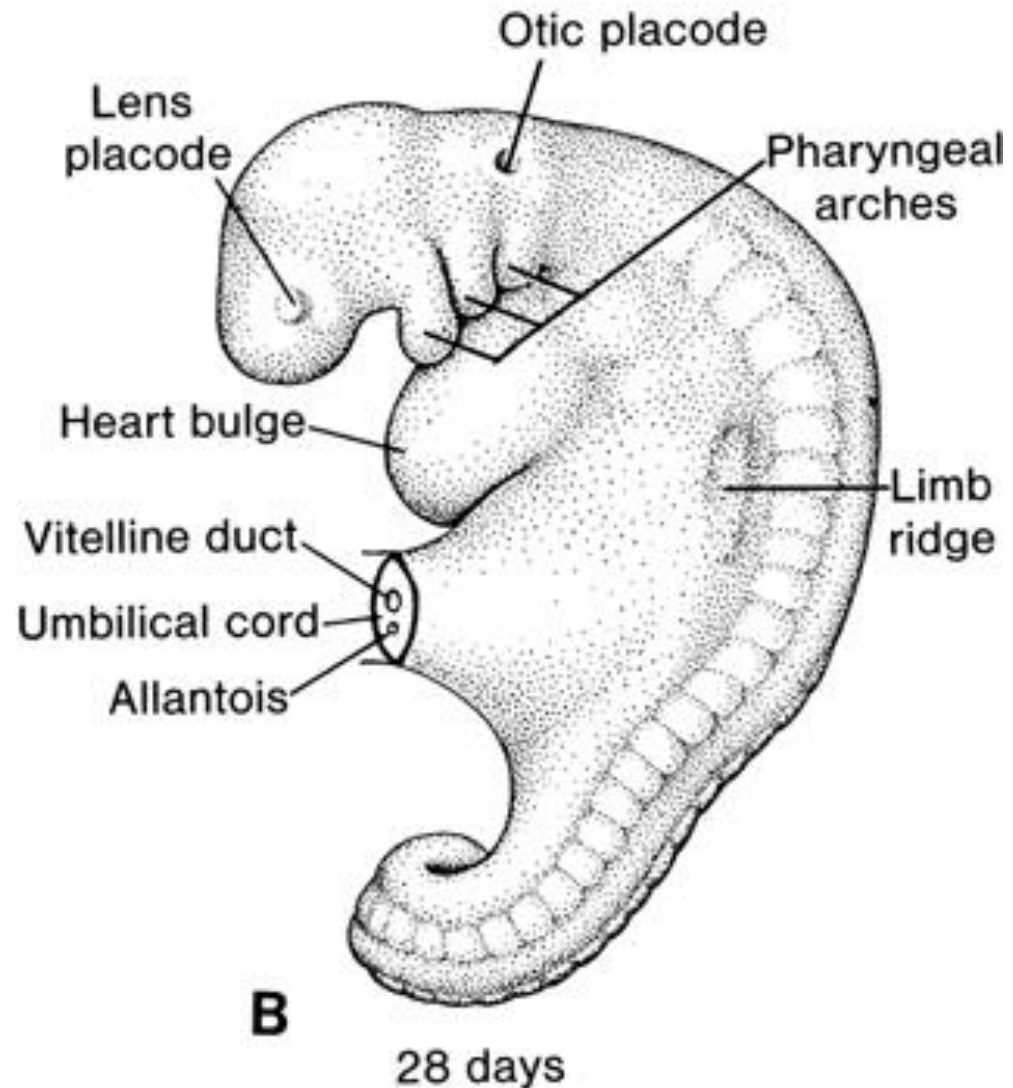
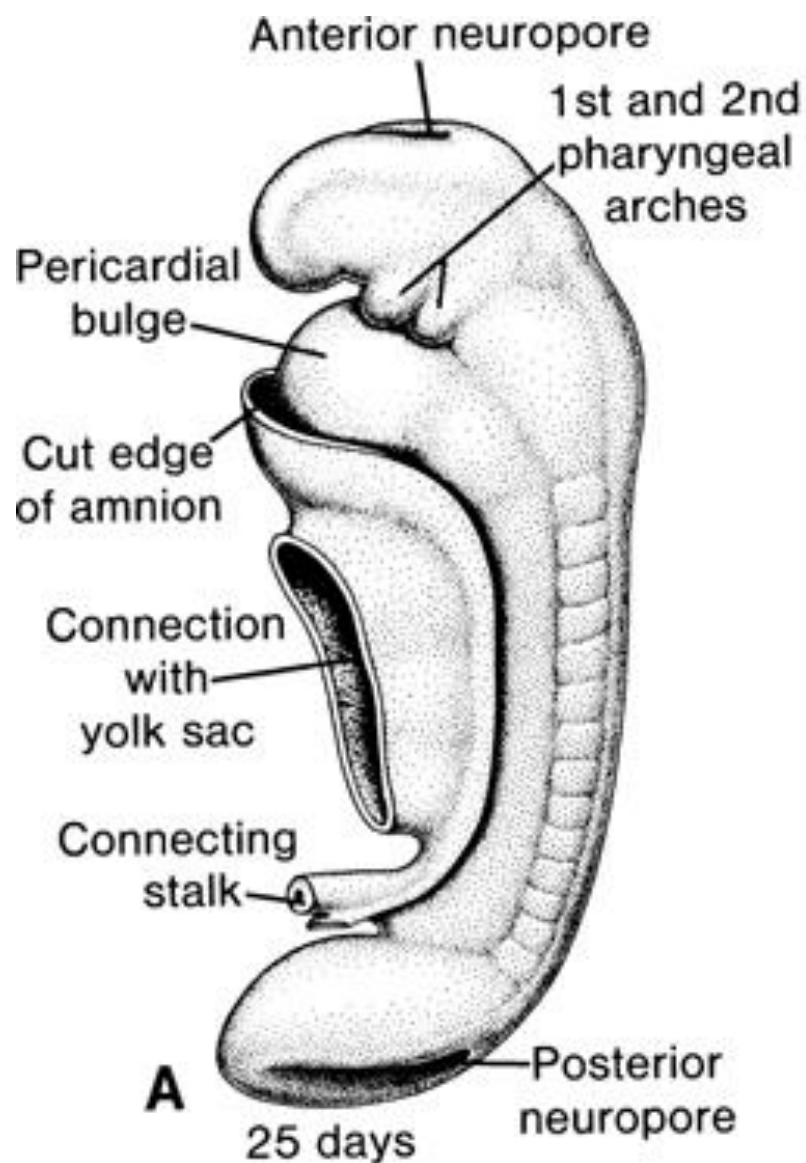


23 days

D

C. Dorsal view of an embryo at approximately **day 23**. Note the pericardial bulge on each side of the midline in the cephalic part of the embryo.

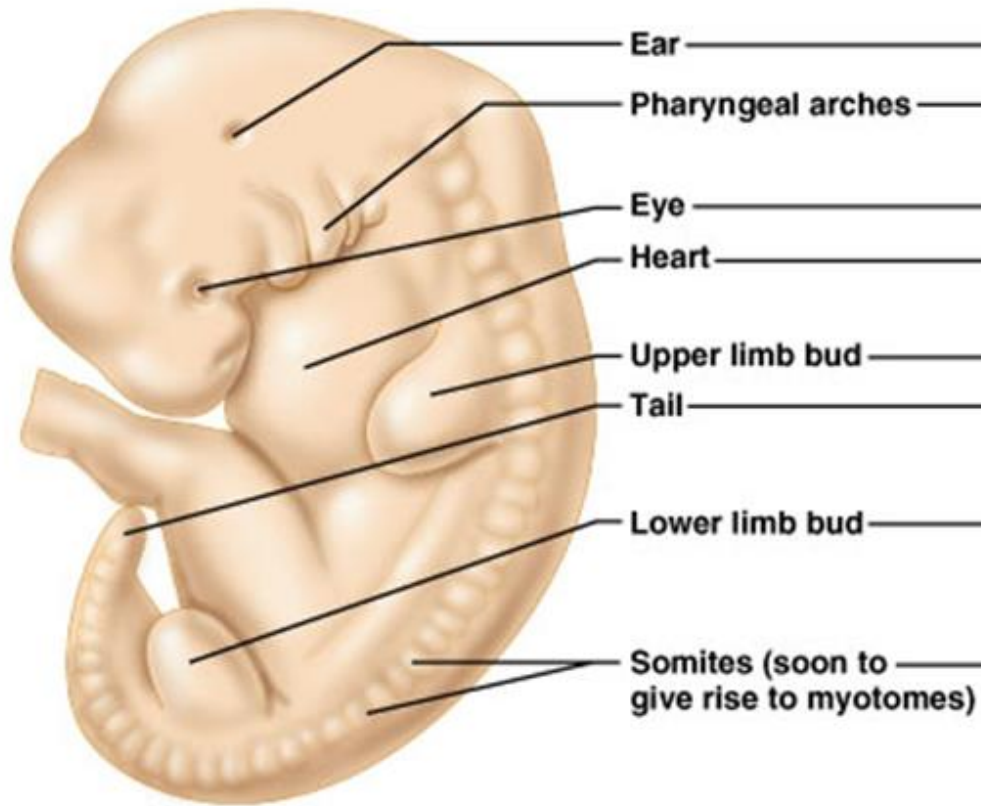
D. Dorsal view of a human embryo at **23 days**.



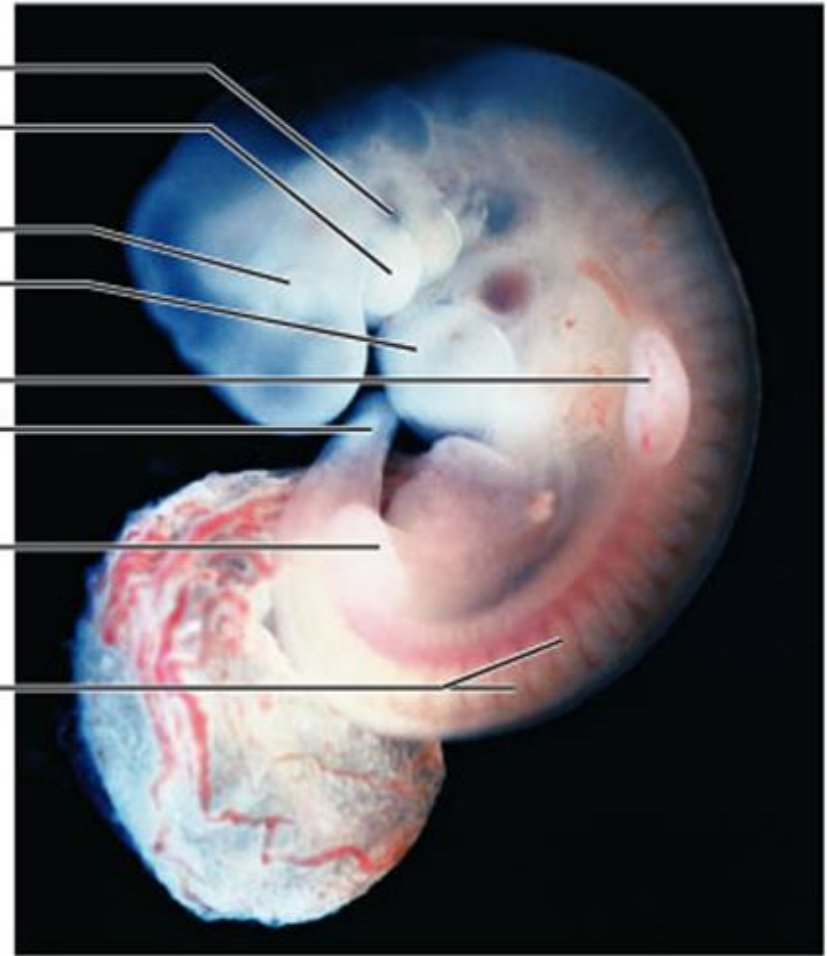
- A. Lateral view of a 14-somite embryo (approximately **25 days**).
 Note the bulging pericardial area and the first and second pharyngeal arches.
- B. The left side of a 25-somite embryo approximately **28 days** old.
 The first three pharyngeal arches and lens and otic placodes are visible.

29 day embryo

→ this is when the heart starts pumping, about 4 weeks



(a)



(b)

- **Somites :**
 - first appear in the future occipital region of the embryo.
 - develop craniocaudally and give rise to most of the *axial skeleton* and *associated musculature* as well as to *the adjacent dermis of the skin*.
 - The first pair of somites appears at the **end of the third** week a short distance caudal to the site at which the otic placode forms.
 - **Cranial somites are the oldest and caudal somites are the youngest.**
 - The ordered progression of segmentation involves a clock mechanism (oscillator) of gene expression, in particular Notch.
 - Also, motor axons from the spinal cord innervate muscle cells in the somites, a process that requires the correct guidance of axons from the spinal cord to the appropriate target cells.

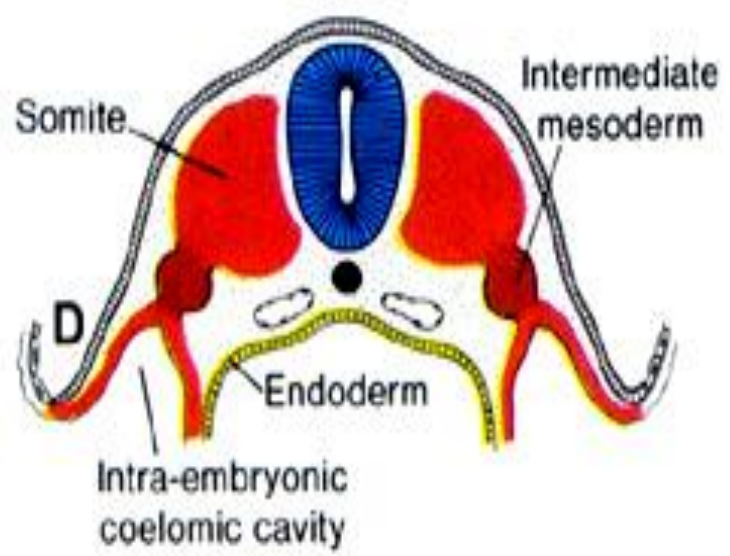
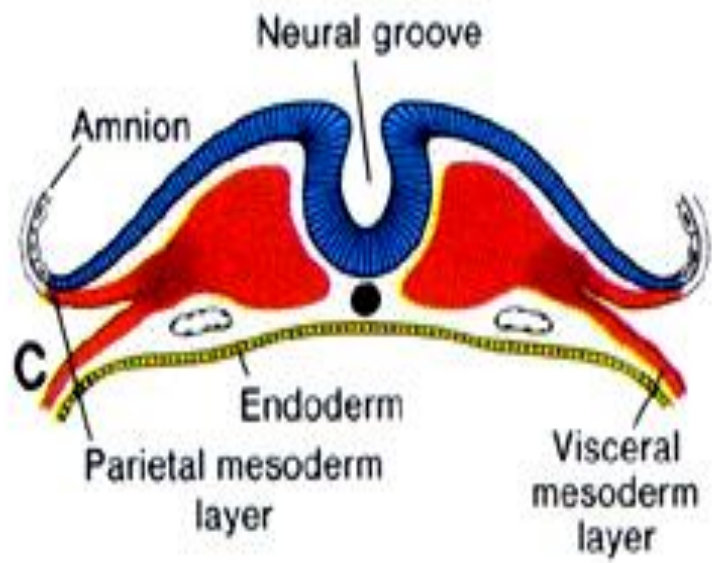
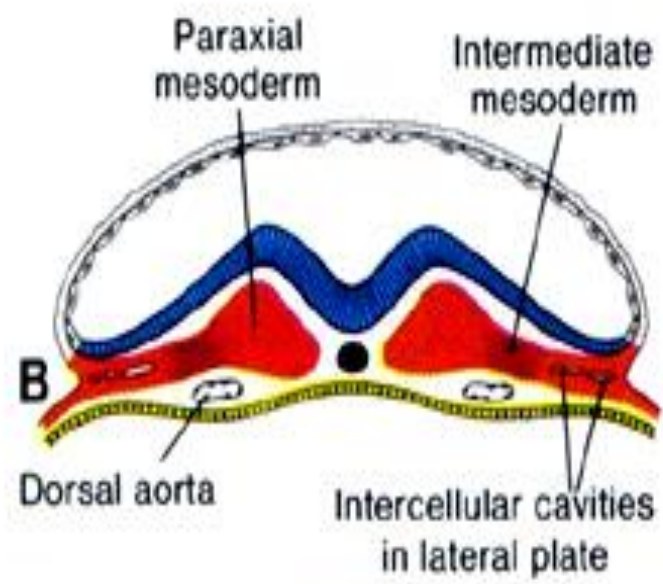
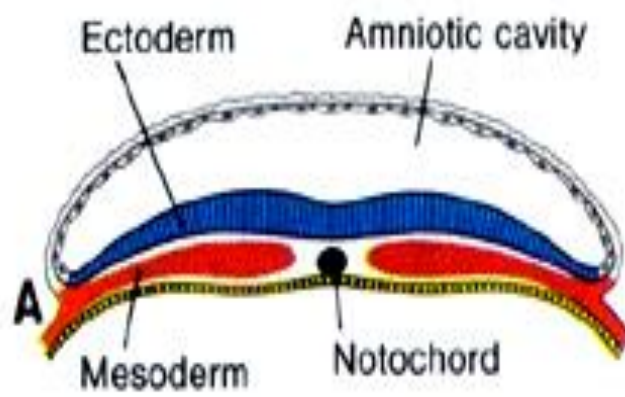
- There are:
 - **four occipital**
 - **eight cervical**
 - **twelve thoracic**
 - **five lumbar**
 - **five sacral, and eight to 10 coccygeal pairs.**
- The **first occipital** and the last **five to seven** coccygeal somites later disappear, while the remaining somites form the axial skeleton

2. INTERMEDIATE MESODERM

- Temporarily connects paraxial mesoderm with the lateral plate mesoderm
- *Differentiates into urogenital structures.*
- In cervical and upper thoracic regions:
 - it forms segmental cell clusters(future nephrotomes)
 - whereas more caudally, it forms an unsegmented mass of tissue, **the nephrogenic cord.**
- Excretory units of the urinary system and the gonads develop from this partly segmented, partly unsegmented intermediate mesoderm

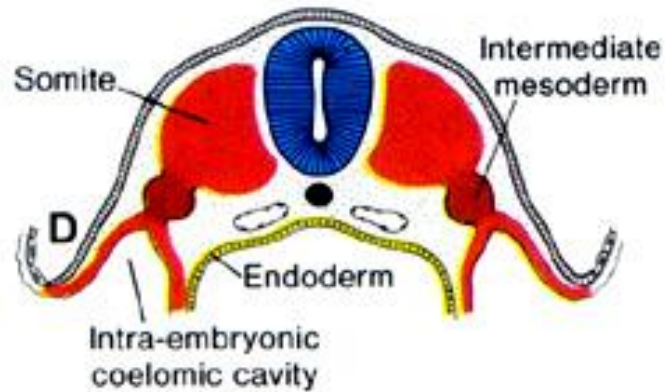
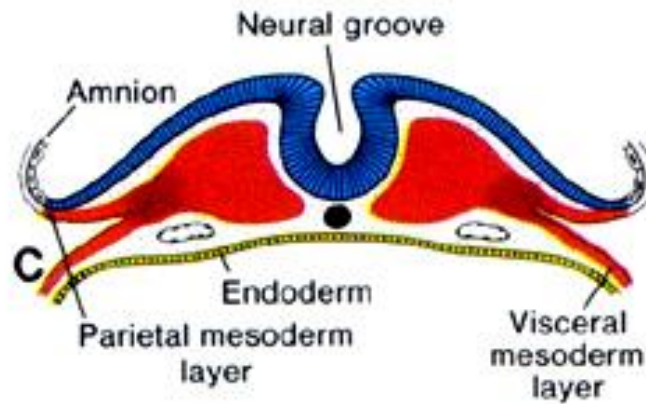
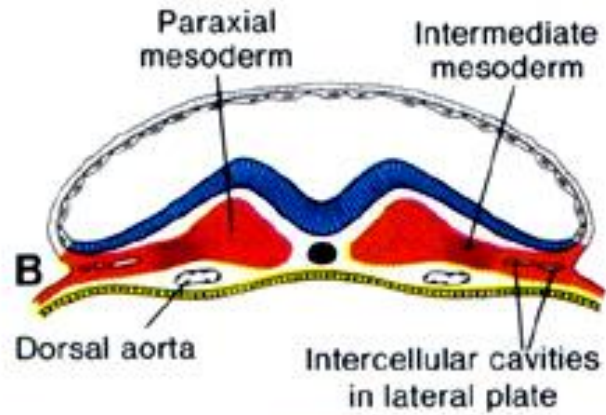
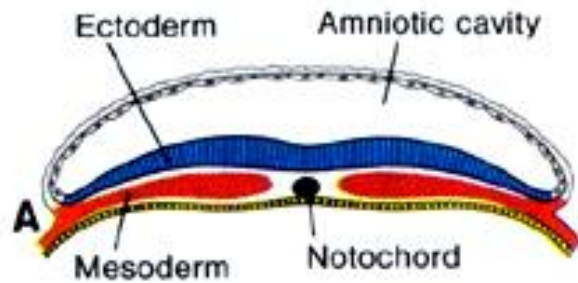
LATERAL PLATE MESODERM

- Splits into parietal and visceral layers, which line the intraembryonic cavity and **surround the organs**, respectively
- *Mesoderm from the **parietal layer**, together with overlying ectoderm, **will form the lateral and ventral body wall**.*
- The **visceral layer** and embryonic endoderm will form the wall of the gut
- Cells of the parietal layer surrounding the intraembryonic cavity will form thin membranes, the **mesothelial membranes, or serous membranes, which will** line the peritoneal, pleural, and pericardial cavities and secrete serous fluid
- *Mesoderm cells of the visceral layer will form a thin serous membrane around each organ*



DEVELOPMENT OF THE INTRAEMBRYONIC COELOM

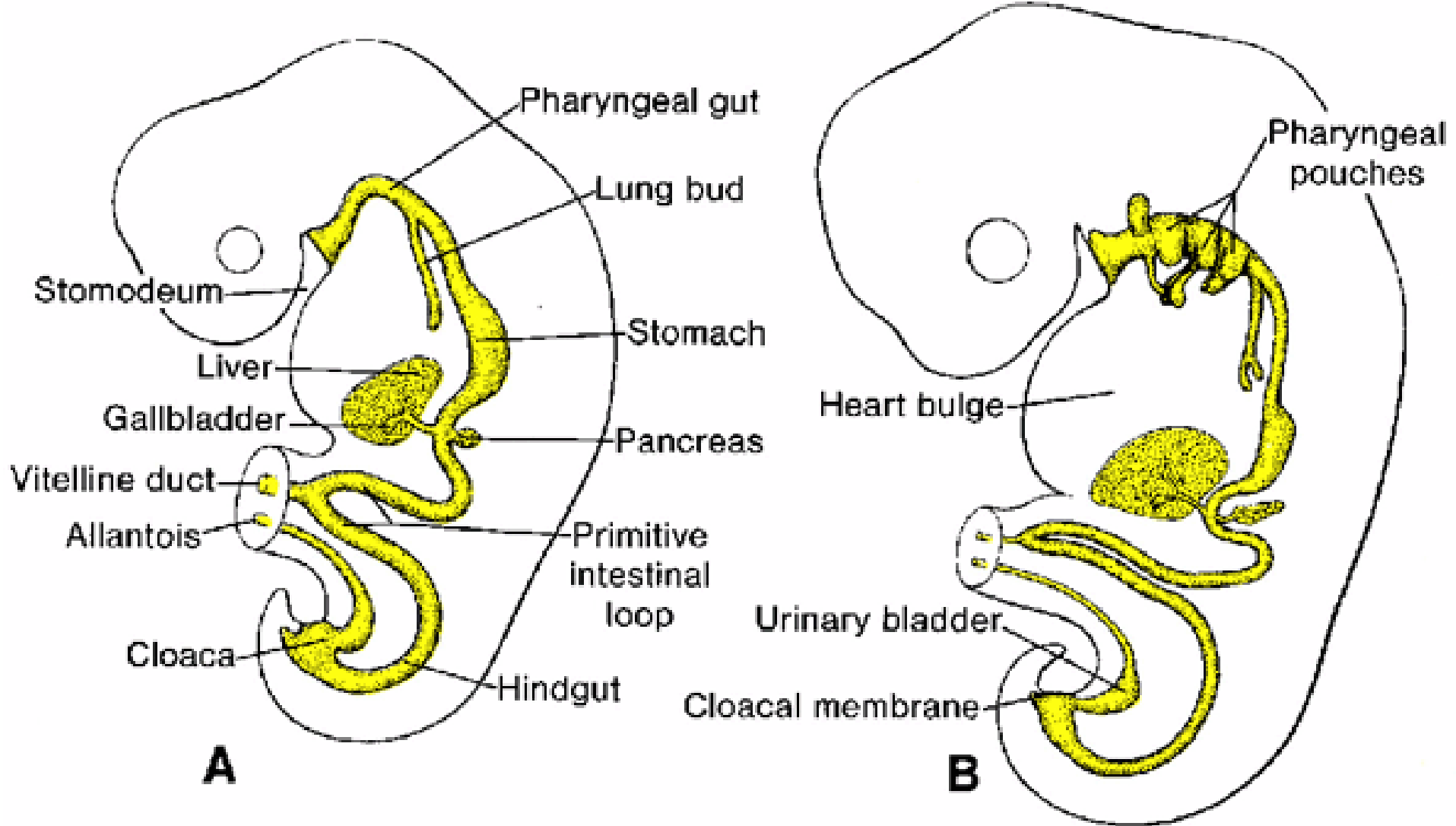
- The primordium of the intraembryonic coelom (embryonic body cavity)
 - appears as isolated **coelomic spaces** in the lateral mesoderm and **cardiogenic (heart-forming) mesoderm**.
 - These spaces soon coalesce to form a single horseshoe-shaped cavity, **the intraembryonic coelom**
 - The cavity divides the lateral mesoderm into two layers:
 - A **somatic or parietal layer** of lateral mesoderm located beneath the ectodermal epithelium and continuous with the extraembryonic mesoderm covering the amnion
 - A **splanchnic or visceral layer** of lateral mesoderm located adjacent to the endoderm and continuous with the extraembryonic mesoderm covering the umbilical vesicle



- The somatic mesoderm and overlying ectoderm form the embryonic body wall or **somatopleure**
- the splanchnic mesoderm and underlying embryonic endoderm form the embryonic gut or **splanchnopleure**.
- During the **2nd month**, the intraembryonic coelom is divided into three body cavities:
 - **Pericardial cavity**
 - **Pleural cavities**
 - **Peritoneal cavity**

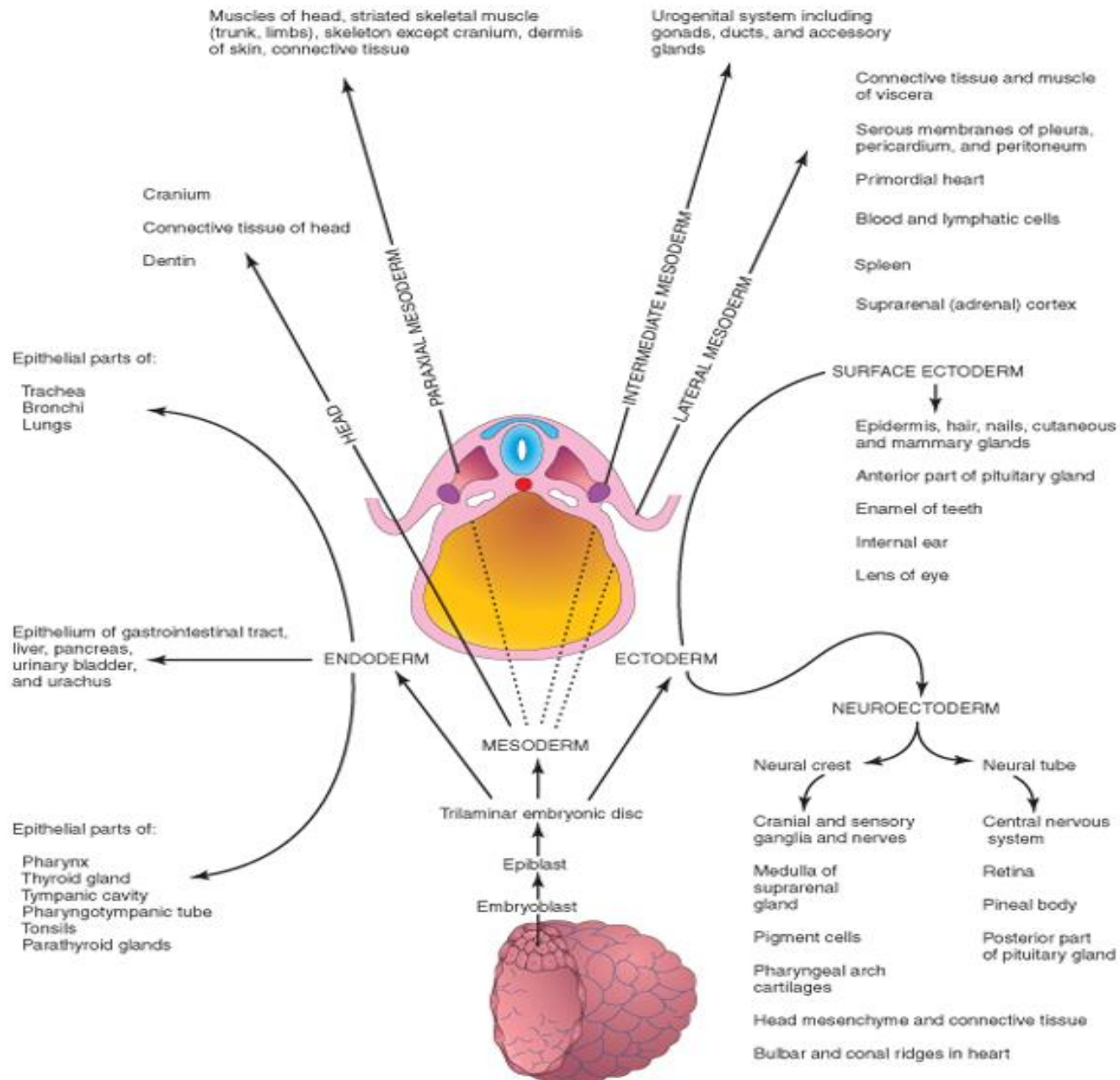
Differentiations of Endoderm:

- gives rise to:
 - the epithelial lining of the gastrointestinal and respiratory tracts
 - parenchyma of the tonsils
 - thyroid and parathyroid glands
 - thymus
 - Liver
 - Pancreas
 - epithelial lining of the urinary bladder and most of the urethra
 - the epithelial lining of the tympanic cavity, tympanic antrum, and pharyngotympanic (auditory) tube



Sagittal sections through embryos showing **derivatives of the endodermal germ layer**.

A. Pharyngeal pouches, epithelial lining of the lung buds and trachea, liver, gallbladder, and pancreas. B. The urinary bladder is derived from the cloaca and, at this stage of development, is in open connection with the allantois.



HIGHLIGHTS OF THE 4th TO 8th WEEKS

The following descriptions summarize the main developmental events and changes in external form of the embryo during the fourth to eighth weeks.

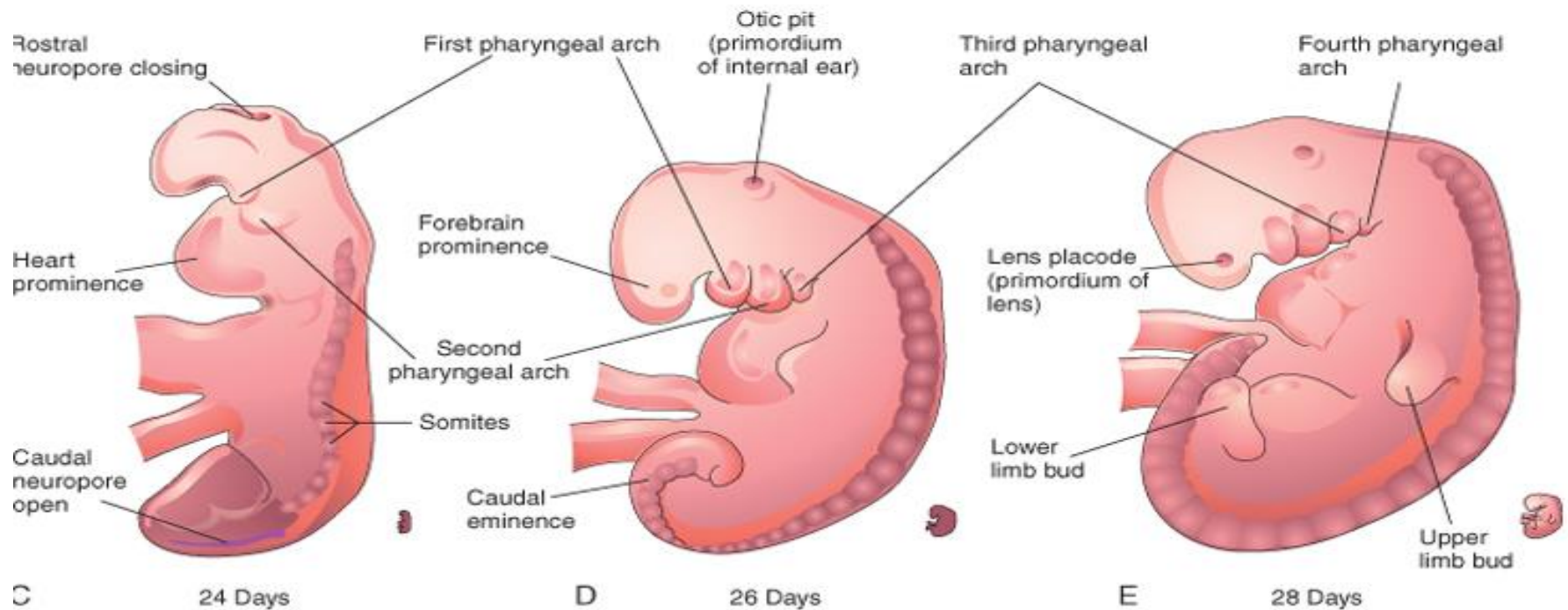
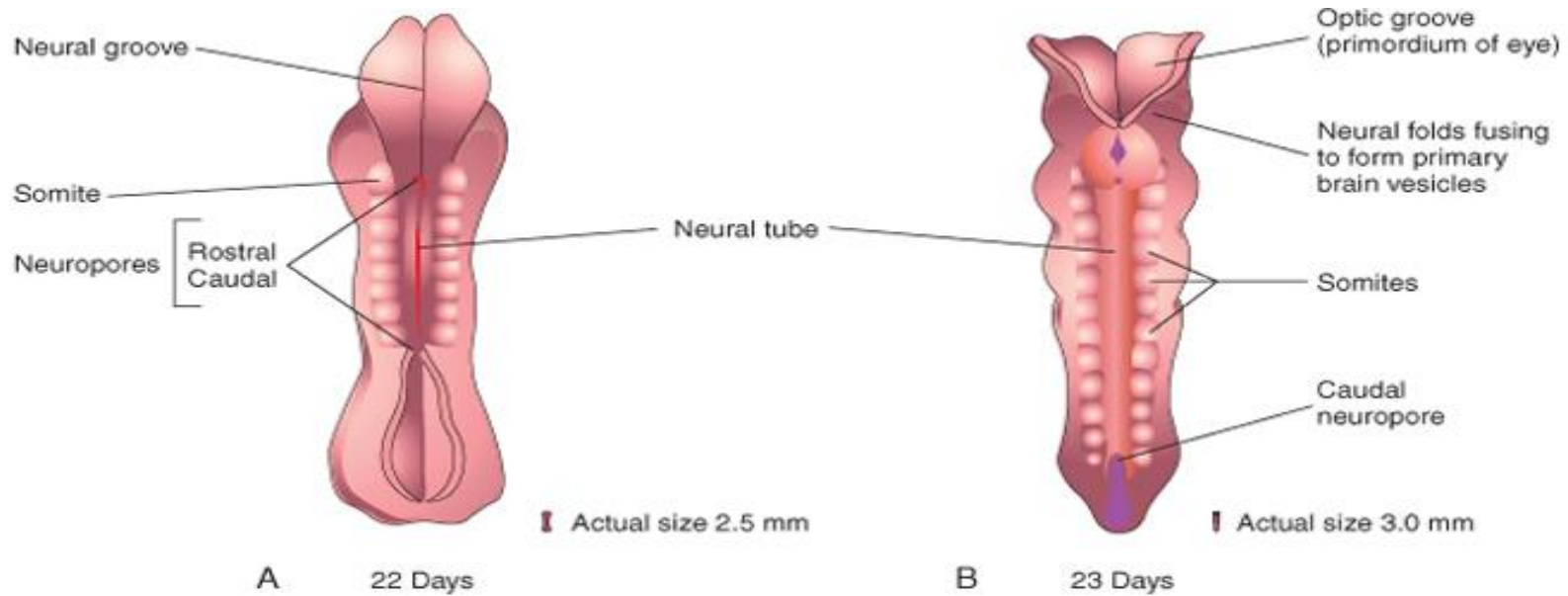
Major events during the 4th Week:

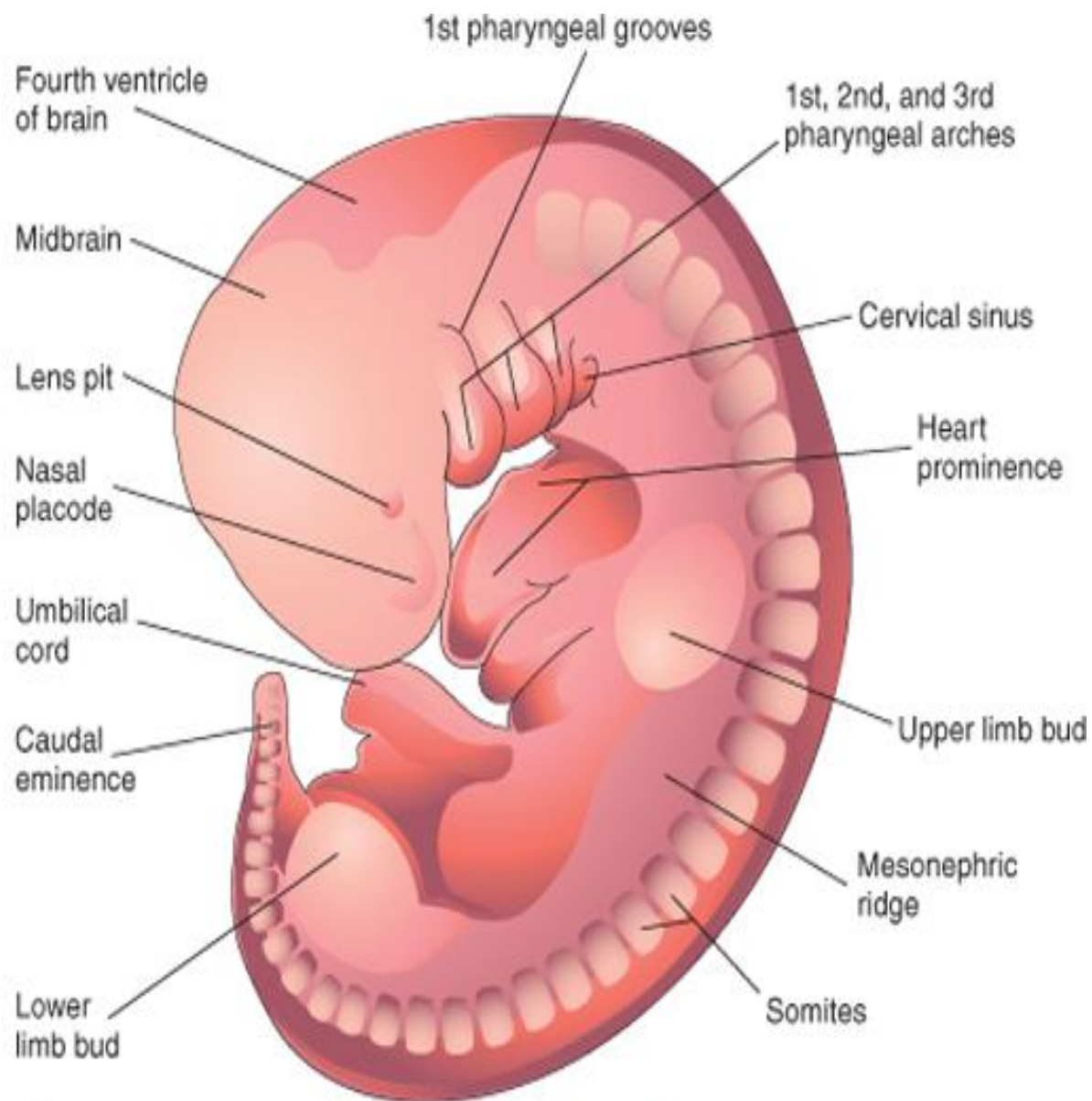
- Major changes in body form occur
- the embryo is almost straight and has **four to 12 somites** that produce conspicuous surface elevations
- The **neural tube** is formed opposite the somites, but it is widely open at **the rostral and caudal neuropores**

- **By 24 days**, the first two pharyngeal arches are visible.
 - The first (mandibular arch) and the second (hyoid arch) are distinct.
 - The major part of the first arch gives rise to the **mandible (lower jaw)**, and a rostral extension of the arch, the maxillary prominence, contributes to **the maxilla (upper jaw)**.
 - The embryo is now slightly curved because of the head and tail folds.
 - The heart produces a large ventral prominence and pumps blood.

- Three pairs of **pharyngeal arches** are visible **by 26 days** , and the rostral neuropore is closed.
- **The forebrain** produces a prominent elevation of the head
- Folding of the embryo has given the embryo **a C-shaped curvature**
- **Upper limb buds** are recognizable by day 26 or 27 as small swellings on the ventrolateral body walls.
- **The otic pits**, the primordia of the internal ears, are also visible.
- Ectodermal thickenings (**lens placodes**) indicating the future lenses of the eyes are visible on the sides of the head.
- The fourth pair of pharyngeal arches and the lower limb buds are visible by the end of the 4th week.
- Toward the end of the fourth week, a long tail-like **caudal eminence** is a characteristic feature.

• *By the end of the fourth week, the caudal neuropore is usually closed.*

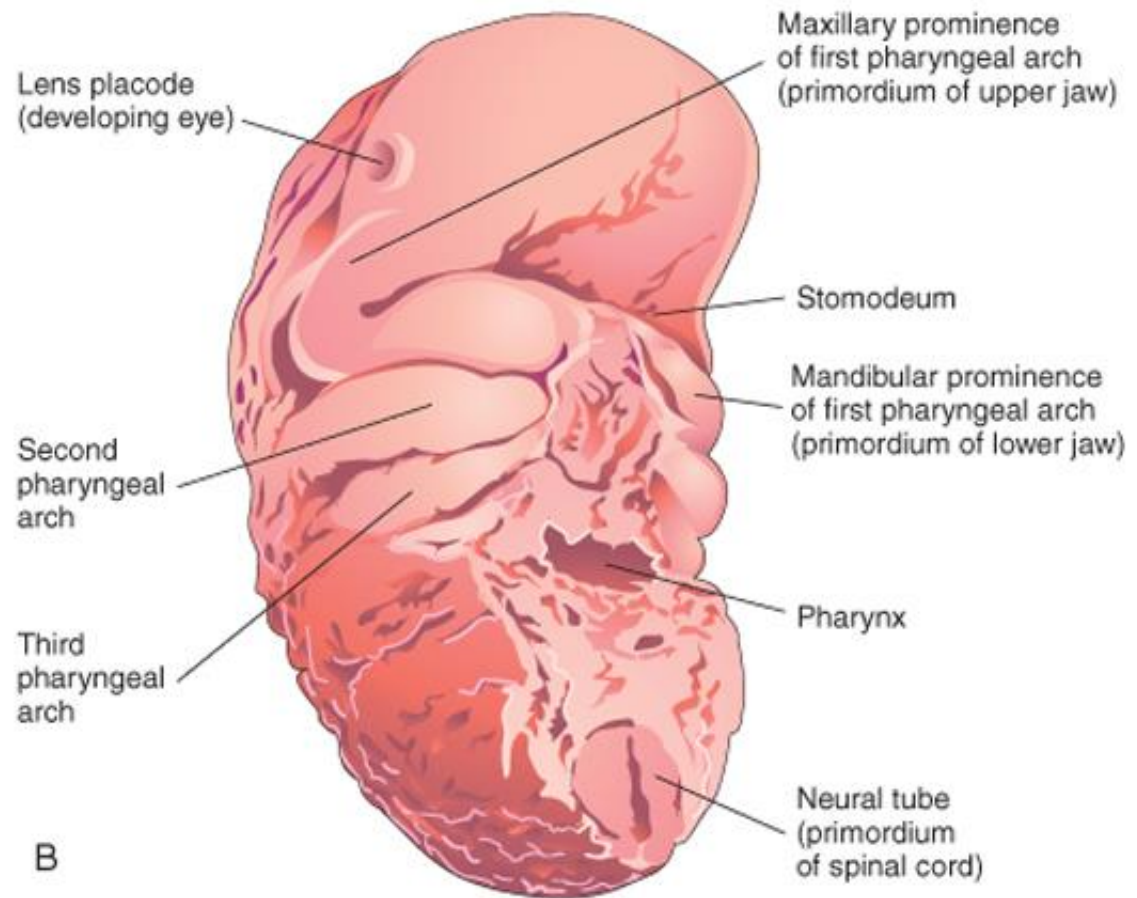




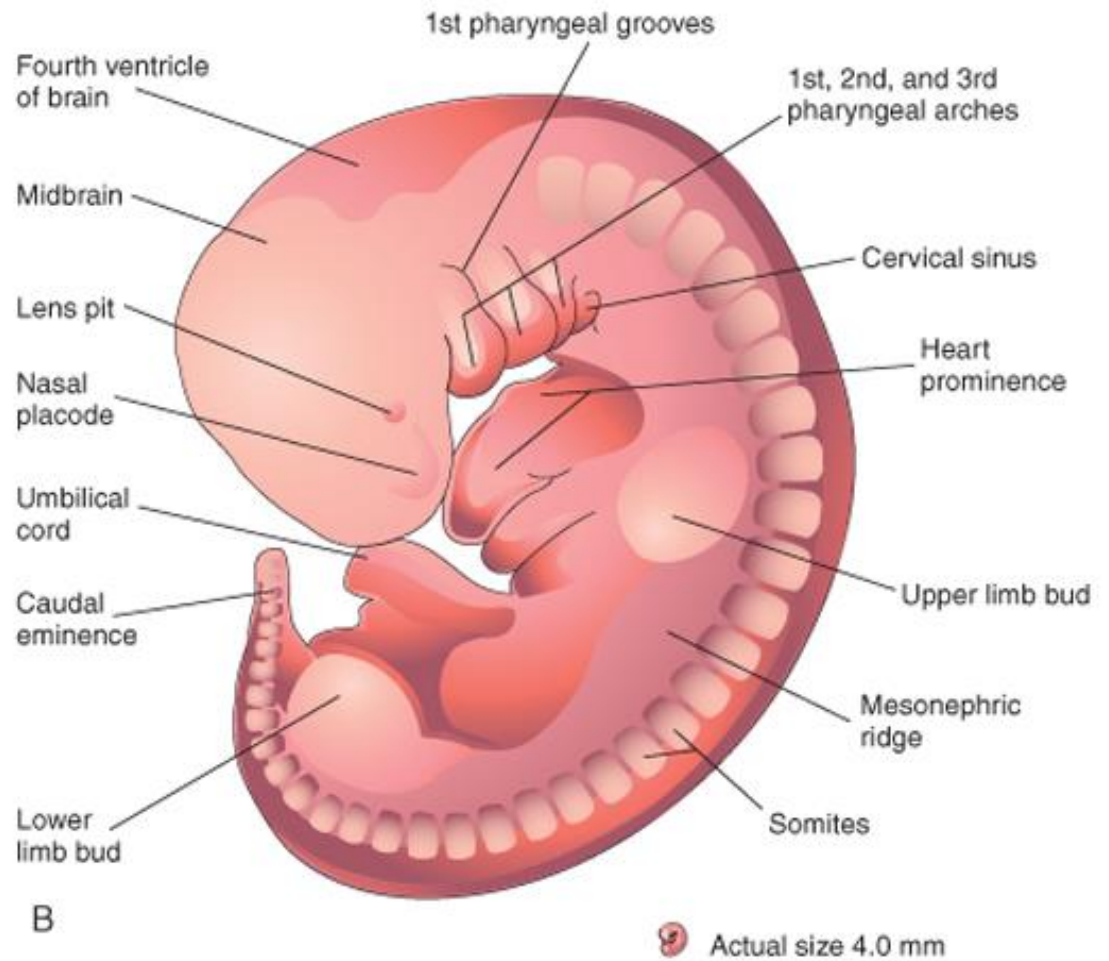
 Actual size 4.0 mm

Major events during the 5th Week

- Changes in body form are minor during the 5th week as compared with those of the 4th week,
- but growth of the **head exceeds** that of other regions
 - Enlargement of the head is caused mainly by **the rapid development of the brain and facial prominences**
- The face soon contacts the **heart prominence**.
- The rapidly growing 2nd pharyngeal arch overgrows the 3rd and 4th arches, forming a lateral ectodermal depression on each side-the **cervical sinus**.
- Mesonephric ridges indicate the site of the mesonephric kidneys, which are interim excretory organs in humans.



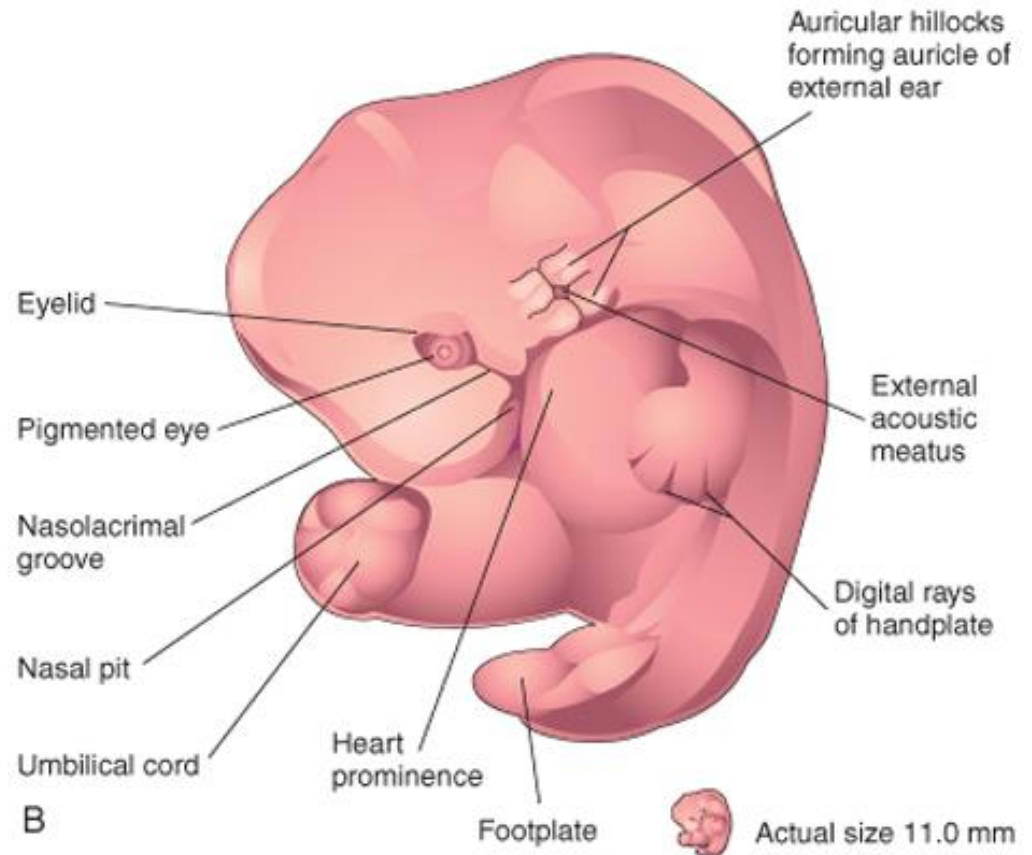
Scanning electron micrograph of the craniofacial region of a human embryo of **approximately 32 days** (Carnegie stage 14, 6.8 mm). Three pairs of pharyngeal arches are present. The maxillary and mandibular prominences of the first arch are clearly delineated. Observe the large mouth located between the maxillary prominences and the fused mandibular prominences



A, Lateral view of an embryo at [Carnegie stage 14](#), approximately 32 days. The second pharyngeal arch has overgrown the third arch, forming a depression known as the cervical sinus. The mesonephric ridge indicates the site of the mesonephric kidney, an interim kidney (see Chapter 12). **B**, Illustration of the structures shown in **A**. The upper limb buds are paddle shaped and the lower limb buds are flipper-like

Sixth Week

- embryos show reflex response to touch
- The upper limbs begin to show regional differentiation as the **elbows and large handplates develop**
- The primordia of the digits (fingers), called **digital rays**, begin to develop in the handplates
- Embryos shows **spontaneous movements**, such as twitching of the trunk and limbs
- Development of the lower limbs occurs **4 to 5 days later** than that of the upper limbs
- Several small swellings-**auricular hillocks**-develop around the pharyngeal groove or cleft between the first two pharyngeal arches.
- This groove becomes the **external acoustic meatus** (external auditory canal).
- The auricular hillocks contribute to the formation of the **auricle, the shell-shaped part of the external ear.**

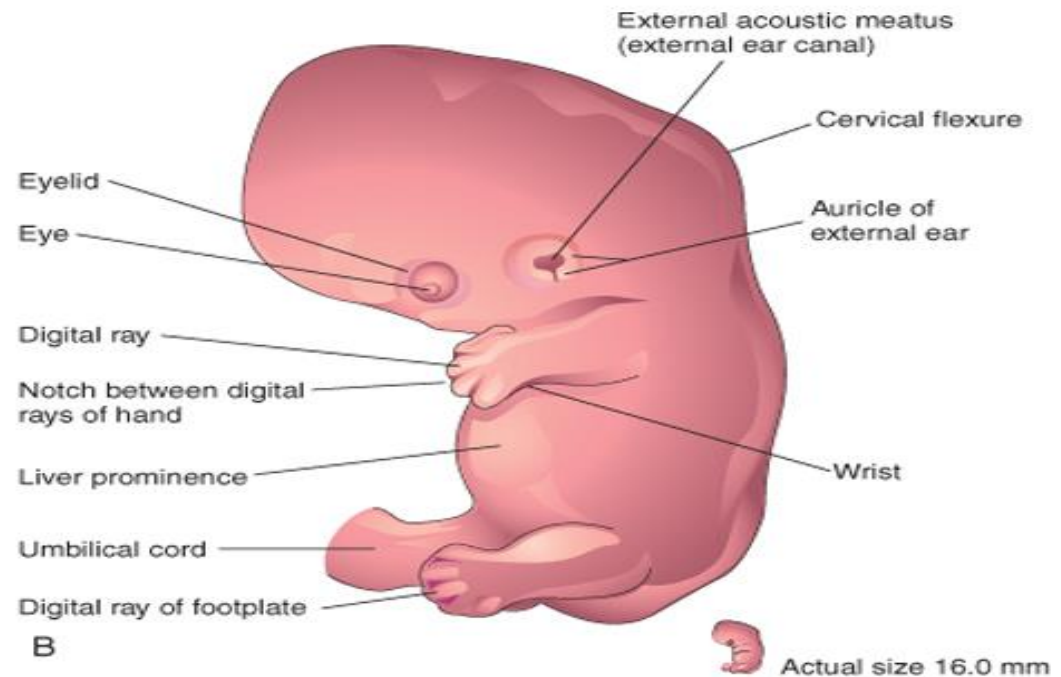


- **A**, Lateral view of an embryo at **Carnegie stage 17, approximately 42 days**.
- Digital rays are visible in the handplate, indicating the future site of the digits.
- **B**, Drawing illustrating the structures shown in **A**. The eye, auricular hillocks, and external acoustic meatus are now obvious.

- **The eye is now obvious** because retinal pigment has largely formed
- The head is now much larger relative to the trunk and is bent **over the heart prominence.**
- This head position results from bending in the cervical (neck) region.
- The trunk and neck have begun to straighten
- The intestines enter the extraembryonic coelom in the proximal part of the umbilical cord
- This **umbilical herniation** is a normal event in the embryo.
- The herniation occurs because the abdominal cavity is too small at this age to accommodate the rapidly growing intestine.

Seventh Week

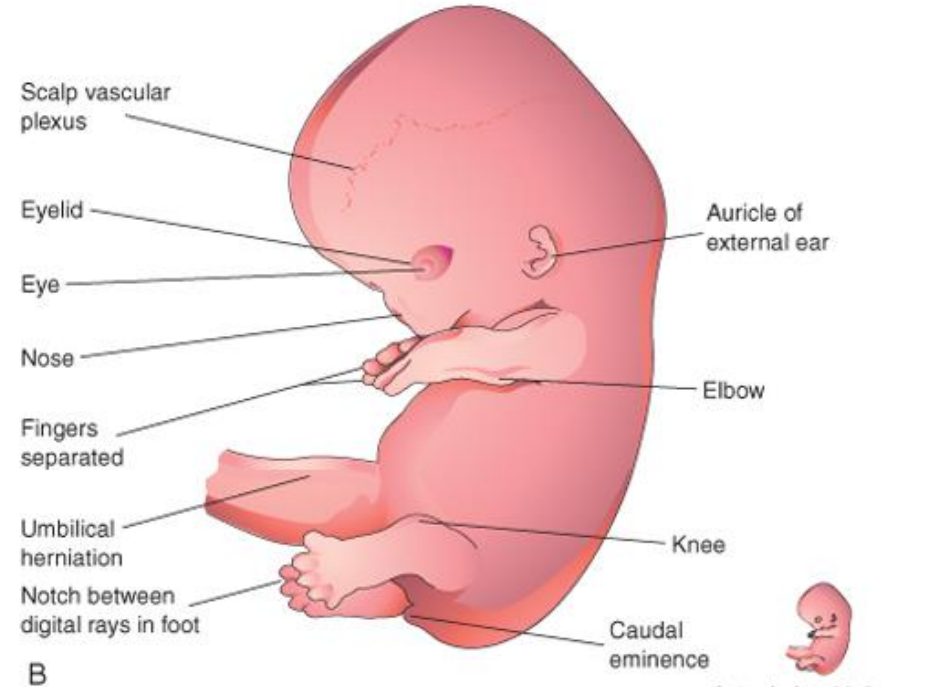
- The limbs undergo considerable change
- Notches appear between the digital rays in the **handplates, clearly indicating the future digits**).
- Communication between the primordial gut and umbilical vesicle is now reduced to a relatively slender duct, the omphaloenteric duct.
- By the end of the **7th week**, ossification of the bones of the upper limbs has begun.



Eighth Week

- this final week of the embryonic period, the digits of the hand are separated but noticeably webbed
- Notches are now clearly visible between the digital rays of the feet.
- The caudal eminence is still present but stubby.
- The **scalp vascular plexus** has appeared and forms a characteristic band around the head

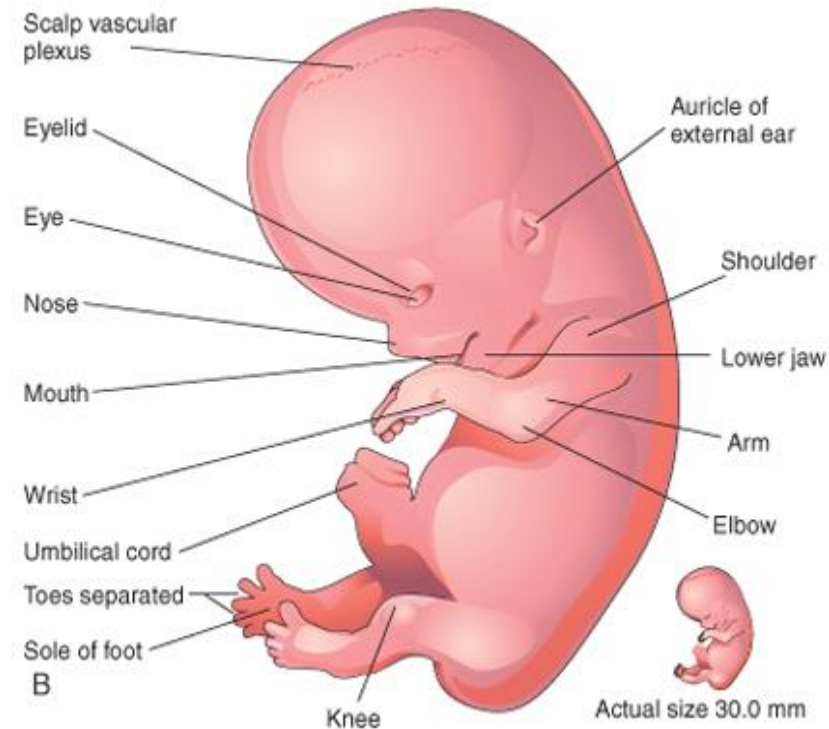
A, Lateral view of an embryo at Carnegie stage 21, approximately 52 days. Note that the feet are fan shaped. The scalp vascular plexus now forms a characteristic band across the head. The nose is stubby and the eye is heavily pigmented



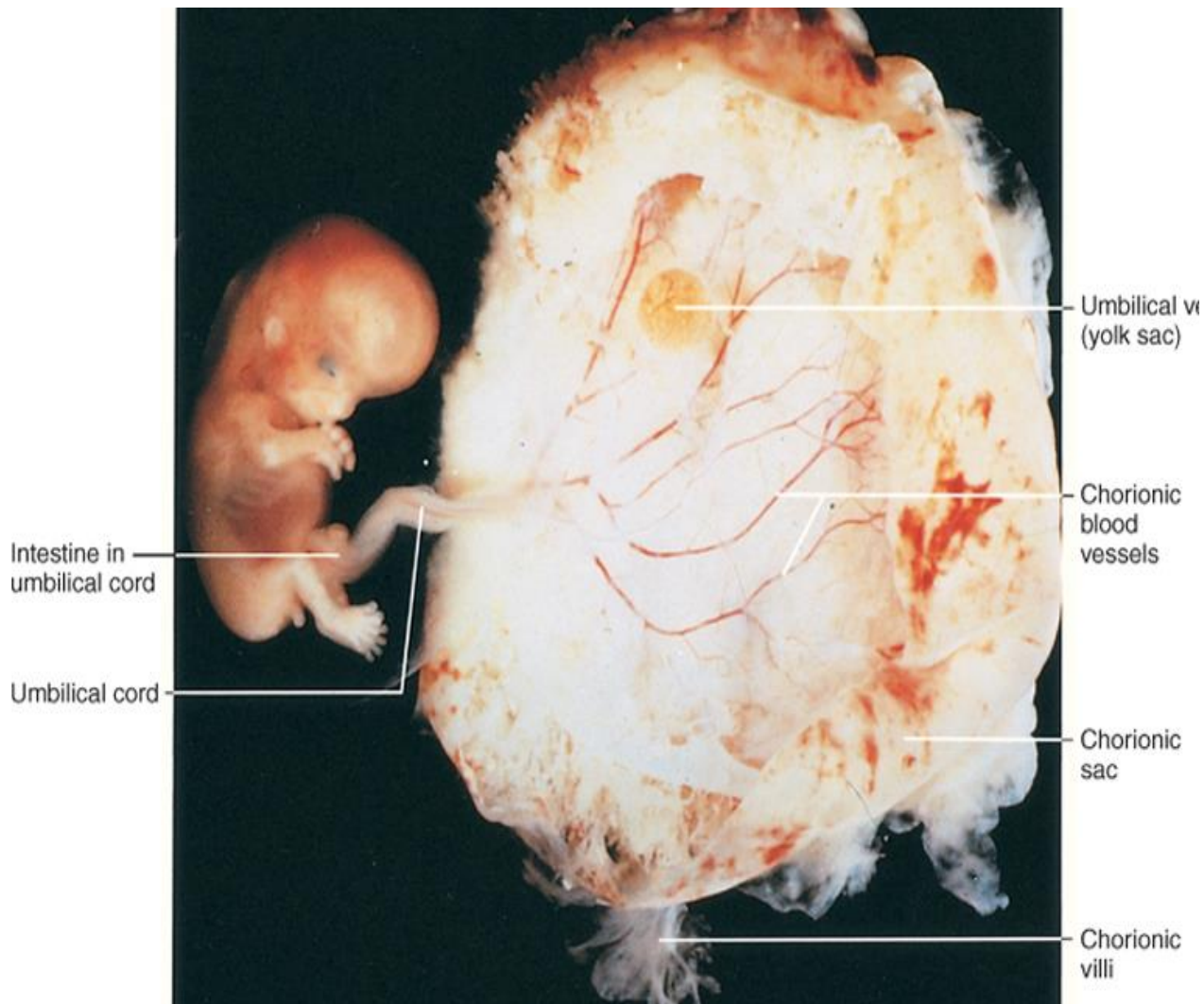
- By the end of the eighth week:

- all regions of the limbs are apparent
- the digits have lengthened and are completely separated
- purposeful limb movements first occur during this week.
- ossification begins in the femur.
- all evidence of the caudal eminence has disappeared
- both hands and feet approach each other ventrally.

Lateral view of an embryo at Carnegie stage 23, approximately 56 days. The embryo has a distinct human appearance. **B**, Illustration of the structures shown in **A**. **C**, A Carnegie stage 23 embryo, approximately 56 days after ovulation, imaged with optical microscopy (*left*) and magnetic resonance microscopy (*right*).



- **At the end of the eighth week:**
 - ✓ the embryo has distinct human characteristics
 - ✓ the head is still **disproportionately large**, constituting almost half of the embryo.
 - ✓ The neck region is established, and the eyelids are more obvious.
 - ✓ The **eyelids are closing and** they begin to unite by epithelial fusion
 - ✓ The intestines are still in the proximal portion of the umbilical cord.
 - ✓ The auricles of the external ears begin to assume their final shape.
 - ✓ Although there are sex differences in the appearance of the external genitalia
 - ✓ they are not distinctive enough to permit accurate sexual identification



Lateral view of an embryo and its chorionic sac at Carnegie stage 23, **approximately 56** days.

Observe the human appearance of the embryo

Table 5-1. Criteria for Estimating Developmental Stages in Human Embryos

AGE (DAYS)	CARNEGIE STAGE	NO. OF SOMITES	LENGTH (MM)*	MAIN EXTERNAL CHARACTERISTICS
20-21	9	1-3	1.5-3.0	<i>Flat embryonic disc. Deep neural groove and prominent neural folds. One to three pairs of somites present.</i> Head fold evident.
22-23	10	4-12	1.0-3.5	<i>Embryo straight or slightly curved.</i> Neural tube forming or formed opposite somites, but widely open at rostral and caudal neuropores. First and second pairs of pharyngeal arches visible.
24-25	11	13-20	2.5-4.5	<i>Embryo curved owing to head and tail folds.</i> Rostral neuropore closing. Otic placodes present. Optic vesicles formed.
26-27	12	21-29	3.0-5.0	<i>Upper limb buds appear.</i> Rostral neuropore closed. Caudal neuropore closing. Three pairs of pharyngeal arches visible. Heart prominence distinct. Otic pits present.
28-30	13	30-35	4.0-6.0	<i>Embryo has C-shaped curve.</i> Caudal neuropore closed. Upper limb buds are flipper-like. Four pairs of pharyngeal arches visible. Lower limb buds appear. Otic vesicles present. Lens placodes distinct. <i>Tail-like caudal eminence present.</i>
31-32	14		5.0-7.0	<i>Upper limbs are paddle shaped.</i> Lens pits and nasal pits visible. Optic cups present.
33-36	15		7.0-9.0	<i>Handplates formed; digital rays visible.</i> Lens vesicles present. Nasal pits prominent. <i>Lower limbs are paddle shaped.</i> Cervical sinuses visible.
37-40	16		8.0-11.0	<i>Footplates formed.</i> Pigment visible in retina. Auricular hillocks developing.
41-43	17		11.0-14.0	<i>Digital rays clearly visible in handplates.</i> Auricular hillocks outline future auricle of external ear. Trunk beginning to straighten. Cerebral vesicles prominent.

AGE (DAYS)	CARNEGIE NO. OF STAGE	SOMITES	LENGTH (MM)*	MAIN EXTERNAL CHARACTERISTICS
44-46	18		13.0-17.0	<i>Digital rays clearly visible in footplates.</i> Elbow region visible. Eyelids forming. Notches between the digital rays in the hands. Nipples visible.
47-48	19		16.0-18.0	<i>Limbs extend ventrally.</i> Trunk elongating and straightening. Midgut herniation prominent.
49-51	20		18.0-22.0	<i>Upper limbs longer and bent at elbows. Fingers distinct but webbed.</i> Notches between the digital rays in the feet. Scalp vascular plexus appears.
52-53	21		22.0-24.0	<i>Hands and feet approach each other. Fingers are free and longer. Toes distinct but webbed.</i>
54-55	22		23.0-28.0	<i>Toes free and longer.</i> Eyelids and auricles of external ears more developed.
56	23		27.0-31.0	<i>Head more rounded and shows human characteristics.</i> External genitalia still have sexless appearance. Distinct bulge still present in umbilical cord, caused by herniation of intestines. <i>Caudal eminence ("tail") has disappeared.</i>

Summary of Key Events during the Embryonic Period

Days	Somites	Length (mm)	Characteristic Features
14–15	0	0.2	Appearance of primitive streak
16–18	0	0.4	Notochordal process appears; hemopoietic cells in yolk sac
19–20	0	1.0–2.0	Intraembryonic mesoderm spread under cranial ectoderm; primitive streak continues; umbilical vessels and cranial neural folds beginning to form
20–21	1–4	2.0–3.0	Cranial neural folds elevated and deep neural groove established; embryo beginning to bend
22–23	5–12	3.0–3.5	Fusion of neural folds begins in cervical region; cranial and caudal neuropores open widely; visceral arches 1 and 2 present; heart tube beginning to fold
24–25	13–20	3.0–4.5	Cephalocaudal folding under way; cranial neuropore closing or closed; optic vesicles formed; otic placodes appear
26–27	21–29	3.5–5.0	Caudal neuropore closing or closed; upper limb buds appear; three pairs of visceral arches
28–30	30–35	4.0–6.0	Fourth visceral arch formed; hindlimb buds appear; otic vesicle and lens placode
31–35		7.0–10.0	Forelimbs paddle-shaped; nasal pits formed; embryo tightly C-shaped
36–42		9.0–14.0	Digital rays in hand and foot plates; brain vesicles prominent; external auricle forming from auricular hillocks; umbilical herniation initiated
43–49		13.0–22.0	Pigmentation of retina visible; digital rays separating; nipples and eyelids formed; maxillary swellings fuse with medial nasal swellings as upper lip forms; prominent umbilical herniation
50–56		21.0–31.0	Limbs long, bent at elbows, knees; fingers, toes free; face more human-like; tail disappears; umbilical herniation persists to end of third month

THANK YOU